ETAS RTA-RTE v12.7.2

Status: FINAL

Error Description

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ETAS RTA-RTE 12.7.2 - Error Description R01 EN – 05.2025

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1 About this Manual

This document is a supplementary reference guide for ETAS RTA-RTE. It lists the messages built into RTA-RTE. For each error message a description of the cause and possible solutions is included.

1.1 Document Conventions

RTA-RTE error messages are tagged with a code of the pattern `<X><Code>` where `<X>` is a single character that indicates the error level (Fatal, Error, Warning, Information); `<Code>` is a unique message identifier.

In this document, messages are ordered by `<Code>`. For example, the error `E001701` can be found simply under `1701`.

2 Error List

This chapter lists all information, warning, error and fatal messages that can be generated by RTA-RTE.

1001:

Element relates to previous occurrence of error

Description:

A model element has been identified as being a possible cause of the referenced error message.

How to Solve:

See the referenced error ID.

Variants:

F001001, E001001, W001001, I001001

1003:

Internal Error

Description:

An internal error occured.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F001003, E001003, W001003, I001003

1005:

Invalid assembly connector

Description:

RTA-RTE has detected an invalid assembly connector that is missing references to provided and required ports.

How to Solve:

An Assembly Connector must reference both provided and required ports.

Variants:

F001005, E001005, W001005, I001005

1006:

Invalid non-local assembly connector

Description:

An Assembly Connector cannot join a port from another composition.

How to Solve:

Ensure that only local `ComponentPrototype` references are configured.

Variants:

F001006, E001006, W001006, I001006

1008:

Invalid delegation connector

Description:

Invalid Delegation connector encountered with a mixture of both Provided and Required ports.

How to Solve:

Ensure that all delegated ports have the same type as the outer port.

Variants:

F001008, E001008, W001008, I001008

1010:

Invalid non-local delegation

Description:

Invalid Delegation connector encountered. The connector attempts to delegate a port from another composition.

How to Solve:

Ensure that only local `ComponentPrototype` references are configured.

Variants:

F001010, E001010, W001010, I001010

1011:

Invalid assembly connector (`PPort`)

Description:

The RTE generator has detected an assembly connected where the provider is not a `PPort`.

How to Solve:

Ensure that the provider of an assembly connector references a `PPort`.

Variants:

F001011, E001011, W001011, I001011

1012:

Invalid assembly connector (`RPort`)

Description:

The RTE generator has detected an assembly connected where the requirer

is not an `RPort`.

How to Solve:

Ensure that the requirer of an assembly connector references an `RPort`.

Variants:

F001012, E001012, W001012, I001012

1013:

Invalid inner port

Description:

RTA-RTE has encountered a delegation connector with an invalid or missing `innerPort` reference.

How to Solve:

Ensure that each delegation connector references an inner port (on an SWC) and an outer port (on the composition).

Variants:

F001013, E001013, W001013, I001013

1014:

Invalid outer port

Description:

RTA-RTE has encountered a delegation connector with an invalid or missing `outerPort` reference.

How to Solve:

Ensure that each delegation connector references an inner port (on an SWC) and an outer port (on the composition).

Variants:

F001014, E001014, W001014, I001014

1020:

Invalid reference

Description:

RTA-RTE has attempted to resolve a reference from a node of an unexpected form.

How to Solve:

Ensure that XML schema validation is enabled and that the input is syntactically correct.

Variants:

F001020, E001020, W001020, I001020

1021:

Invalid type reference

Description:

RTA-RTE has attempted to resolve a type reference from a node of an unexpected form.

How to Solve:

Ensure that XML schema validation is enabled and that the input is syntactically correct.

Variants:

F001021, E001021, W001021, I001021

1022:

Invalid destination

Description:

RTA-RTE has resolved a reference to an unexpected element type.

How to Solve:

Ensure that the reference resolves to one of the expected element types, that XML schema validation is enabled and that the input is syntactically correct.

Variants:

F001022, E001022, W001022, I001022

1023:

Invalid destination attribute (R4.0)

Description:

RTA-RTE has resolved a reference but the resolved element does not match the references `dest` attribute.

How to Solve:

Ensure that the reference contains the correct `dest` attribute, that XML schema validation is enabled and that the input is syntactically correct.

Variants:

F001023, E001023, W001023, I001023

1024:

Invalid BSW referenced (R4.0)

Description:

RTA-RTE has encountered a invalid BSW reference.

How to Solve:

Ensure that referenced `BswSchedulableEntity` elements are defined as `SCHEDULED` and `TASK`, referenced `BswInterruptEntity`s are defined as

`INTERRUPT` with the correct `InterruptCategory` and that referenced `BswCalledEntity` elements are defined as `REGULAR` or `CALLBACK`.

Variants:

F001024, E001024, W001024, I001024

1046:

Expected `RPort` not found

Description:

The RTE generator has detected a reference to a provide port in a context where a require port was expected.

How to Solve:

Ensure that specified reference is to an `RPort`.

Variants:

F001046, E001046, W001046, I001046

1047:

Connected `NvBlockSwComponentType`s encountered

Description:

The RTE generator has detected a connection between ports on two `NvBlockSwComponentType`s. Such a connection is invalid.

How to Solve:

Ensure that ports on instances of `NvBlockSwComponentType` are not connected.

Variants:

F001047, E001047, W001047, I001047

1048:

`ApplicationValueSpecification` missing `SwValueCont`

Description:

No `SwValueCont` container was found in an `ApplicationValueSpecification` initializer.

How to Solve:

Ensure that there is a `SwValueCont` container within the `ApplicationValueSpecification`.

Variants:

F001048, E001048, W001048, I001048

1050:

License checkout fail

Description:

RTA-RTE failed to checkout the specified license feature.

How to Solve:

Please use the ETAS License Manager to request/install a valid license.

Variants:

F001050, E001050, W001050, I001050

1065:

License checkout failed

Description:

RTA-RTE could not check out a suitable licence. *Note: The licensing shared library might display dialogs in this situation, but RTA-RTE does not respond to the button clicks. It is recommended to set* `FLEXLM_BATCH=1` *in your environment to suppress these dialogs*.

How to Solve:

Contact your toolchain supplier or ETAS GmbH to ensure a valid license is installed.

Variants:

F001065, E001065, W001065, I001065

1067:

No INI file

Description:

RTA-RTE failed to load the specified INI file.

How to Solve:

Ensure that the file exists (in the same folder as the RTE generator executable) and is readable by the current user.

Variants:

F001067, E001067, W001067, I001067

1072:

RTE library code template failed to load

Description:

RTA-RTE failed to load the specified file needed by the RTE library code template processor.

How to Solve:

Check that the file is present and is readable by the current user.

Variants:

F001072, E001072, W001072, I001072

1074:

Error during XML parser initialization

Description:

RTA-RTE has detected an error while setting up the XML parser.

How to Solve:

This error can occur while setting the options on the XML parser, or while caching the schemas. The error message contains the text from the parser library. Check the message and review your schemas.

Variants:

F001074, E001074, W001074, I001074

1075:

`TimingEvent` with zero or missing period

Description:

RTA-RTE has encountered a `TimingEvent` with a zero period in the input XML.

How to Solve:

A `TimingEvent` with a zero period (or an omitted period) is not a valid configuration. Please ensure that each `TimingEvent` has a non-zero, positive, period.

Variants:

F001075, E001075, W001075, I001075

1079:

Errors in RTE library code template

Description:

RTA-RTE failed to correctly process all library templates. Specific errors were placed in the output files as `#Error` directives.

How to Solve:

Check that you are using the correct Library Templates as shipped by ETAS. If you are using custom templates (not recommended), then look in the generated library files for `#Error` directives, and correct the errors in the corresponding `.template` file.

Variants:

F001079, E001079, W001079, I001079

1100:

Invalid subfile

Description:

RTA-RTE failed to open the specified command subfile for reading.

How to Solve:

Ensure that the file exists and that the current user has read permission.

Variants:

F001100, E001100, W001100, I001100

1101:

Invalid optimization parameter

Description:

RTA-RTE has detected an invalid optimization strategy either on the command line or within the ECUC file.

How to Solve:

Ensure that the optimization strategy is one of the supported values.

Variants:

F001101, E001101, W001101, I001101

1102:

Invalid COM version parameter

Description:

RTA-RTE has detected an invalid COM version.

How to Solve:

Ensure that the command line specified a valid COM version.

Variants:

F001102, E001102, W001102, I001102

1103:

Unknown option

Description:

RTA-RTE has encountered an unknown option.

How to Solve:

Ensure that the option is listed within the *RTA-RTE Reference Manual* as a valid option. To pass an XML file beginning with a hyphen to the RTE generator use the "`--`" option to stop option processing.

Variants:

F001103, E001103, W001103, I001103

1104:

Invalid verbosity

Description:

RTA-RTE has encountered an invalid verbosity level on the command line.

How to Solve:

Ensure that the specified verbosity is 0, 1, 2 or 3.

Variants:

F001104, E001104, W001104, I001104

1106:

Invalid or missing RTE generation phase

Description:

RTA-RTE failed to detect the generation phase from the specified command-line options.

How to Solve:

Ensure that the command line specifies the generation phase, eg using the `--contract` or `--rte` options.

Variants:

F001106, E001106, W001106, I001106

1107:

Invalid or missing ECU instance reference

Description:

RTA-RTE could not detect the ECU instance for which the RTE is to be generated.

How to Solve:

Ensure that in RTE generation phase an ECU Instance is specified using the `--rte` option.

Variants:

F001107, E001107, W001107, I001107

1108:

Invalid or missing SWC reference

Description:

RTA-RTE could not locate the SWC type for which contract phase has been requested.

How to Solve:

Ensure that a valid reference to an SWC type (or, for R4.0, a BSW implementation) is included on the command line.

Variants:

F001108, E001108, W001108, I001108

1110:

Invalid or missing ECU instance reference

Description:

RTA-RTE could not detect the ECU instance for which the RTE is to be generated.

How to Solve:

Ensure that the reference is specified as an absolute reference of the form '`/package/ecu_instance_name`'.

Variants:

F001110, E001110, W001110, I001110

1111:

Missing filename

Description:

RTA-RTE failed to detect a filename for the `--makedep` command-line option. The requested dependency information cannot be emitted.

How to Solve:

Ensure a valid filename is specified.

Variants:

F001111, E001111, W001111, I001111

1113:

Invalid fast-init option

Description:

RTA-RTE detected an invalid reference when processing a `--fast-init` option.

How to Solve:

Ensure that the reference must be an absolute reference to either an atomic SWC type or a `ModeSwitchEvent` within an atomic Component type

Variants:

F001113, E001113, W001113, I001113

1114:

Invalid boolean switch

Description:

RTA-RTE detected an invalid boolean switch on the command line.

How to Solve:

Ensure that boolean option uses "`0`" (false) or "`1`" (true).

Variants:

F001114, E001114, W001114, I001114

1115:

Invalid or missing calibration method

Description:

RTA-RTE could not detect the calibration method.

How to Solve:

Ensure that a valid method is specified either in the ECUC file or on the command line.

Variants:

F001115, E001115, W001115, I001115

1116:

Invalid calibration disable option

Description:

Invalid reference for calibration disable using the `--disable-calprm` option.

How to Solve:

Ensure that the SWC reference must be an absolute reference to either an atomic SWC type or a Calibration Parameter Component type.

Variants:

F001116, E001116, W001116, I001116

1117:

Invalid main function period

Description:

RTA-RTE has detected an invalid main function period.

How to Solve:

Ensure that the period must be at least 1 microsecond. Time values are specified in seconds. The default period is 10 milliseconds.

Variants:

F001117, E001117, W001117, I001117

1120:

Invalid calibration instantiation method

Description:

RTA-RTE has detected an invalid calibration instantiation method.

How to Solve:

The accepted values are `import` and `allocate`.

Variants:

F001120, E001120, W001120, I001120

1121:

Invalid SWS parameter

Description:

RTA-RTE has detected an invalid SWS specified on the command line.

How to Solve:

In invoking backend processors directly make sure that the correct executable is invoked.

Variants:

F001121, E001121, W001121, I001121

1122:

Message identifier not recognized

Description:

The provided argument is not a valid warning or info message identifier.

How to Solve:

Ensure that the argument provided to `--disable-warning` is valid warning or info message identifier. See the *Reference Manual* for information on valid formats. `--disable-warning` cannot be used to disable error or fatal messages.

Variants:

F001122, E001122, W001122, I001122

1123:

Disabled option

Description:

The specified option has been disabled in the INI file and cannot be used.

How to Solve:

If use of the option is required it must be re-enabled in the INI file.

Variants:

F001123, E001123, W001123, I001123

1126:

Invalid `--strict-unconnected-rport-check` option

Description:

The Strict Unconnected R-Port Check option has been passed an invalid parameter.

How to Solve:

Consult the reference manual and supply a valid parameter

Variants:

F001126, E001126, W001126, I001126

1129:

Invalid EA optimization policy option

Description:

The EA Optimization policy option has been passed an invalid parameter.

How to Solve:

Ensure that the parameter is 'enable', 'disable', or 'asw'. Select 'enable' to enable optimization based on task mapping and 'disable' to disable optimization. Parameter 'asw' enables optimization for non-synchronized EAs only.

Variants:

F001129, E001129, W001129, I001129

1131:

Invalid error

Description:

RTA-RTE has detected an attempt to convert an invalid 'error' message to a warning.

How to Solve:

Ensure that the 'error-as-warning' option is only used to inhibit error messages - it cannot be used to demote a fatal message. This option is intended as a diagnostic tool and that that the RTE generator may function incorrectly after an error message has been treated as a warning.

Variants:

F001131, E001131, W001131, I001131

1132:

Invalid error reporter option

Description:

RTA-RTE did not recognise the argument as a valid error reporter.

How to Solve:

Ensure that the 'error-report' option parameter is one of the following 'console', 'file' or 'xml'.

Variants:

F001132, E001132, W001132, I001132

1134:

Invalid `--strict-config-check` option

Description:

The Strict Config Check option has been passed an invalid parameter.

How to Solve:

Consult the reference manual and supply a valid parameter

Variants:

F001134, E001134, W001134, I001134

1135:

Invalid rounding behavior

Description:

The specified rounding behavior is not one of those that is supported.

How to Solve:

Consult the reference manual and supply a valid parameter

Variants:

F001135, E001135, W001135, I001135

1136:

Unknown argument given for a command-line option

Description:

A command-line option was presented to RTA-RTE with a unknown argument.

How to Solve:

Ensure that the option is provided with valid arguments. See the RTA-RTE reference manual or run –help to see which arguments are valid.

Variants:

F001136, E001136, W001136, I001136

1140:

Multiple calibration methods

Description:

RTA-RTE has detected that the global calibration method has been set on both the command line and the ECU configuration file. The value specified on the command line has been used.

How to Solve:

Ensure that all specifications of the global calibration method are consistent.

Variants:

F001140, E001140, W001140, I001140

1141:

Multiple optimization types

Description:

RTA-RTE has detected that the optimization method has been set on both the command line and the ECU configuration file. The value specified on the command line has been used.

How to Solve:

Ensure that all specifications of the optimization method are consistent.

Variants:

F001141, E001141, W001141, I001141

1142:

Multiple VFB trace enable values

Description:

RTA-RTE has detected that VFB trace enable has been set on both the command line and the ECU configuration file. The value specified on the command line has been used.

How to Solve:

Ensure that all specifications of VFB trace enable are consistent.

Variants:

F001142, E001142, W001142, I001142

1143:

Multiple measurement enable

Description:

RTA-RTE has detected that measurement enable been set on both the command line and the ECU configuration file. The value specified on the command line has been used.

How to Solve:

Ensure that all specifications of measurement enable are consistent.

Variants:

F001143, E001143, W001143, I001143

1144:

Force-basic conflict

Description:

The RTA-RTE has detected conflicting forced-basic configurations for the specified task.

How to Solve:

Ensure that all specifications of force-basic for a task are consistent.

Variants:

F001144, E001144, W001144, I001144

1145:

Multiple tool-chain significant character length values

Description:

RTA-RTE has detected that the tool-chain significant character length has been set on both the command line and the ECU configuration file. The value specified on the command line has been used.

How to Solve:

Ensure that all specifications of the tool-chain significant character length are consistent.

Variants:

F001145, E001145, W001145, I001145

1200:

No configuration plug-ins available

Description:

No configuration plug-in(s) are available. Emitted configuration files will be incomplete.

How to Solve:

Check that the application INI file contains a valid `ConfigDLLs` section.

Variants:

F001200, E001200, W001200, I001200

1201:

Multiple plug-ins accepted port instance

Description:

Two (or more) primary plug-ins processed the same port instance.

How to Solve:

Ensure that any user-implemented plug-ins do not conflict with standard RTA-RTE plug-ins. If no user-specified plug-ins are in use then please contact ETAS GmbH for support.

Variants:

F001201, E001201, W001201, I001201

1202:

No output plug-ins available

Description:

No output plug-in(s) are available. No output files will be generated.

How to Solve:

Check that the application INI file contains a valid `OutputDLLs` section.

Variants:

F001202, E001202, W001202, I001202

1203:

Unable to access folder

Description:

The specified folder does not exist or the current user does not have the correct permissions within the folder.

How to Solve:

Check the existence and user permissions of the folder.

Variants:

F001203, E001203, W001203, I001203

1204:

No plug-ins accepted port instance

Description:

No primary plug-ins processed the specified port instance. This error can be raised when a previous error has caused processing to be aborted by a primary plug-in so, ultimately, no lug-in fully handles the port instance.

How to Solve:

Eliminate all errors that could cause processing to be aborted. Ensure that any user-implemented plug-ins do not conflict with standard RTA-RTE plug-ins. If the error persists and no user-specified plug-ins are in use then please contact ETAS GmbH for support.

Variants:

F001204, E001204, W001204, I001204

1205:

No ports

Description:

The specified SWC types does not declare any provided or required ports.

How to Solve:

This warning message is for information only but does indicate that the SWC cannot communicate and therefore the configuration may not be as intended. The message can be suppressed if this is the expected behavior.

Variants:

F001205, E001205, W001205, I001205

1206:

Duplicate task name

Description:

The specified task is defined multiple times in the input configuration; this is invalid

How to Solve:

Remove the duplication.

Variants:

F001206, E001206, W001206, I001206

1207:

Incomplete RTE/BSW Event information in Contract phase

Description:

During Contract phase processing RTA-RTE has detected one or more runnable entities in a SWC type without an associated RTE event that starts the runnable. This will mean the contract for the runnable (prototype and related APIs) will be generated.

How to Solve:

Ensure that all runnables are referenced by at least one RTE Event. If not required with warning can be dismissed.

Variants:

F001207, E001207, W001207, I001207

1209:

No RTE Configuration

Description:

RTA-RTE cannot locate an RTE module configuration.

How to Solve:

RTA-RTE requires an RTE module configuration to be able to determine RTE event mappings, etc. Ensure that the input contains an ECU configuration that references an RTE module configuration.

Variants:

F001209, E001209, W001209, I001209

1210:

Multiple module configurations

Description:

RTA-RTE has encountered multiple module configurations for the specified module. This is invalid according to the AUTOSAR multiplicity rules.

How to Solve:

Ensure that at most one such module exists in the ECU configuration processed by RTA-RTE.

Variants:

F001210, E001210, W001210, I001210

1211:

No ECU Configuration

Description:

RTA-RTE cannot locate an ECU configuration for the specified ECU instance.

How to Solve:

Ensure that the input contains an ECU configuration. The ECU configuration must reference an OS and RTE configuration and, optionally, a COM configuration.

Variants:

F001211, E001211, W001211, I001211

1212:

Invalid RTE generation mode

Description:

The processed RTE configuration defines an invalid RTE generation mode.

How to Solve:

Ensure that the generation mode is one of the listed supported values.

Variants:

F001212, E001212, W001212, I001212

1213:

Invalid mode configuration

Description:

RTA-RTE has detected an invalid mode configuration when processing the specified port and `ModeDeclarationGroupPrototype`. The error message indicates how many names RTA-RTE has attempted to assign to the mode instance.

How to Solve:

Ensure that each mode instance has exactly one mode provider.

Variants:

F001213, E001213, W001213, I001213

1214:

Non-local mode configuration

Description:

RTA-RTE has detected an invalid mode configuration when processing the

specified port and `ModeDeclarationGroupPrototype`. Non-local connections (where either the mode manager and mode user mapped to different ECUs or one/both of the SWC instances is not mapped to an ECU instance) are not permitted.

How to Solve:

Ensure all mode connections are between SWC instances located on the same ECU.

Variants:

F001214, E001214, W001214, I001214

1215:

Multiplicity check failure for `RteGeneration`

Description:

RTA-RTE has encountered an invalid number of `RteGeneration` containers according to the AUTOSAR multiplicity rules which state that there must be exactly one such container.

How to Solve:

Ensure that exactly one `RteGeneration` container exists in the RTE module configuration processed by RTA-RTE.

Variants:

F001215, E001215, W001215, I001215

1218:

Invalid reference to SWC type

Description:

When specifying the contract phase SWC the reference passed on the command line is not an SWC reference.

How to Solve:

The reference must be an absolute reference to an SWC type.

Variants:

F001218, E001218, W001218, I001218

1219:

Invalid queue length

Description:

RTA-RTE has detected an invalid queue length.

How to Solve:

Ensure all configured queues have at least one element.

Variants:

F001219, E001219, W001219, I001219

1220:

Invalid client invocation

Description:

RTA-RTE has detected invalid client invocation that is both synchronous and asynchronous.

How to Solve:

Ensure that each client is invoked either synchronously or asynchronously – but not both.

Variants:

F001220, E001220, W001220, I001220

1221:

Invalid call-point timeout

Description:

RTA-RTE has detected multiple server call-points for the same client that contain different timeouts. This is invalid.

How to Solve:

Ensure that all `ServerCallPoint`s that reference the same operation specify the same timeout value.

Variants:

F001221, E001221, W001221, I001221

1222:

Initial values for queued data

Description:

RTA-RTE has detected the invalid configuration of initial values for queued data items.

How to Solve:

Initial values can only be specified for unqueued data.

Variants:

F001222, E001222, W001222, I001222

1223:

Too many modes

Description:

RTA-RTE has encountered a mode declaration group that contains too many mode declarations. The maximum supported number of modes per mode declaration group is 32.

How to Solve:

Reduce the number of modes within the group.

Variants:

F001223, E001223, W001223, I001223

1224:

Invalid duplication

Description:

RTA-RTE has encountered an element that has a duplicate name. Elements and packages with the same name are merged and all sub-elements within the namespace formed by the package/element must have unique names.

How to Solve:

Check that all elements within a namespace are either themselves splittable across multiple files or have unique names. Only the contents splittable elements, eg packages, software components, etc., are mergeable. This error can also occur if an XML file is specified multiple times on the command line since this will create many duplicates of non-splittable elements.

Variants:

F001224, E001224, W001224, I001224

1225:

Invalid client-server receive mode

Description:

RTA-RTE has detected an inconsistent client-server receive mode.

How to Solve:

It is not permitted to use a client with a blocking poll (wake up of wait point) and activation of runnable entity of a result runnable.

Variants:

F001225, E001225, W001225, I001225

1226:

Invalid sender-receiver receive mode

Description:

RTA-RTE has detected an inconsistent sender-receiver receive mode. Reception can be blocking or activation of runnable entity but not both.

How to Solve:

It is not permitted to use a receive with a blocking poll (wake up of wait point) and activation of runnable entity of a receiver runnable. Check that no `DataReceiveEvent` that references the port/data item both starts a runnable-entity and is referenced by a `WaitPoint`.

Variants:

F001226, E001226, W001226, I001226

1227:

Feedback event Invalid sender-receiver feedback mode

Description:

RTA-RTE has detected a `DataSendCompletedEvent` or `ModeSwitchedAckEvent` that is configured both to start a `RunnableEntity` and to wake a `WaitPoint`. It is not possible to do both.

How to Solve:

Remove either the reference from the `WaitPoint` to the `RteEvent` or remove the reference from the `RteEvent` to the `RunnableEntity`.

Variants:

F001227, E001227, W001227, I001227

1228:

Invalid runnable reference

Description:

RTA-RTE has encounter a `startOnEventRef` that does not reference a runnable entity

How to Solve:

Ensure the element specifies a runnable entity within the SWC type.

Variants:

F001228, E001228, W001228, I001228

1229:

Multiple PDAV lists

Description:

RTA-RTE has detected multiple `PortDefinedArgumentList` elements for the specified port.

How to Solve:

Ensure that at most one `PortDefinedArgumentList` element is specified for a port.

Variants:

F001229, E001229, W001229, I001229

1231:

Missing or invalid SWC to ECU mapping

Description:

RTA-RTE cannot locate a `SwCompToECUMapping` element within the `System` that references the ECU instance for which an RTE has been requested. No RTE can be generated.

How to Solve:

Ensure that a valid SWC to ECU mapping is provided

Variants:

F001231, E001231, W001231, I001231

1232:

Invalid ECU instance

Description:

The RTE Generator could not identify the applicable ECU Instance.

How to Solve:

Check the path placed on the RTE Generator command line. RTA-RTE supports command-line ECU instance selection by either a reference to an `EcuInstance` or a reference to an `EcuConfiguration` that references a system extract.

Variants:

F001232, E001232, W001232, I001232

1234:

Invalid internal triggering point

Description:

RTA-RTE has encountered an internal triggering point that does not contain a valid `startOnEventRef` reference.

How to Solve:

Ensure that the specified internal triggering point references an internal trigger event.

Variants:

F001234, E001234, W001234, I001234

1235:

Invalid SWC reference

Description:

RTA-RTE has encountered a reference that should refer to an application or calibration SWC but does not.

How to Solve:

Ensure that the reference specified in the error points to the correct type of SWC.

Variants:

F001235, E001235, W001235, I001235

1236:

Missing or invalid root composition reference

Description:

RTA-RTE cannot locate the root (top-level) composition for the system. No RTE can be generated.

How to Solve:

Ensure that the `System` contains a valid `RootSwCompositionPrototype` with a valid `SoftwareCompositionTref` to a valid `CompositionSwComponentType`.

Variants:

F001236, E001236, W001236, I001236

1237:

Malformed variation point

Description:

RTA-RTE has detected a variation point without a required `shortLabel`.

How to Solve:

Ensure that the specified variation point includes a short label element.

Variants:

F001237, E001237, W001237, I001237

1238:

Call point references a provide port

Description:

RTA-RTE has detected a server call point that references a provide port, not a require port.

How to Solve:

Ensure that the call point references a require port on the same software component.

Variants:

F001238, E001238, W001238, I001238

1239:

Call point references a port on the wrong SWC

Description:

RTA-RTE has detected a server call point that references a port that is not on the same software component.

How to Solve:

Ensure that the call point references a require port on the same software component.

Variants:

F001239, E001239, W001239, I001239

1251:

Invalid SWC Implementation

Description:

RTA-RTE has detected a `SwcImplementationMapping` element where the referenced internal behavior is not associated with the same SWC type as the referenced component prototype.

How to Solve:

Ensure that the SWC-implementation mapping references an internal behavior associated with the same SWC type as the referenced component prototype.

Variants:

F001251, E001251, W001251, I001251

1252:

DSMQ queue length extended

Description:

RTA-RTE has detected a shared-mode queue with insufficient queue length – at least one entry per mode instance is required to handle initialization events.

How to Solve:

Ensure all the shared-mode queue length is at least as large as the number of mode instances. If not required this warning can be dismissed.

Variants:

F001252, E001252, W001252, I001252

1260:

Mode disabling specified for operation invoked

Description:

RTA-RTE has detected an `OperationInvokedEvent` with mode dependencies. This configuration is invalid.

How to Solve:

Remove the mode disabling dependencies. The recommended AUTOSAR equivalent is for the server to respond within an appropriate application error when the server operation is not supported in a mode.

Variants:

F001260, E001260, W001260, I001260

1261:

Multiple mode managers

Description:

RTA-RTE has detected multiple mode managers (provide ports) specified for a mode instance; a mode user (require port) cannot have more than one mode manager (provide port).

How to Solve:

Ensure that each mode instance has at most one mode manager.

Variants:

F001261, E001261, W001261, I001261

1262:

Incomplete RTE/BSW Event mapping

Description:

RTA-RTE has detected an incomplete RTE/BSW Event mapping container. Missing or invalid reference to an RTE or BSW Event.

How to Solve:

Ensure that all RTE/BSW Event mapping containers reference an RTE/BSW Event.

Variants:

F001262, E001262, W001262, I001262

1263:

Invalid mode disabling specified for mode switch

Description:

RTA-RTE has detected a `ModeSwitchEvent` with mode dependencies that reference the same mode declaration as the event's mode. This configuration is invalid since it leaves the mode switch event unable to trigger.

How to Solve:

With any mode disabling dependencies that reference the same mode(s) as the `ModeSwitchEvent`.

Variants:

F001263, E001263, W001263, I001263

1265:

Malformed RTE Event mapping

Description:

RTA-RTE has detected a malformed RTE Event mapping; only mappings for `TimingEvent` or `BackgroundEvent` can reference an `OsAlarm` or an `OsSchTblExpiryPoint`.

How to Solve:

Remove the `OsAlarm` or `OsSchTblExpiryPoint` reference from the RTE Event mapping

Variants:

F001265, E001265, W001265, I001265

1266:

Invalid RTE Event mapping

Description:

RTA-RTE has detected an invalid RTE Event mapping that specifies both an `OsAlarm` and `OsSchTblExpiryPoint`.

How to Solve:

Ensure that the specified RTE Event mapping specifies either an `OsAlarm` and `OsSchTblExpiryPoint` but not both

Variants:

F001266, E001266, W001266, I001266

1267:

Invalid mode RTE event mapping

Description:

RTA-RTE has detected an invalid RTE event mapping; all runnables triggered by a `ModeSwitchEvent` for the same mode instance must be mapped to the same task.

How to Solve:

Ensure that all specified `ModeSwitchEvent`s trigger runnables mapped to the same task.

Variants:

F001267, E001267, W001267, I001267

1269:

Invalid RTE Event mapping

Description:

RTA-RTE has encountered an invalid RTE event mapping for a mode instance. To ensure correct termination of `ON_EXIT` runnables before invocation of `ON-ENTRY` runnables AUTOSAR requires that for a particular mode instance, all runnables triggered by a `ModeSwitchEvent` on `ON-ENTRY` to a mode must have a higher `rtePositionInTask` any runnables triggered on `ON-EXIT` from a mode within the mode instance.

How to Solve:

Ensure that the specified RTE Event is mapped after runnables triggered on `ON-ENTRY`.

Variants:

F001269, E001269, W001269, I001269

1272:

File open failed

Description:

RTA-RTE failed to open the specified file for writing.

How to Solve:

Check that all path components exist and that the current user has the correct permissions to create and write to files in the folder.

Variants:

F001272, E001272, W001272, I001272

1273:

ECU configuration mismatch

Description:

No ECU configuration was found matching the specified ECU instance name. Using the only existing configuration.

How to Solve:

Ensure that names match, or specify the ECU configuration on the command line.

Variants:

F001273, E001273, W001273, I001273

1276:

File open failed

Description:

RTA-RTE failed to open the specified file for reader.

How to Solve:

Check that all path components exist and that the current user has the correct permissions to read files in the folder.

Variants:

F001276, E001276, W001276, I001276

1277:

No report-template root element

Description:

RTA-RTE failed to locate the start tag when processing the specified report template file.

How to Solve:

Ensure that the start tag must be present in the template file to ensure generated output is inserted at the correct point.

Variants:

F001277, E001277, W001277, I001277

1278:

Conflicting exclusive area method

Description:

RTA-RTE has detected multiple, conflicting, implementation methods for an exclusive areas.

How to Solve:

Ensure that all such methods are consistent.

Variants:

F001278, E001278, W001278, I001278

1279:

Shared OS resource

Description:

RTA-RTE has detected a `ExclusiveAreaImplementation` for the specified SWCI that specifies a resource which is already used by an different SWC instance.

How to Solve:

SWC instances are not permitted to share OS Resources in this way. Consider creating additional OS Resources and/or correcting or removing `ExclusiveAreaOsResourceRef` configuration items within `ExclusiveAreaImplementation` containers.

Variants:

F001279, E001279, W001279, I001279

1280:

`ModeSwitchEvent` for Synchronous Mode Switch is not mapped to an `OsTask`

Description:

All `ModeSwitchEvent`s for a Synchronous mode machine instance must be mapped to a single task.

How to Solve:

For a Synchronous Mode Machine Instance, all `ModeSwitchEvent`s must be mapped to a single task. The priority of that task must be no higher than the lowest priority task containing Runnables with a `ModeDisablingDependency` on that `ModeMachineInstance`. Alternatively, to allow an Asynchronous mode switch ensure that all mode users explicitly enable it and all the

`ModeSwitchEvent`s of the mode users are unmapped. Note that asynchronous mode switch cannot be used when the manager and users are in different `OsApplication`s.

Variants:

F001280, E001280, W001280, I001280

1281:

Conflicting constant mappings

Description:

RTA-RTE has detected conflicting mappings for the specified physical constant.

How to Solve:

Ensure that the `ConstantValueMappingRef`s of the SWC's `InternalBehavior` are consistent.

Variants:

F001281, E001281, W001281, I001281

1282:

Invalid option parameter

Description:

RTA-RTE detected an invalid option parameter

How to Solve:

Ensure that option parameter is a valid integer.

Variants:

F001282, E001282, W001282, I001282

1283:

Invalid SWCI name

Description:

RTA-RTE cannot set the SWC instance internal name to the specified value since the name must be globally unique.

How to Solve:

Check that all `FlatInstanceDescriptor`s specify unique names.

Variants:

F001283, E001283, W001283, I001283

1285:

Report template file missing

Description:

A report has been requested but no template file set.

How to Solve:

Specify a report template or explicitly indicate no template is required by using the template name 'null'.

Variants:

F001285, E001285, W001285, I001285

1286:

Chapter appears more than once in template

Description:

Chapter appears more than once in template.

How to Solve:

Each chapter tag (like <?rtarte MY-CHAPTER-HERE?> can only appear once in any template. Edit the template.

Variants:

F001286, E001286, W001286, I001286

1287:

Unknown processing instruction in template

Description:

Template contains a processing instruction starting <?rtarte and containing only lower-case characters, but the instruction did not indicate any known report-processing command

How to Solve:

Each chapter tag (like <?rtarte MY-CHAPTER-HERE?> can only contain a recognized report-processing instruction or a chapter tag. Chapter tags are uppercase, instructions are lowercase. Edit the template.

Variants:

F001287, E001287, W001287, I001287

1288:

Unsupported `ModeSwitchedAckEvent` mapping

Description:

RTA-RTE has encountered a `ModeSwitchedAckEvent` mapped to a task such that a subsequent mode transition may start before acknowledgement is complete.

How to Solve:

Ensure that the specified `ModeSwitchedAckEvent` is mapped to a task at higher priority than all `ModeSwitchEvent`s for the mode instance or is mapped to the mode switch task before all the `ModeSwitchEvent`s for the mode instance. For

a mode machine instance with distributed tasks the `ModeSwitchedAckEvent` must not be mapped to any of the mode switch tasks, so must be mapped to a task at higher priority than all the mode switch tasks.

Variants:

F001288, E001288, W001288, I001288

1289:

Unsupported use of a `ModeSwitchedAckEvent` on a cross-core mode machine instance

Description:

RTA-RTE has encountered a `ModeSwitchedAckEvent` on a cross-core mode machine instance. Acknowledgement will race with any subsequent transition.

How to Solve:

RTA-RTE does not yet support cross-core synchronisation of mode switch acknowledgement with the rest of the mode machine instance.

Variants:

F001289, E001289, W001289, I001289

1290:

File hash failed

Description:

RTA-RTE failed to generate a hash value for the specified file. Incremental build cannot be completed.

How to Solve:

Ensure that the file exists and can be read by the current user.

Variants:

F001290, E001290, W001290, I001290

1291:

File operation failed

Description:

The operating system has indicated that a file operation has failed.

How to Solve:

The error message includes further details as to the cause.

Variants:

F001291, E001291, W001291, I001291

1292:

Missing report section in template

Description:

Some XML tags could not be placed in the report because there was no section with a matching tag.

How to Solve:

Edit the template file to add tags like those listed in the error message at appropriate places for output in the report.

Variants:

F001292, E001292, W001292, I001292

1293:

SWC to ECU mapping ignored

Description:

RTA-RTE has ignored the `SwCompToECUMapping` element(s) within the `System` since it has `category ECU_EXTRACT`. Instead all component prototypes within the root SW-composition are used for RTE generation.

How to Solve:

Remove the SWC to ECU mapping within the ECU extract. Alternatively this warning can be suppressed.

Variants:

F001293, E001293, W001293, I001293

1294:

Implicit access combined with `enableUpdate`

Description:

The specified implicit access (Read Access) is to a data item for which `enableUpdate` has been specified. This combination is invalid; only explicit access is permitted.

How to Solve:

Ensure that only explicit access (read by value or by argument) is used for variable data prototypes for which `enableUpdate` is specified.

Variants:

F001294, E001294, W001294, I001294

1295:

Direct server call crosses trust boundary

Description:

The specified server call has been implemented as a direct function call but the client and server partitions are not both trusted.

How to Solve:

To prevent the call from being optimized to a direct function call, map the `OperationInvokedEvent` to a task.

Variants:

F001295, E001295, W001295, I001295

1296:

Missing RTE Event mapping

Description:

When processing modes, RTA-RTE found an event without a `rtePositionInTask`.

How to Solve:

Mode events must all have corresponding `rtePositionInTask`. Even for asynchronous tasks, which do not need an `rteMappedToTaskRef`, there must be an `RteEventToTaskMapping` containing a `rtePositionInTask`.

Variants:

F001296, E001296, W001296, I001296

1297:

Legacy support for ECU extraction invoked

Description:

AUTOSAR specifies that an `ECU_EXTRACT` should be supplied to the RTE Generator for the generation step. RTA-RTE also contains support for the hierarchical model view used in former times. This message indicates that RTA-RTE is working with a complete system so is effectively creating an ECU extract before generation.

How to Solve:

Ensure that RTA-RTE has been provided with an ECU extract and set the `category` of the `System` element to '`ECU_EXTRACT`'.

Variants:

F001297, E001297, W001297, I001297

1298:

No ECU Configuration present

Description:

No ECU configuration was found in all the input files.

How to Solve:

Please supply an `EcucValueCollection` on the input. RTA-RTE expects to find one `EcucValueCollection` containing an `ecuExtractRef` element and `moduleRef` elements for the RTE and OS.

Variants:

F001298, E001298, W001298, I001298

1299:

Multiple ECU Configurations present

Description:

There were multiple ECU configurations on the input. RTA-RTE was unable to determine the ECU to build.

How to Solve:

Either supply exactly one `EcucValueCollection` on the input, or specify the ECU to generate using the argument to the `--rte` option.

Variants:

F001299, E001299, W001299, I001299

1300:

No Swc mapping

Description:

RTA-RTE was unable to locate an `SwCompToECUMapping` element within the `System`.

How to Solve:

Ensure that a valid SWC mapping is supplied. In particular, check that the correct `System` element is referenced from the `EcuConfiguration`.

Variants:

F001300, E001300, W001300, I001300

1301:

`EcucValueCollection` does not reference a `System` element

Description:

The `EcucValueCollection` did not reference a `System` element via its `EcuExtractRef`. An `EcucValueCollection` must contain an `EcuExtractRef`.

How to Solve:

Ensure that the `EcuConfiguration` refers to exactly one `System` element via its `EcuExtractRef`. If the `EcuConfiguration` is split across multiple files, at least one fragment must contain an `EcuExtractRef`.

Variants:

F001301, E001301, W001301, I001301

1302:

Ambiguous ECU configuration

Description:

RTA-RTE failed to identify a single, unambiguous `EcuConfiguration` that refers to a `System` element.

How to Solve:

Ensure that the RTE generator can locate the required system. Alternatively consider referencing the required ECU configuration on the command line.

Variants:

F001302, E001302, W001302, I001302

1303:

Multiple ECU instances

Description:

RTA-RTE has detected multiple ECU instances referenced within the referenced system.

How to Solve:

Identify ECU instance on the command line

Variants:

F001303, E001303, W001303, I001303

1304:

No ECU instance

Description:

The RTE Generator could not identify the applicable ECU Instance.

How to Solve:

The absolute path to an `EcuInstance` or `EcuConfiguration` should be placed on the RTE Generator command line.

Variants:

F001304, E001304, W001304, I001304

1306:

Inter-ECU connection to `EcuAbstractionSwComponent`

Description:

RTA-RTE has detected an attempt to connect a port of an `EcuAbstractionSwComponentPrototype` to a `SoftwareComponent` on a different ECU.

How to Solve:

Either remove the `SwConnector` or map the connected Sensor- or `ActuatorSwComponent`s to the same ECU as the `EcuAbstractionSwComponent`.

Variants:

F001306, E001306, W001306, I001306

1307:

No `DataReceivedEvent` found in the context of NV store-immediate semantics

Description:

Store-immediate semantics have been requested for an `NvBlockDescriptor` but there is no `DataReceivedEvent` associated with the `NvBlockDescriptor`. It is mandatory to have a `DataReceivedEvent` that triggers the immediate writing of the `NvData` that is mapped to this `NvBlockDescriptor` when store-immediate semantics are requested.

How to Solve:

Ensure that there is a `DataReceivedEvent` that points to the `NvData` that is subject to store-immediate semantics.

Variants:

F001307, E001307, W001307, I001307

1308:

No `TimingEvent` found in the context of NV store-cyclic semantics

Description:

Store-cyclic semantics have been requested for an `NvBlockDescriptor` but there is no `TimingEvent` associated with the `NvBlockDescriptor`. It is mandatory to have a `TimingEvent` that triggers the periodic potential writing of the `NvData` that is mapped to this `NvBlockDescriptor`. Please refer to `constr_1309`.

How to Solve:

Ensure that the `NvBlockDescriptor` references a `TimingEvent` via its `timingEventRef`.

Variants:

F001308, E001308, W001308, I001308

1309:

NVRAM writing semantics are disabled because `supportsDirtyFlag` is not set

Description:

At least one of `storeImmediate`, `storeCyclic` or `storeAtShutdown` is set on the `NvBlockDescriptor`. `nvBlockNeeds`, but `NvBlockDescriptor`. `supportsDirtyFlag` has not been set to `true`.

How to Solve:

As well as specifying the writing strategy via `NvBlockNeeds`, `NvBlockDescriptor`. `supportsDirtyFlag` should be set to `true`.

Variants:

F001309, E001309, W001309, I001309

1310:

`NvBlockDescriptor`. `timingEvent` without `storeCyclic` = `true`

Description:

An `NvBlockDescriptor` references a `TimingEvent`, but does not have store-cyclic semantics enabled. This contravenes `constr_1309`.

How to Solve:

Either enable store-cyclic semantics for the `NvBlockDescriptor`, or remove its `timingEventRef`.

Variants:

F001310, E001310, W001310, I001310

1311:

`DataReceivedEvent` relates to an `NvBlockDescriptor` without `storeImmediate = true`

Description:

A `NvBlockDescriptor` that is mapped to `NvData` referenced by a `DataReceivedEvent` does not have store-Immediate semantics enabled.

How to Solve:

Either enable store-Immediate semantics for the `NvBlockDescriptor`, or remove the `DataReceivedEvent`.

Variants:

F001311, E001311, W001311, I001311

1312:

1:N NVRAM communication

Description:

A `VariableDataPrototype` in the role `nvData` is associated with more than one `NvBlockDescriptor`. This contravenes constr_1404.

How to Solve:

Ensure that all `NvBlockDataMappings` referencing a given `nvData` in a given `NvBlockSwComponentType`. `Port` reference the same `NvBlockDescriptor`.

Variants:

F001312, E001312, W001312, I001312

1313:

`PortDefinedArgumentValue` is not used for `BlockID`

Description:

When implementing an NVRAM writing strategy, eg store-immediate semantics, RTA-RTE found a `portDefinedArgumentValue` relating to the port. It may be that this `portDefinedArgumentValue` is intended to set the `BlockID` but this use of `portDefinedArgumentValue` is deprecated in AUTOSAR and is not supported in RTA-RTE.

How to Solve:

Specify the `BlockID` using the ECU Value File and either remove the `portDefinedArgumentValue` or suppress the warning.

Variants:

F001313, E001313, W001313, I001313

1314:

`NvMBlockDescriptor` missing for `NvBlockDescriptor` with dirty flag

Description:

An NVRAM writing strategy is enabled for this `NvBlockDescriptor` but RTA-RTE was unable to determine the symbol to use for the `blockId` argument to the relevant NvM API. This error has arisen because there is either no `RteNvRamAllocation` aggregated in the relevant `RteSwComponentInstance` in the `EcucValueCollection`, or there are multiple `RteNvRamAllocation`s that reference the `NvBlockDescriptor`.

How to Solve:

The `RteSwComponentInstance` relating to the `NvBlockSwComponent` should contain exactly one `RteNvRamAllocation` for each `RteSwNvBlockDescriptor`, associating it with an `RteNvmBlock` belonging to the NvM module.

Variants:

F001314, E001314, W001314, I001314

1315:

`RTEEvent` in `NvblockSwComponentType` not associated with any `NvBlockDescriptor`

Description:

In an `NvBlockSwComponentType`, a `RunnableEntity` is triggered by an `RTEEvent` of type `TimingEvent` that is not referenced by any `NvBlockDescriptor`, or, of type `DataReceivedEvent` that does not indirectly reference an `NvBlockDescriptor` by way of an `NvBlockDataMapping`.

How to Solve:

Ensure that each `RunnableEntity`s is started by `RTEEvent` that is associated with an `NvBlockDescriptor`.

Variants:

F001315, E001315, W001315, I001315

1316:

Model too large

Description:

The model requires the generation of more of the given artefacts than RTA-RTE supports. An internal counter has overflowed and some of the generated names may collide.

How to Solve:

Reduce the complexity of the software configuration for the ECU if possible or contact ETAS if support for larger models is required.

Variants:

F001316, E001316, W001316, I001316

1330:

Invalid or missing mode request type map

Description:

RTA-RTE has failed to detect a `ModeRequestTypeMap` within the specified SWC context.

How to Solve:

Ensure that a `ModeRequestTypeMap` exists for each used mode declaration group and that it is referenced from the SWC's internal behavior.

Variants:

F001330, E001330, W001330, I001330

1331:

Invalid implementation type

Description:

RTA-RTE is unable to determine the implementation type from the specified reference.

How to Solve:

Ensure that the reference is correct and refers to a valid element.

Variants:

F001331, E001331, W001331, I001331

1332:

Mode group conflict

Description:

RTA-RTE has detected that the SWC uses multiple mode declaration groups that have the same `shortName` but declare non-identical mode sets.

How to Solve:

Ensure mode groups with the same `shortName` that are used by an SWC are consistent.

Variants:

F001332, E001332, W001332, I001332

1333:

Invalid schedule point

Description:

RTA-RTE has detected an RTE event mapping that specified an invalid `RteOsSchedulePoint`.

How to Solve:

Ensure that only valid enumeration values are used within the schedule point.

Variants:

F001333, E001333, W001333, I001333

1335:

Unmapped application type

Description:

RTA-RTE has failed to find a mapping to an implementation data type for the specified application type.

How to Solve:

Ensure that all application types used by an SWC are mapped to implementation types.

Variants:

F001335, E001335, W001335, I001335

1336:

SWC in BSW generation

Description:

RTA-RTE has detected a SWC definition when running as the BSW Scheduler Generator.

How to Solve:

Ensure that no SWC definitions are present.

Variants:

F001336, E001336, W001336, I001336

1337:

Unmapped application (physical) value

Description:

An `initialValue` or `invalidValue` is specified in the physical domain for a `DataPrototype` of `ApplicationDataType` without a mapping to a value specification in the implementation domain. RTA-RTE cannot currently derive the equivalent value for a physical value in the implementation domain automatically. The value specified has been used without conversion so this warning can be ignored if the value has been converted manually.

How to Solve:

Either specify the `initialValue` or `invalidValue` using an `ApplicationValueSpecification` or using a `ConstantReference` to a `ConstantSpecification` that is mapped by a `ConstantSpecificationMappingSet` referenced from the SWC to another `ConstantSpecification` for the corresponding `ImplementationDataType`.

Variants:

F001337, E001337, W001337, I001337

1338:

Data Conversion precision/range constraints exceeded

Description:

RTA-RTE cannot represent the conversion between the two types specified in the error message, because precision or range constraints were exceeded when performing functional composition.

How to Solve:

Consider reducing the number of significant figures in a `factorSIToUnit`, `offsetSIToUnit`, or `compuRationalCoefficient` that is involved in the data conversion path.

Variants:

F001338, E001338, W001338, I001338

1339:

`ImplmentationDataType` references `CompuMethod` of invalid `category`

Description:

RTA-RTE has detected that an `ImplementationDataType` or `ImplementationDataTypeElement` referenced in its `SwDataDefProps` a `CompuMethod` with a `category` other than `TEXTTABLE`, `BITFIELD_TEXTTABLE`, `SCALE_LINEAR_AND_TEXTTABLE`, `SCALE_RATIONAL_AND_TEXTTABLE`. The `CompuMethod` has been ignored.

How to Solve:

If the `ImplementationDataType` is meant to express an Enumerated type, then it should reference a `CompuMethod` having `category` `TEXTTABLE`, `BITFIELD_TEXTTABLE`, `SCALE_LINEAR_AND_TEXTTABLE`, or `SCALE_RATIONAL_AND_TEXTTABLE`; and having point-range `CompuScale`s expressing the enumeration constants. If the `ImplementationDataType` does not represent an Enumeration, remove the `CompuMethodRef` from its `SwDataDefProps`.

Variants:

F001339, E001339, W001339, I001339

1340:

Non-unique type name

Description:

RTA-RTE has detected non-unique SWC type names.

How to Solve:

AUTOSAR requires that all SWC types must have unique names irrespective of the package hierarchy.

Variants:

F001340, E001340, W001340, I001340

1341:

Conflicting application errors

Description:

RTA-RTE has detected conflicting application errors declared by interfaces with the same `shortName`.

How to Solve:

Ensure that all similarly named interfaces used within a SWC have application errors with compatible names and error codes.

Variants:

F001341, E001341, W001341, I001341

1342:

Invalid texttable connection

Description:

RTA-RTE has detected that a `TextTableMapping` was provided for a connection between the specified data types but the types' `CompuMethod`s are of `Linear category`s. `TextTableMappings` should be used with types whose `CompuMethod`s are of `category TEXTTABLE`. The `TextTableMapping` has been ignored.

How to Solve:

Ensure that valid compumethods are used.

Variants:

F001342, E001342, W001342, I001342

1343:

Unexpected or invalid tag

Description:

RTA-RTE has detected an invalid, or missing, `ValueSpec` element.

How to Solve:

RTA-RTE supports `NumericalValueSpecification`, `TextValueSpecification`,

`RecordValueSpecification`, `ArrayValueSpecification` and `ConstantReference`.

Variants:

F001343, E001343, W001343, I001343

1344:

Invalid `AutosarVariableIref`

Description:

RTA-RTE has detected an `AutosarVariableIref` element which does not reference a Ram block typed by an AUTOSAR `ApplicationDataType`.

How to Solve:

If a `NvBlockDataMapping` needs to reference a Ram block typed by an `ImplementationDataType`, use an `AutosarVariableInImplDatatype` element.

Variants:

F001344, E001344, W001344, I001344

1345:

Untyped application value

Description:

RTA-RTE is trying to convert an `ApplicationValueSpec` into internal representation, but the type of the target (eg `DataPrototype` or some sub-element thereof) could not be determined.

How to Solve:

Check that all locations using the identified `ApplicationValueSpecification` reference some `ApplicationPrimitiveDataType` or reference a `CompositeDataType` with some sub-element having `ApplicationPrimitiveDataType` in a corresponding location to the location of the identified `ApplicationValueSpecification` within its parent `ValueSpecification`. See feature TFS 351344 in the Release Note for more information.

Variants:

F001345, E001345, W001345, I001345

1347:

Too many initializers were found in the Application Value Spec

Description:

An `ApplicationValueSpec` was encountered with more initializers than the corresponding `ApplicationDataType` has members.

How to Solve:

Ensure that the initializer shape matches the underlying `ApplicationDataType`.

Variants:

F001347, E001347, W001347, I001347

1348:

`swArraySize` missing from `ApplicationValueSpecification`

Description:

When a compound `ApplicationValueSpecification` is specified, the `swArraySize` values are compulsory.

How to Solve:

Ensure that the compound `ApplicationValueSpecification` contains a valid `swArraySize`.

Variants:

F001348, E001348, W001348, I001348

1351:

Duplicate Offset in Schedule Table Expiry Points

Description:

This is a problem with the OS configurations, but RTA-RTE relies on Expiry points in a Schedule Table having unique offsets. RTA-RTE has lost some information and will likely fail to generate.

How to Solve:

Check the offsets in the Schedule Table for duplicates

Variants:

F001351, E001351, W001351, I001351

1352:

Inter-`OsApplication` access to IRV

Description:

Information message to indicate that the RTE generator has detected Inter-`OsApplication` access to an IRV. The message indicates the accessing `OsApplications`.

How to Solve:

Message is for information only.

Variants:

F001352, E001352, W001352, I001352

1353:

Shared OS spinlock

Description:

RTA-RTE has detected a `ExclusiveAreaImplementation` for the specified SWCI that specifies a spinlock which is already used by an different SWC instance.

How to Solve:

SWC instances are not permitted to share OS `SpinLocks` in this way. Consider creating additional OS Spinlocks and/or correcting or removing `rteExclusiveAreaOsSpinlockRef` configuration items within `ExclusiveAreaImplementation` containers.

Variants:

F001353, E001353, W001353, I001353

1354:

Multiple instances of split SWC

Description:

RTA-RTE has detected that a SWC has at least one instance on the target ECU that is split across `OsApplication`s and multiple instances on that ECU.

How to Solve:

Multiple instances of split SWCs are not permitted.

Variants:

F001354, E001354, W001354, I001354

1355:

Split BSW detected

Description:

RTA-RTE has detected that a BSW module is split across `OsApplication`s on the target ECU.

How to Solve:

Split BSW modules are not permitted.

Variants:

F001355, E001355, W001355, I001355

1357:

Multiple events start a data-accessing Runnable in a split SWCI

Description:

When a runnable is part of a split SWCI and accesses port data or IRVs then it may not be started by more than one RteEvent. This constraint has been violated.

How to Solve:

Start the runnable with a single event or remove its data accesses.

Variants:

F001357, E001357, W001357, I001357

1358:

Unmapped server detected in split SWCI

Description:

RTA-RTE has detected an unmapped server runnable in a SWCI which is split across `OsApplications`.

How to Solve:

Unmapped servers are disallowed in split SWCIs.

Variants:

F001358, E001358, W001358, I001358

1359:

Unmapped server detected in split SWCI

Description:

RTA-RTE has detected an unmapped server runnable in a SWCI which is split across `EcuPartitions`.

How to Solve:

Unmapped servers are disallowed in split SWCIs.

Variants:

F001359, E001359, W001359, I001359

1360:

Sender/Receiver `DataItem` accessed from different `OsApplications`

Description:

RTA-RTE has detected a Sender/Receiver `DataItem` being accessed by `RunnableEntitys` in (or directly invoked from) different `OsApplications`.

How to Solve:

`DataItem` accesses in split SWCIs must be from `Runnables` mapped to the same `OsApplication` and `Runnables` with `DataItem` accesses must not be configured for invocation by direct function call for client-server, mode switch or trigger connections from multiple `OsApplications`.

Variants:

F001360, E001360, W001360, I001360

1361:

Client/Server `Operation` accessed from different `OsApplications`

Description:

RTA-RTE has detected a Client/Server `Operation` in a split SWCI is being accessed by Runnables in different `OsApplications`.

How to Solve:

Calls to `Operations` in split SWCIs must be from the same `OsApplication`.

Variants:

F001361, E001361, W001361, I001361

1362:

`ModeGroup` accessed from different `OsApplications`

Description:

RTA-RTE has detected a `ModeGroup` in a split SWCI is being accessed by Runnables in different `OsApplications`.

How to Solve:

`ModeGroup` accesses in split SWCIs must be from the same `OsApplication`.

Variants:

F001362, E001362, W001362, I001362

1363:

Trigger accessed from different `OsApplications`

Description:

RTA-RTE has detected a Trigger in a split SWCI is being accessed by Runnables in different `OsApplications`.

How to Solve:

Trigger accesses in split SWCIs must be from the same `OsApplication`.

Variants:

F001363, E001363, W001363, I001363

1364:

IRV read from different `OsApplications`

Description:

RTA-RTE has detected an IRV which is read by its writer as well as a remote reader.

How to Solve:

IRVs cannot be read both locally and remotely.

Variants:

F001364, E001364, W001364, I001364

1365:

IRV accessed from `OsApplication`s in M:N configuration

Description:

RTA-RTE has detected an IRV which is written to from multiple `OsApplication`s and is read from multiple `OsApplication`s.

How to Solve:

Inter-partition IRVs must not have multiple reading `OsApplication`s and multiple writing `OsApplication`s (M:N).

Variants:

F001365, E001365, W001365, I001365

1409:

No SWC node

Description:

RTA-RTE has been unable to location the named SWC node.

How to Solve:

Internal error; please contact ETAS GmbH for support. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F001409, E001409, W001409, I001409

1411:

Not provided port

Description:

RTA-RTE has attempted to perform an action which can only operate on a provided port with a require port.

How to Solve:

Internal error; please contact ETAS GmbH for support. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F001411, E001411, W001411, I001411

1413:

Unexpected exception

Description:

RTA-RTE has detected an exception which has not been handled at the correct level. The error message gives further information as to the cause.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F001413, E001413, W001413, I001413

1420:

Invalid `RTEEvent` map

Description:

RTA-RTE has detected an invalid (NULL) RTE event map.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F001420, E001420, W001420, I001420

1422:

No container

Description:

RTA-RTE has been unable to find container for runnable entity instance

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F001422, E001422, W001422, I001422

1424:

No instance wrapper

Description:

RTA-RTE has been unable to find an instance wrapper for a runnable entity

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F001424, E001424, W001424, I001424

1430:

NULL value

Description:

RTA-RTE has detected an unexpected NULL value.

How to Solve:

Internal error; please contact ETAS GmbH for support. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F001430, E001430, W001430, I001430

1431:

Unable to create XML document

Description:

Unable to create XML document object when parsing input.

How to Solve:

Internal error; please contact ETAS GmbH for support. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F001431, E001431, W001431, I001431

1432:

Unable to locate current XML node

Description:

While parsing an XML document, RTA-RTE was unable to locate the current node.

How to Solve:

Internal error; please contact ETAS GmbH for support. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F001432, E001432, W001432, I001432

1433:

Object lookup failed

Description:

RTA-RTE failed to find identifiable object when resolving the specified reference. The error message indicates the found object type (if any) and the expected type of the reference destination element.

How to Solve:

Check that all references refer to an XML element that exists and is of the expected type.

Variants:

F001433, E001433, W001433, I001433

1435:

Object lookup for invalid node

Description:

RTA-RTE attempted to find an identifiable object corresponding to an invalid XML node.

How to Solve:

Internal error; please contact ETAS GmbH for support. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F001435, E001435, W001435, I001435

1437:

Bad allocation exception

Description:

RTA-RTE has detected a bad allocation exception and has terminated.

How to Solve:

Please attempt to reduce the model size by only supplying the ECU Extract.

Variants:

F001437, E001437, W001437, I001437

1500:

Missing XML element

Description:

The specified XML element could not be located. The error message indicates the location where the element was not found.

How to Solve:

Ensure that XML schema validation is enabled and that the input is syntactically correct.

Variants:

F001500, E001500, W001500, I001500

1501:

Expected child missing

Description:

RTA-RTE was unable to locate the indicated child element.

How to Solve:

Ensure that XML schema validation is enabled and that the input is syntactically correct.

Variants:

F001501, E001501, W001501, I001501

1502:

XML element not within expected context

Description:

RTA-RTE was unable to locate the owning element with the given tag for the element at the location specified. Elements must be correctly nested in the right contexts as defined by the XML schema.

How to Solve:

Ensure that XML schema validation is enabled and that the input is syntactically correct.

Variants:

F001502, E001502, W001502, I001502

1522:

Duplicate type

Description:

RTA-RTE has attempted to add a duplicate named type to the internal types database. All such types must have unique names and therefore this configuration is not supported.

How to Solve:

RTA-RTE attempts to filter duplicate types before their addition to the database. Please contact ETAS GmbH for support. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F001522, E001522, W001522, I001522

1523:

Reserved typename

Description:

RTA-RTE has detected an attempt has been made to declare a type with a reserved name.

How to Solve:

BSW (platform) types cannot be re-declared in the input configuration and therefore a different typename must be used.

Variants:

F001523, E001523, W001523, I001523

1530:

Zero-size array

Description:

While processing the internal types database, RTA-RTE encountered an array type with an array-size of zero.

How to Solve:

Ensure the array type has a definition for its element, and that the array type has an arraysize greater than zero.

Variants:

F001530, E001530, W001530, I001530

1532:

Invalid type declaration

Description:

The declaration for the specified type is neither an array nor a record. An array must have exactly one `ImplementationDataTypeElement` whereas a record may not have an `ArraySize` in any of its `ImplementationDataTypeElement` elements.

How to Solve:

Ensure that XML schema validation is enabled and that the input is syntactically correct.

Variants:

F001532, E001532, W001532, I001532

1533:

Invalid or missing array size

Description:

RTA-RTE was unable to determine the size of the specified array type.

How to Solve:

Ensure that XML schema validation is enabled and that the input is syntactically correct.

Variants:

F001533, E001533, W001533, I001533

1534:

Empty structure or union

Description:

While processing the internal types database, RTA-RTE encountered a structure or union type with no member elements. This is not supported.

How to Solve:

Ensure all structure and union types declare at least one member element.

Variants:

F001534, E001534, W001534, I001534

1535:

Expected type information not present

Description:

RTA-RTE has be unable to retreive expected type information from its internal database when generating an XML report.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F001535, E001535, W001535, I001535

1536:

No native declaration (array)

Description:

RTA-RTE has encountered an element definition for the specified array that has `category VALUE` but the referenced `SwBaseType` has no `nativeDeclaration`, which means no valid C declaration can be generated.

How to Solve:

Consider using a `TYPE_REFERENCE` in the `ImplementationDataTypeElement`.

Variants:

F001536, E001536, W001536, I001536

1537:

No native declaration (struct)

Description:

RTA-RTE has encountered a member definition that has `category VALUE` but the referenced `SwBaseType` has no `nativeDeclaration`, which means no valid C declaration can be generated.

How to Solve:

Consider using a `TYPE_REFERENCE` in the `ImplementationDataTypeElement`.

Variants:

F001537, E001537, W001537, I001537

1545:

No `AsyncServerCallReturns`

Description:

RTA-RTE was unable to find the expected `AsynchronousServerCallReturnsEvent` that references the specified for operation.

How to Solve:

Ensure that an `AsynchronousServerCallReturnsEvent` is specified for the indicated port and operation.

Variants:

F001545, E001545, W001545, I001545

1546:

No entry point

Description:

RTA-RTE was unable to determine the entry point for the specified runnable.

How to Solve:

Ensure that the runnable entity declares an entry point using the `symbol` element.

Variants:

F001546, E001546, W001546, I001546

1547:

Invalid time

Description:

RTA-RTE has detected an error when converting the specified time value.

How to Solve:

Ensure that all time values must be specified in seconds and no characters (units) are permitted after the time.

Variants:

F001547, E001547, W001547, I001547

1550:

Partial per-instance memory declaration

Description:

RTA-RTE has encountered a per-instance memory declaration that omits the specified configuration element.

How to Solve:

Ensure that the declaration of C-typed per-instance memories includes both a `type` and `typeDefinition` element.

Variants:

F001550, E001550, W001550, I001550

1551:

Invalid type definition

Description:

RTA-RTE has encountered a per-instance memory section type-definition that attempts to define a function pointer. This is invalid.

How to Solve:

Modify the specified type-definition to remove the pointer definition.

Variants:

F001551, E001551, W001551, I001551

1560:

Non-mapped RTE event

Description:

RTA-RTE has detected an unidentified RTE Event mapped to a task. The event could not be located in any instance of a SWC mapped to the current ECU instance.

How to Solve:

Check that all events and runnable entity names within the SWC's internal behavior are unique.

Variants:

F001560, E001560, W001560, I001560

1561:

Invalid RTE event mapping

Description:

RTA-RTE has detected an RTE event that is mapped to a task yet the event does not trigger a runnable. Such events must not be mapped.

How to Solve:

Ensure that only RTE events that start a runnable entity are mapped to tasks.

Variants:

F001561, E001561, W001561, I001561

1562:

Multiple RTE event mapping

Description:

RTA-RTE has detected multiple RTE event mappings for the specified

instance.

How to Solve:

Ensure that each RTE event instance is uniquely mapped to a task.

Variants:

F001562, E001562, W001562, I001562

1563:

Unmapped RTE event

Description:

RTA-RTE cannot locate an RTE Event mapping for the specified instance.

How to Solve:

Ensure that all RTE Events that can activate a runnable entity are individually mapped to a task.

Variants:

F001563, E001563, W001563, I001563

1565:

Possibly unsafe optimization to invocation by direct function call

Description:

RTA-RTE cannot locate an RTE Event mapping for the specified instance but the triggered runnable entity is not marked as supporting concurrent invocation. The runnable will be invoked by a direct function call, which might be unsafe if any of the calling tasks can preempt each other.

How to Solve:

Check whether preemption of the calling tasks can occur and if necessary map the RTE Event to a task.

Variants:

F001565, E001565, W001565, I001565

1570:

Multiple `RunnableEntity`s mapped to the same position within a task

Description:

The input configuration contained multiple runnables that were mapped to the same `OsTask` with the same task position (ie using the same value for `RtePositionInTask` or `RteBswPositionInTask`. AUTOSAR constraint [constr_9082] requires that all runnables within a single `OsTask` use unique task position values so that the task body can be predictably generated.

How to Solve:

Ensure that all runnables mapped to a particular `OsTask` are associated with a different `RtePositionInTask` value. The following info messages 1001 provide more details of the `RteEventToTaskMapping`s or `RteBswEventToTaskMapping`s

that are mapped to the same task and position.

Variants:

F001570, E001570, W001570, I001570

1571:

Invalid task reference

Description:

RTA-RTE was unable to resolve the specified task reference to a valid `OsTask`.

How to Solve:

Ensure the reference references a task in the correct ECU Configuration.

Variants:

F001571, E001571, W001571, I001571

1572:

RTE Event with no runnable

Description:

RTA-RTE has detected that the specified RTE Event does not directly start a runnable and has no associated wait-point.

How to Solve:

This message indicates a possible configuration problem – check that the configuration is correct and complete. If the configuration is correct then this message can be suppressed.

Variants:

F001572, E001572, W001572, I001572

1573:

Invalid partition reference

Description:

RTA-RTE was unable to resolve the partition reference to a valid `Partition`.

How to Solve:

Ensure the reference references a partition in the correct ECU Configuration.

Variants:

F001573, E001573, W001573, I001573

1574:

Missing RTE event position

Description:

RTA-RTE has detected that the position of an event in a task is missing.

How to Solve:

Ensure that all RTE events mapped to a task have a position specified.

Variants:

F001574, E001574, W001574, I001574

1580:

Invalid multiple instantiation

Description:

RTA-RTE has detected multiple instances of the specified software-component but it does not support multiple instantiation.

How to Solve:

Ensure that at most one instance of the SWC is mapped to the ECU instance for which the RTE is being generated.

Variants:

F001580, E001580, W001580, I001580

1581:

Multiple internal behaviors

Description:

RTA-RTE has detected multiple internal behaviors for the specified SWC type. This is invalid.

How to Solve:

Ensure that exactly one internal behavior is configured for the SWC type. For AUTOSAR R4.0 check that exactly one variation point enables an internal behavior.

Variants:

F001581, E001581, W001581, I001581

1583:

No SWC type reference

Description:

RTA-RTE could not locate an SWC type referenced within a component prototype.

How to Solve:

Ensure that XML schema validation is enabled and that the input is syntactically correct.

Variants:

F001583, E001583, W001583, I001583

1584:

Unable to locate SWC type

Description:

RTA-RTE could not locate an SWC type associated with an internal behavior.

How to Solve:

Ensure that XML schema validation is enabled and that the input is syntactically correct. Ensure that the internal behavior's component reference is present and correct.

Variants:

F001584, E001584, W001584, I001584

1587:

Invalid operation parameter direction

Description:

RTA-RTE could not recognize the specified client-server operation parameter direction. The error message indicates the valid directions.

How to Solve:

Ensure that XML schema validation is enabled and that the input is syntactically correct.

Variants:

F001587, E001587, W001587, I001587

1591:

Missing or invalid activation condition

Description:

RTA-RTE was unable to locate the activation condition for a mode switch event. An entry' condition is assumed.

How to Solve:

Ensure that XML schema validation is enabled and that the input is syntactically correct.

Variants:

F001591, E001591, W001591, I001591

1592:

Invalid `NvBlockSwc` multiple instantiation

Description:

RTA-RTE has detected a `NvBlockSwComponentType` whose internal behavior supports multiple instantiation. This configuration is invalid since it contravenes constr_2015.

How to Solve:

Ensure that the internal behavior is marked as Not supporting multiple instantiation.

Variants:

F001592, E001592, W001592, I001592

1593:

Invalid `MemoryAllocationKeywordPolicy` on `ExecutableEntity`'s `SwAddrMethod`

Description:

RTA-RTE has detected that an `ExecutableEntity` has a `SwAddrMethod` with a `MemoryAllocationKeywordPolicy` of AddrMethodShortNameAndAlignment.

How to Solve:

Ensure that the `SwAddrMethod` of an executable entity has a memoryAllocationKeywordPolicy of AddrMethodShortName.

Variants:

F001593, E001593, W001593, I001593

1595:

Inconsistent SWC-scoped constant

Description:

RTA-RTE has detected a mismatch of values when defining SWC-scoped constants.

How to Solve:

Ensure that XML schema validation is enabled and that the input is syntactically correct. If still unable to resolve the problem please contact ETAS GmbH for support. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F001595, E001595, W001595, I001595

1600:

Read-only IRV

Description:

The specified inter-runnable variable is only read; it is never written.

How to Solve:

This warning message indicates an unusual configuration detected by RTA-RTE. It can be suppressed if the behavior is as intended.

Variants:

F001600, E001600, W001600, I001600

1612:

Inter-ECU service connection

Description:

RTA-RTE has detected a connector that joins an application SWC and a service where the two are not located on the same ECU instance. This is an invalid configuration.

How to Solve:

Ensure that connectors to service are local to an ECU instance.

Variants:

F001612, E001612, W001612, I001612

1613:

Conflicting SWC implementations

Description:

RTA-RTE has detected conflicting `SwcToImplementation` mappings for the specified SWC.

How to Solve:

Ensure that all such mappings are consistent.

Variants:

F001613, E001613, W001613, I001613

1620:

Invalid `SwcImplementation`

Description:

RTA-RTE was unable to locate a `behaviorRef` element for an SWC implementation.

How to Solve:

Ensure that XML schema validation is enabled and that the input is syntactically correct.

Variants:

F001620, E001620, W001620, I001620

1630:

Invalid record type reference

Description:

RTA-RTE expected a reference to a complex type but failed to find it.

How to Solve:

Ensure that the indicated reference is to the expected type.

Variants:

F001630, E001630, W001630, I001630

1632:

No record elements

Description:

While processing signal mappings, RTA-RTE encountered a record type with no elements.

How to Solve:

A record type must have at least one member element.

Variants:

F001632, E001632, W001632, I001632

1633:

Missing array or record signal mapping

Description:

The specified array or record signal mapping was not found by RTA-RTE at the indicated location.

How to Solve:

Ensure the signal mapping is complete. For complex types each sub-element (record member or array element) must be mapped to a separate signal.

Variants:

F001633, E001633, W001633, I001633

1635:

Missing signal mapping

Description:

The specified signal mapping element was expected but could not be found by RTA-RTE at the indicated location.

How to Solve:

Ensure the signal mapping is complete. For complex types each sub-element (record member or array element) must be mapped to a separate signal.

Variants:

F001635, E001635, W001635, I001635

1640:

Shared runnable entry point

Description:

The specified runnable entities share a symbol (entry point).

How to Solve:

Ensure that all runnable entities have a unique symbol so that their imple-

mentationscan be unambiguously referenced.

Variants:

F001640, E001640, W001640, I001640

1641:

Invalid SWCI reference

Description:

RTA-RTE was unable to resolve the Software Component Instance reference within the specified ECU configuration `SWCInstance` container.

How to Solve:

Check that the reference is valid and that the component is mapped to the ECU for which the RTE is being generated.

Variants:

F001641, E001641, W001641, I001641

1660:

Interface mismatch (operation)

Description:

RTA-RTE has detected an interface mismatch in the specified `Operation` Instance reference. The referenced port prototype is categorized by a one interface whereas the referenced operation prototype is contained within a different interface.

How to Solve:

Ensure that the port and operation references within an instance reference are consistent.

Variants:

F001660, E001660, W001660, I001660

1661:

Interface mismatch (data item)

Description:

RTA-RTE has detected an interface mismatch in the specified Data Item reference. The referenced port prototype is categorized by a one interface whereas the referenced data item is contained within a different interface.

How to Solve:

Ensure that the port and data item references within an instance reference are consistent.

Variants:

F001661, E001661, W001661, I001661

1662:

In

Description:

RTA-RTE has detected an invalid `DataItemInstance` reference; the data item prototype's port prototype reference must be to the port prototype encapsulating the Com-spec element.

How to Solve:

Ensure that the port and data item references within an instance reference are consistent.

Variants:

F001662, E001662, W001662, I001662

1663:

Mode interface mismatch

Description:

RTA-RTE has detected an interface mismatch in a Mode Instance reference. The referenced port prototype is categorized by a different interface to the one containing the referenced mode declaration group prototype.

How to Solve:

Ensure that the two references are consistent.

Variants:

F001663, E001663, W001663, I001663

1664:

Invalid mode instance reference

Description:

RTA-RTE has detected an invalid Mode Instance reference; the mode declaration group prototype reference specifies one mode group but the referenced mode declaration is contained within a different mode group.

How to Solve:

Ensure that the two references are consistent.

Variants:

F001664, E001664, W001664, I001664

1665:

Invalid mode instance reference - port prototype not on owning SWC

Description:

RTA-RTE has detected an invalid instance reference; the port prototype referenced is within one SWC type yet the instance reference is within an internal behavior that references a different SWC type.

How to Solve:

Ensure that the two references are consistent.

Variants:

F001665, E001665, W001665, I001665

1666:

Invalid mode switch event instance reference (entry/exit)

Description:

RTA-RTE has detected an invalid mode instance reference for a `ModeSwitchEvent` that is activated on `ON-ENTRY` or `ON-EXIT` from a mode.

How to Solve:

Ensure that an on-entry/exit mode switch event references exactly one mode.

Variants:

F001666, E001666, W001666, I001666

1667:

Invalid mode switch event instance reference (transition)

Description:

RTA-RTE has detected an invalid mode instance reference for a `ModeSwitchEvent` that is activated on `TRANSITION`.

How to Solve:

Ensure that a transition mode switch event references exactly two modes.

Variants:

F001667, E001667, W001667, I001667

1670:

No signal

Description:

RTA-RTE failed to find the specified signal.

How to Solve:

Ensure that the specified signal exists and is correctly referenced.

Variants:

F001670, E001670, W001670, I001670

1671:

No signal instance

Description:

RTA-RTE was unable to find a frame instance containing the specified

system signal.

How to Solve:

This could be because no `PDUTypes` contain the signal, or because there are no instances of the `PDUType` in frames, or because there are no frames with the appropriate sender / receiver ECU.

Variants:

F001671, E001671, W001671, I001671

1680:

Incorrect client-server signal mapping

Description:

RTA-RTE has detected an incorrect client-server signal group mappings for the specified operation.

How to Solve:

Ensure that there should be exactly two mappings, one for the request and one for the response.

Variants:

F001680, E001680, W001680, I001680

1681:

Missing request or response signal group

Description:

RTA-RTE could not locate a request or response signal group in client-server mappings for the specified operation.

How to Solve:

Ensure that the configuration contains valid client-server mappings.

Variants:

F001681, E001681, W001681, I001681

1682:

Missing sequence counter signal

Description:

RTA-RTE was unable to locate an expected sequence counter signal in client-server mappings for the specified operation

How to Solve:

Ensure that sequence counter signals are supplied for both the request and the response mappings, or for neither.

Variants:

F001682, E001682, W001682, I001682

1683:

NO client-server data mapping

Description:

RTA-RTE could not determine the client-server data mapping for the specified operation.

How to Solve:

ensure that the client-server data mapping for the operation contains either signal mappings for the parameters or an empty signal.

Variants:

F001683, E001683, W001683, I001683

1690:

Empty record mapping

Description:

RTA-RTE has detected a mapping for a record type with no elements.

How to Solve:

Ensure that all records have at least one element.

Variants:

F001690, E001690, W001690, I001690

1691:

Invalid array mapping

Description:

RTA-RTE has detected a mapping for an array type that references a type that is not an array.

How to Solve:

Ensure that array mappings reference array types.

Variants:

F001691, E001691, W001691, I001691

1692:

Invalid array type

Description:

RTA-RTE has detected an array type without a number of elements specification.

How to Solve:

Ensure that all arrays have a size specification.

Variants:

F001692, E001692, W001692, I001692

1693:

Too few mappings for an array

Description:

RTA-RTE has detected a data mapping for an array that has too few elements.

How to Solve:

Ensure that all array elements are mapped.

Variants:

F001693, E001693, W001693, I001693

1694:

Nested mappings not found

Description:

RTA-RTE could not find the appropriate data mappings for the nested record or array type.

How to Solve:

Ensure that the complex type is fully mapped.

Variants:

F001694, E001694, W001694, I001694

1697:

Unsupported binding time attribute

Description:

RTA-RTE does not support `bindingTime` set to `LINK-TIME` or `SYSTEM-DESIGN-TIME`.

How to Solve:

ensure that the `bindingTime` is set to one of `PRE-COMPILE-TIME` or `CODE-GENERATION-TIME`.

Variants:

F001697, E001697, W001697, I001697

1698:

Missing binding time attribute

Description:

RTA-RTE was unable to locate a `bindingTime` attribute. This must be supplied.

How to Solve:

Ensure that the XML attribute is supplied.

Variants:

F001698, E001698, W001698, I001698

1699:

Duplicate attribute

Description:

RTA-RTE found duplicate definitions of an Autosar attribute within a split-table container. Attributes should be supplied in only one location or must have identical values.

How to Solve:

Ensure that the attributes are identically defined in all locations.

Variants:

F001699, E001699, W001699, I001699

1700:

Unable to locate required package

Description:

RTA-RTE cannot locate a package named "`AUTOSAR_Platform`". This package is standardized by AUTOSAR and is required.

How to Solve:

RTA-RTE includes an example package; please ensure this is included in the input.

Variants:

F001700, E001700, W001700, I001700

1701:

Missing type substitution

Description:

RTA-RTE has detected a missing type in the "`AUTOSAR_Platform`" package and replaced it with the specified type.

How to Solve:

Ensure the specified missing type is included.

Variants:

F001701, E001701, W001701, I001701

1702:

Missing type

Description:

RTA-RTE has detected a missing type in the "`AUTOSAR_Platform`" package.

How to Solve:

Ensure the specified missing type is included.

Variants:

F001702, E001702, W001702, I001702

1710:

Invalid application error value

Description:

RTA-RTE has detected an invalid value for an application error.

How to Solve:

The valid range for an application error is 1..63 inclusive. 0 is accepted if and only if the short name is `E_OK`.

Variants:

F001710, E001710, W001710, I001710

1721:

Wait-point in exclusive area

Description:

RTA-RTE has detected that the specified runnable entry contains a `WaitPoint` but is configured to run inside an exclusive area.

How to Solve:

This configuration is not valid; either remove the wait point or the exclusive area reference.

Variants:

F001721, E001721, W001721, I001721

1730:

Invalid service connector

Description:

RTA-RTE has detected that a service connector connects two applications SWC instances.

How to Solve:

It is forbidden to connect two application SWCs with a service connector.

Variants:

F001730, E001730, W001730, I001730

1745:

A Network Representation referenced no `SwBaseType`

Description:

RTA-RTE was unable to resolve any `swBaseTypeRef` within the `NetworkRepresentation`

How to Solve:

Ensure that a complete Network Representation is provided.

Variants:

F001745, E001745, W001745, I001745

1746:

32-bit exceeded

Description:

RTA-RTE has encountered a value that cannot be expressed as a 32-bit integer

How to Solve:

Ensure the value is in the range -2147483648 to 4294967295.

Variants:

F001746, E001746, W001746, I001746

1747:

Value is not an integer

Description:

RTA-RTE has encountered a value that cannot be expressed as an integer

How to Solve:

Ensure the expression evaluates to an integer.

Variants:

F001747, E001747, W001747, I001747

1748:

Value is not an unsigned integer

Description:

RTA-RTE has encountered a value that cannot be expressed as an unsigned-integer

How to Solve:

Ensure the expression evaluates to an unsigned integer.

Variants:

F001748, E001748, W001748, I001748

1750:

Duplicate connectors

Description:

RTA-RTE has detected multiple connectors between the same pair of ports.

How to Solve:

Ensure that only one connector exists (or is active) between the relevant ports.

Variants:

F001750, E001750, W001750, I001750

1751:

Duplicate connections

Description:

RTA-RTE has detected multiple connection routes between the same pair of ports. This could be the result of a duplicate connector or the result of multiple assembly connectors between delegation fan-out and fan-in.

How to Solve:

Ensure that only one connection exists (or is active) between the relevant ports.

Variants:

F001751, E001751, W001751, I001751

1770:

Invalid data type map

Description:

RTA-RTE has detected an invalid data type map.

How to Solve:

Ensure that the specified data type map references an application and an implementation data type.

Variants:

F001770, E001770, W001770, I001770

1771:

No data type mappings

Description:

RTA-RTE could not locate any data type mappings for the specified SWC.

How to Solve:

Either there is no `DataTypeMappingRef` or the referenced `DataTypeMapping` contains no `DataTypeMap` elements. This is likely to be the cause of any errors about types for this SWC.

Variants:

F001771, E001771, W001771, I001771

1772:

Invalid text table mapping direction

Description:

RTA-RTE failed to locate a valid direction for the text table mapping.

How to Solve:

Ensure the direction is one of `FIRST-TO-SECOND`, `SECOND-TO-FIRST` or `BIDIRECTIONAL`.

Variants:

F001772, E001772, W001772, I001772

1773:

Invalid identity text table mapping

Description:

RTA-RTE has detected an identity `TextTableMapping` (`identicalMapping` set to true) that also contains `ValuePair` elements.

How to Solve:

Remove either the identify declaration or the value pairs.

Variants:

F001773, E001773, W001773, I001773

1774:

Invalid non-identity text table mapping

Description:

RTA-RTE has failed to locate any `ValuePair` elements within the specified mapping.

How to Solve:

Ensure that all non-identify declarations declare appropriate value pairs.

Variants:

F001774, E001774, W001774, I001774

1780:

Invalid calibration value

Description:

RTA-RTE has detected an invalid `CalibrationValue` due to a missing or invalid `InitializedParameterRef`.

How to Solve:

Ensure the `InitializedParameterRef` reference is included and valid.

Variants:

F001780, E001780, W001780, I001780

1781:

Invalid per-instance calibration value

Description:

RTA-RTE has detected an invalid `CalibrationValue`.

How to Solve:

Ensure that the referenced `FlapMapInstance` includes both reference a calibration parameter and must itself be referenced by the ECU extract.

Variants:

F001781, E001781, W001781, I001781

1790:

BSW Trigger task mismatch

Description:

RTA-RTE has detected that the specified released BSW trigger has `BswTriggerDirectImplementation` enabled but is mapped to an incorrect task.

How to Solve:

Ensure the event is mapped to the expected task.

Variants:

F001790, E001790, W001790, I001790

1791:

Inconsistent trigger activation

Description:

RTA-RTE has detected that the specified BSW trigger has `BswTriggerDirectImplementation` enabled but is also issued.

How to Solve:

Ensure the trigger activations are consistent.

Variants:

F001791, E001791, W001791, I001791

1792:

No BSW trigger

Description:

RTA-RTE is unable to find the specified BSW trigger.

How to Solve:

Ensure that the trigger exists and is correctly referenced.

Variants:

F001792, E001792, W001792, I001792

1793:

No BSW mode declaration prototype

Description:

RTA-RTE is unable to find the specified BSW mode declaration prototype.

How to Solve:

Ensure that the mode declaration prototype exists and is correctly referenced.

Variants:

F001793, E001793, W001793, I001793

1796:

Missing type information in `NetworkRepresentation`

Description:

`NetworkRepresentation`. `SwBaseType` has no `nativeDeclaration` and does not name an existing type. If a `SwBaseType` referenced by a `NetworkRepresentation` has no `nativeDeclaration`, then its `shortName` must the be name of an AUTOSAR Platform Type or Standard Type. (`constr_1010`). The type is needed to declare the buffer to hold the interrim value for data conversion.

How to Solve:

Either set the `nativeDeclaration` to a C-language type expression, or set the `shortName` to the name of an AUTOSAR PlatformType or StandardType. As an extension to AUTOSAR, RTA-RTE will accept the name of any used `ImplementationDataType` as the `shortName`.

Variants:

F001796, E001796, W001796, I001796

1797:

An M:N connection configuration has been detected and these are not supported

Description:

RTA-RTE has detected that a connection between two data items in connected ports has implied an M:N connection configuration which is not allowed by AUTOSAR.

How to Solve:

Ensure that a data item in a port is not connected in such a way as to produce an M:N connection configuration.

Variants:

F001797, E001797, W001797, I001797

2003:

Invalid XML tag

Description:

RTA-RTE has encountered an unexpected XML tag.

How to Solve:

Ensure that XML schema validation is enabled and that the input configuration is correct.

Variants:

F002003, E002003, W002003, I002003

2004:

Error detected during parsing of XML

Description:

RTA-RTE has detected an error when parsing an XML file. This message typically occurs when incorrect input XML fails the schema validation. The error message gives details on the specific error cause.

How to Solve:

Ensure that the input configuration is syntactically correct and conformant with the schema – for AUTOSAR XML pay particular attention to the order of elements in a sequence.

Variants:

F002004, E002004, W002004, I002004

2005:

DOMException detected during parsing of XML

Description:

RTA-RTE has detected an error while parsing an XML file. This message indicates an unexpected exception in the XML parser while loading the XML. The error message contains the XERCES exception number and the file.

How to Solve:

If the test of the message does not provide sufficient information to fix the problem, please contact support.

Variants:

F002005, E002005, W002005, I002005

2006:

Unable to locate SWC type

Description:

RTA-RTE has been unable to locate a SW component type for a port prototype.

How to Solve:

Ensure XML validation with an AUTOSAR schema is enabled. If this does not indicate an error in the XML please contact support.

Variants:

F002006, E002006, W002006, I002006

2007:

Unable to locate referenced interface

Description:

RTA-RTE has been unable to locate the interface categorizing the specified port prototype.

How to Solve:

Ensure XML validation with an AUTOSAR schema is enabled and that the reference within the port prototype is valid. If this does not indicate an error in the XML please contact support.

Variants:

F002007, E002007, W002007, I002007

2008:

Multiple SWCI mapping

Description:

RTA-RTE has detected that the specified SWC Instance is mapped to the same ECU Instance via multiple `SwCompToEcuMapping` component prototype references.

How to Solve:

Additional mappings are ignored. The message can be suppressed if not required.

Variants:

F002008, E002008, W002008, I002008

2009:

Inconsistent initial values

Description:

RTA-RTE has detected inconsistent initial values for the specified data element instance

How to Solve:

Ensure that initial value specifications are consistent.

Variants:

F002009, E002009, W002009, I002009

2011:

Invalid RTE events

Description:

RTA-RTE has detected that the specified runnable entity is triggered by an `OperationInvokedEvent` and at least one other RTE Event.

How to Solve:

Ensure that all runnable entities triggered by an `OperationInvokedEvent` are not also triggered by any other RTE Event type

Variants:

F002011, E002011, W002011, I002011

2012:

Conflicting metaclasses in splittable pattern

Description:

The `atpSplitable` stereotype means that an AUTOSAR configuration object may appear in multiple files, where each contains a partial definition of the object. This error arises when XML elements at the same path represent different metaclasses, and therefore cannot correctly represent a single object subject to the splittable pattern.

How to Solve:

Determine how many distinct configuration objects are intended, and ensure each has a unique path, either by changing its name or its location in the package heirarchy.

Variants:

F002012, E002012, W002012, I002012

2013:

Invalid numerical value

Description:

RTA-RTE detected an invalid numerical value in a `V` or `VF` element or a floating-point value in a context where only integer values are allowed.

How to Solve:

Ensure that the indicated value is, or evaluates to, a valid numerical value. If the value is in a `swArraySize` element then it must be an integer.

Variants:

F002013, E002013, W002013, I002013

2021:

Invalid minimum start interval

Description:

RTA-RTE has detected an invalid minimum start interval period.

How to Solve:

Ensure that the minimum start interval time is an integral multiple of the RTE MainFunction period.

Variants:

F002021, E002021, W002021, I002021

2022:

Minimum start interval for server

Description:

RTA-RTE has detected the specification of a minimum start interval for a server.

How to Solve:

It is not permitted to declare a minimum start interval for a server.

Variants:

F002022, E002022, W002022, I002022

2023:

Invalid EA method

Description:

RTA-RTE has detected an invalid exclusive area implementation method.

How to Solve:

Ensure the method is one of the indicated value values.

Variants:

F002023, E002023, W002023, I002023

2024:

Invalid runnable entity mapping

Description:

RTA-RTE has detected an invalid mapping of a runnable entity to multiple tasks.

How to Solve:

Runnable entity cannot be triggered by multiple RTE Events mapped to different (preempt-able) tasks unless it is marked as subject to concurrent invocation.

Variants:

F002024, E002024, W002024, I002024

2032:

Minimum start interval too long

Description:

RTA-RTE has detected a minimum start interval that cannot be expressed within the available constraints given the specified main function period.

How to Solve:

Either increase the main function period or reduce the minimum start interval. Each main function period is one "tick" of the RTE and a minimum start interval can be at most 65535 ticks.

Variants:

F002032, E002032, W002032, I002032

2033:

Exclusive area accessed via direct call

Description:

RTA-RTE has detected that the specified runnable entities are invoked by direct function call (are started by `RTEEvent`s that are not mapped to tasks) and also use the specified `ExclusiveArea`. RTA-RTE is not currently able to determine whether the calling patterns allow the `ExclusiveArea` to be safely optimized and hence no optimization has been performed. Furthermore, interrupt blocking has been used as the implementation strategy even if OS resource locking was requested.

How to Solve:

This message does not indicate a functional defect, merely a possibly sub-optimal implementation.

Variants:

F002033, E002033, W002033, I002033

2050:

Unmapped SWC instance (error)

Description:

RTE module configuration contains information about the specified SWC instance yet this is not mapped to the ECU instance being generated.

How to Solve:

Ensure all required SWC instances are mapped. Alternative explicitly enable unmapped SWC instances uses the appropriate command line option.

Variants:

F002050, E002050, W002050, I002050

2051:

Unmapped SWC instance (warning)

Description:

RTE module configuration contains information about the specified SWC

instance yet this is not mapped to the ECU instance being generated.

How to Solve:

This warning indicates that an unmapped SWC instance has been encountered but the command-line options have been set to allow RTE generation. This warning can be suppressed if required.

Variants:

F002051, E002051, W002051, I002051

2052:

Missing parameter

Description:

RTA-RTE has failed to find the specified parameter within the ECUC file.

How to Solve:

Ensure that the parameter is present and contains a valid value.

Variants:

F002052, E002052, W002052, I002052

2070:

Invalid enumeration

Description:

RTA-RTE has detected an invalid enumeration value.

How to Solve:

Ensure that the specified parameter uses valid enumeration values.

Variants:

F002070, E002070, W002070, I002070

2080:

Multiple SWC instances referenced from a flatmap

Description:

RTA-RTE has detected that flatmap instance references multiple component prototypes.

How to Solve:

Check that the ECU extract reference corresponds to a component prototype typed by single atomic SWC.

Variants:

F002080, E002080, W002080, I002080

2081:

Multiple flatmap instances encountered for same object

Description:

RTA-RTE has detected multiple `FlatInstanceDescriptor`s for the same object but with different names. This can lead to hard to diagnose errors within the configuration and is thus forbidden.

How to Solve:

Check that all `FlatInstanceDescriptor`s referenced unique objects (ie one instance descriptor per object).

Variants:

F002081, E002081, W002081, I002081

2090:

No reference base

Description:

RTA-RTE has encountered a relative reference with neither a `base` attribute nor a default reference base.

How to Solve:

Either include a default `ReferenceBase` within the specified package or add a `base` attribute to the reference.

Variants:

F002090, E002090, W002090, I002090

2091:

Undefined or invalid referenced base

Description:

RTA-RTE has detected that a relative reference uses an invalid or undefined `base`.

How to Solve:

Ensure that the package includes a `ReferenceBase`.

Variants:

F002091, E002091, W002091, I002091

2092:

No package found when building reference

Description:

RTA-RTE is unable to locate a package element starting at the specified node.

How to Solve:

Ensure that XML schema validation is enabled and that the input XML us syntactically correct.

Variants:

F002092, E002092, W002092, I002092

2093:

Multiple reference bases

Description:

RTA-RTE has detected multiple default reference bases for a package.

How to Solve:

Ensure that at most one `ReferenceBase` is marked as the default.

Variants:

F002093, E002093, W002093, I002093

2094:

Duplicate short labels for reference base

Description:

RTA-RTE has encountered duplicate short labels within a reference base specification.

How to Solve:

Ensure that all reference bases within a package must have unique short labels.

Variants:

F002094, E002094, W002094, I002094

2095:

Ignored `McDataInstance` referenced in input

Description:

RTA-RTE has encountered a `mcDataInstanceVarRef` or `mcDataInstanceRef` in the input configuration. This has been ignored because such references are only valid in the `McSupportData` output.

How to Solve:

Generated `McSupportData` output should not be fed back to the RTE generator as input. If the reference was intended to identify the `VariableDataPrototype` of an input variable use an `AutosarVariable` element containing an `AutosarVariableIref` or `LocalVariableRef`. If the reference was intended to identify the `ParameterDataPrototype` of a shared axis use an `ArParameter` element containing an `AutosarParameterIref` or `LocalParameterRef`.

Variants:

F002095, E002095, W002095, I002095

2097:

`SwCalprmAxis` instance-relevant properties set at type level

Description:

RTA-RTE has encountered a `SwCalprmAxis` on a `DataType` that has properties set that are only valid at the `ParameterAccess` or `InstantiationDataDefProps` level.

How to Solve:

Ensure that input variable references or shared axis calibration parameter references are configured in the corresponding `SwCalprmAxis` on a `ParameterAccess` or `InstantiationDataDefProps`, not on the `DataType`.

Variants:

F002097, E002097, W002097, I002097

2098:

`SwCalprmAxis` type-relevant properties set at instance level

Description:

RTA-RTE has encountered a `SwCalprmAxis` on a `ParameterAccess` or `InstantiationDataDefProps` that has properties set that are only valid at the `DataType` level.

How to Solve:

Ensure that only input variable references or shared axis calibration parameter references are configured in a `SwCalprmAxis` on a `ParameterAccess` or `InstantiationDataDefProps`. Other properties should be set in the corresponding `SwCalprmAxis` on the `DataType`.

Variants:

F002098, E002098, W002098, I002098

2099:

`SwCalprmAxisSet` present on a `DataPrototype`

Description:

RTA-RTE has encountered a `SwCalprmAxisSet` on a `DataPrototype`. This is not permitted by AUTOSAR and is not longer supported.

How to Solve:

Ensure that input variable references or shared axis calibration parameter references are configured in a `SwCalprmAxis` on a `ParameterAccess` or `InstantiationDataDefProps`.

Variants:

F002099, E002099, W002099, I002099

2103:

Invalid port context

Description:

RTA-RTE has detected that a referenced port is not declared within the context of the SWC type referenced by the BSW mapping.

How to Solve:

Ensure that in the `SwcBswMapping` element each referenced provide port is declared within the referenced SWC.

Variants:

F002103, E002103, W002103, I002103

2104:

Queued NV-Data item encountered

Description:

A variable data prototype within an `NvDataInterface` cannot be queued.

How to Solve:

Remove the queued status for the NV-Data element

Variants:

F002104, E002104, W002104, I002104

2105:

An `NvData RPort` is not correctly initialized

Description:

An unconnected `RPort` in an application SWC that is categorized by an `NvDataInterface` must have initial values declared for all data items.

How to Solve:

Ensure that all data items have an initial value specified (using an `NvRequireComSpec`) or connect the `RPort` to a provider.

Variants:

F002105, E002105, W002105, I002105

2106:

Invalid port model for `NvBlockSwComponentType` port

Description:

The RTE generator has detected a port within a `NvBlockSwComponentType` that is not categorized by either an `NvDataInterface` or a `ClientServerInterface`.

How to Solve:

Ensure that all ports within a `NvBlockSwComponentType` are either categorized by an `NvDataInterface` or a `ClientServerInterface`.

Variants:

F002106, E002106, W002106, I002106

2107:

Invalid event type for `NvBlockSwComponentType` internal behavior

Description:

AUTOSAR restricts `NvBlockSwComponentTypes` to the configuration of `OperationInvokedEvents`, `DataReceivedEvents` or `TimingEvents` only.

How to Solve:

Ensure only `OperationInvokedEvents`, `DataReceivedEvents` or `TimingEvents` are declared within `NvBlockSwComponentTypes`.

Variants:

F002107, E002107, W002107, I002107

2108:

Invalid runnable entity for `NvBlockSwComponentType` internal behavior

Description:

Runnable entities within the internal behavior of an `NvBlockSwComponentTypes` must declare the mandatory attributes `symbol` and `canBeInvokedConcurrently`.

How to Solve:

Ensure that the specified runnable declares all mandatory attributes.

Variants:

F002108, E002108, W002108, I002108

2109:

Invalid port reference for read `NvData`

Description:

The RTE generator has detected an invalid port referenced from a read-variable within an `NvBlockDataMapping`. A written-variable reference must reference a provide port within the SWC-type containing the parent `NvBlockDescriptor`.

How to Solve:

Ensure that only provided ports within the SWC-type containing the parent `NvBlockDescriptor` are referenced by read-variable declarations. The provided Nv-data is "read" by the application SWC and therefore must be provided by the `NvBlockSwComponentType`.

Variants:

F002109, E002109, W002109, I002109

2110:

Invalid port reference for written `NvData`

Description:

The RTE generator has detected an invalid port referenced from a writ-

ten-variablewithin an `NvBlockDataMapping`. A written-variable reference must reference a require port within the SWC-type containing the parent `NvBlockDescriptor`.

How to Solve:

Ensure that only required ports within the SWC-type containing the parent `NvBlockDescriptor` are referenced by written-variable declarations. The Nv-data is "provided" by the application SWC and therefore must be required by the `NvBlockSwComponentType`.

Variants:

F002110, E002110, W002110, I002110

2111:

Invalid `NvBlock` fan-out

Description:

The declaration of multiple `NvBlock` data mappings for the same port/data item is not supported. Multiple mappings to the same ram-block (or sub-element) are supports as longs as each mapping reference a different port/datum.

How to Solve:

Declare one port/data item per `NvBlock` data mapping.

Variants:

F002111, E002111, W002111, I002111

2112:

Non-`SchedulableEntity`s not allowed to run inside an Exclusive Area

Description:

Neither `BswCalledEntity`s nor `BswInterruptEntity`s are allowed to use an `ExclusiveArea` in the role `runsInsideExclusiveArea`.

How to Solve:

Remove the exclusive area in the role `runsInsideExclusiveArea`.

Variants:

F002112, E002112, W002112, I002112

2113:

Runnable Entities cannot be synchronized with `category` 1 Interrupt Entities

Description:

The RTE Generator has detected that a Runnable Entity has been synchronized with `category` 1 BSW Interrupt Entity. This is not permitted by AUTOSAR.

How to Solve:

Do not synchronize a SWC Runnable Entity with a `category` 1 BSW Interrupt Entity.

Variants:

F002113, E002113, W002113, I002113

2115:

Inconsistent `SwAddrMethod`s associated with NVRAM Ramblock

Description:

The RTE generator has found more than one `SwAddrMethod` associated with ports mapped to the ram-block. A `SwAddrMethod` from a mapped port is used when no `SwAddrMethod` has been associated directly with the ram-block.

How to Solve:

Specify a `SwAddrMethod` for the ram-block to override the conflict, or ensure each port mapped to the ram-block uses the same `SwAddrMethod`.

Variants:

F002115, E002115, W002115, I002115

2116:

Inconsistent `SwAddrMethod`s for connected ports

Description:

The RTE generator has found more than one `SwAddrMethod` associated with provide `Port`s connected to the same require `Port` or with require `Port`s connected to the same provide `Port`. An `SwAddrMethod` on a data item in a provide `Port` overrides any `SwAddrMethod` on the connected data item in the require `Port` and therefore the specifications must be consistent. When no `SwAddrMethod` is associated with a data item in a provide `Port` for which separate receive buffers for the connected require ports cannot be used (eg NVRAM and calibration parameter connections) then it inherits any `SwAddrMethod` on the connected data items in the require `Ports` and therefore their specifications must be consistent. Note that in the case of a `Port` on an NV component an `SwAddrMethod` can be specified on the data item in the `Port`, on the `RamBlock` to which the data item is mapped or via some other port data mapped to the `RamBlock`.

How to Solve:

Ensure that the data items in all connected `Port`s either specify the same `SwAddrMethod` or do not specify an `SwAddrMethod`. For data items in `Port`s on NV components ensure that the mapped `RamBlock`s either specify `SwAddrMethod`s that are consistent with the other `Port`s or do not specify `SwAddrMethod`s.

Variants:

F002116, E002116, W002116, I002116

2117:

Partition implemented by more than one OS application

Description:

RTA-RTE has detected that more than one `OsApplication` is marked as implementing the given `Partition`.

How to Solve:

Ensure that each `Partition` is implemented by exactly one `OsApplication`.

Variants:

F002117, E002117, W002117, I002117

2118:

Duplicate BSW partitions

Description:

RTA-RTE has detected that more than one `Partition` is marked as being the one containing the BSW.

How to Solve:

Ensure that only one `Partition` is marked as containing the BSW.

Variants:

F002118, E002118, W002118, I002118

2119:

Partition not implemented by any OS application

Description:

RTA-RTE has detected that the specified partition is not implemented by any OS application.

How to Solve:

Ensure that for every `Partition` there is one `OsApplication` marked as implementing that `Partition`.

Variants:

F002119, E002119, W002119, I002119

2120:

Missing partition configuration

Description:

RTA-RTE has detected that `OsApplications` are used without any corresponding `Partitions` configured but the specified software component instance has no `RTEEvents` mapped to tasks and therefore cannot be associated with any `OsApplication`.

How to Solve:

Ensure that `Partitions` are configured and every SWC instance is referenced from some `Partition`.

Variants:

F002120, E002120, W002120, I002120

2121:

No BSW partition

Description:

RTA-RTE has detected that the ECU is partitioned and that there are inter-ECU connections (requiring the use of COM) but that no partition is marked as containing the BSW for the ECU.

How to Solve:

Ensure that `Partition`s are defined in the ECUC and that one `Partition` is marked as containing the BSW.

Variants:

F002121, E002121, W002121, I002121

2122:

BSW partition restartable

Description:

RTA-RTE has detected that the `Partition` that contains the BSW for the ECU is marked as being restartable.

How to Solve:

Ensure that the `Partition` that is marked as containing the BSW is not marked as being able to restart.

Variants:

F002122, E002122, W002122, I002122

2123:

BSW partition not implemented by trusted OS application

Description:

RTA-RTE has detected that the `Partition` that contains the BSW for the ECU is not implemented by a trusted `OsApplication`.

How to Solve:

Ensure that a trusted `OsApplication` is defined and marked as implementing the `Partition` that is marked as containing the BSW.

Variants:

F002123, E002123, W002123, I002123

2124:

OS application not assigned to any core

Description:

RTA-RTE has detected that the `OsApplication` is not assigned to a particular core in a system that is multi-core; other `OsApplication`s are assigned to cores.

How to Solve:

Ensure that either all `OsApplication`s have a core assignment or none do.

Variants:

F002124, E002124, W002124, I002124

2125:

No BSW partition

Description:

RTA-RTE has detected that the ECU is partitioned and that there are BSW modules but that no partition is marked as containing the BSW for the ECU.

How to Solve:

Ensure that `Partition`s are defined in the ECUC and that one `Partition` is marked as containing the BSW.

Variants:

F002125, E002125, W002125, I002125

2126:

Missing partition configuration

Description:

RTA-RTE has detected that `OsApplication`s are used without any corresponding `Partition`s configured but the specified BSW module instance has no BSW Events mapped to tasks and therefore cannot be associated with any `OsApplication`.

How to Solve:

Ensure that `Partition`s are defined in the ECUC and that one `Partition` is marked as containing the BSW.

Variants:

F002126, E002126, W002126, I002126

2127:

Data reference (pointer) type passed between application software components

Description:

RTA-RTE has detected that a data reference (pointer) type is passed between two application software components via a port interface. Data reference types may only be passed to or from service components, `NvBlock` component, ECU abstraction components or complex device drivers.

How to Solve:

Ensure that ports on application software components that have data reference types in the interface are connected to service components, `NvBlock` components, ECU abstraction components or complex device drivers.

Variants:

F002127, E002127, W002127, I002127

2128:

`DATA_REFERENCE` type used as an OUT or INOUT parameter

Description:

RTA-RTE has detected that a data reference (pointer) type is used as an 'out' or 'inout' parameter of a client-server operation.

How to Solve:

Client-server operation arguments may only be typed with data reference (pointer) types when they are 'in' parameters.

Variants:

F002128, E002128, W002128, I002128

2129:

No EcuC Module Configuration

Description:

RTA-RTE has failed to detect an EcuC module configuration within the ECUC value collection. The module is necessary for partition definition.

How to Solve:

Ensure that a valid EcuC module configuration is referenced from the appropriate ECUC value collection.

Variants:

F002129, E002129, W002129, I002129

2130:

`VariableDataPrototype` referenced as a read local variable is not an IRV

Description:

RTA-RTE has detected that the given `VariableAccess` in the role `ReadLocalVariable` references a `VariableDataPrototype` that is not an inter-runnable variable.

How to Solve:

Ensure that all `VariableAccess`s in the `ReadLocalVariable` role reference a `VariableDataPrototype` in the role `ImplicitInterRunnableVariable` or `ExplicitInterRunnableVariable`.

Variants:

F002130, E002130, W002130, I002130

2131:

`VariableDataPrototype` referenced as a written local variable is not an IRV

Description:

RTA-RTE has detected that the given `VariableAccess` in the role `WrittenLocalVariable` references a `VariableDataPrototype` that is not an inter-runnable variable.

How to Solve:

Ensure that all `VariableAccess`s in the `WrittenLocalVariable` role reference a `VariableDataPrototype` in the role `ImplicitInterRunnableVariable` or `ExplicitInterRunnableVariable`.

Variants:

F002131, E002131, W002131, I002131

2132:

No `NvBlockDataMapping` for `PPort`

Description:

The RTE generator has failed to locate a `NvBlockDataMapping` for a provided `VariableDataPrototype` in an `NvRamSwcComponentType`.

How to Solve:

Ensure that all provided ports have an associated `NvBlockDataMapping`.

Variants:

F002132, E002132, W002132, I002132

2133:

Missing reference to accessed variable or calibration parameter

Description:

A `VariableAccess` or `ParameterAccess` contains no reference to the variable or calibration parameter to be accessed and has hence been ignored by RTA-RTE.

How to Solve:

Ensure that the mentioned `VariableAccess` or `ParameterAccess` references the data prototype to be accessed.

Variants:

F002133, E002133, W002133, I002133

2134:

Wrong type of data prototype referenced as written

Description:

A `DataSendPoint` or `DataWriteAccess` references a data prototype that is not

within a port or that is within a require port. This is not valid.

How to Solve:

Ensure that the mentioned `DataSendPoint` or `DataWriteAccess` references a provide port and a variable data prototype within the sender-receiver or NV-data interface of that port.

Variants:

F002134, E002134, W002134, I002134

2135:

Wrong type of data prototype referenced as read

Description:

A `DataReceivePoint` or `DataReadAccess` references a data prototype that is not within a port or that is within a provide port. This is not valid.

How to Solve:

Ensure that the mentioned `DataReceivePoint` or `DataReadAccess` references a require port and a variable data prototype within the sender-receiver or NV-data interface of that port.

Variants:

F002135, E002135, W002135, I002135

2150:

RTA-RTE inferred the `SwcInstance` for a `SwcBswMappingRef`

Description:

This message indicates that a `SwcBswMapping` was not referenced by any relevant `SwcInstance`, but RTA-RTE was able to use the referenced `InternalBehavior` to infer the `SwcInstance` and continue processing.

How to Solve:

To prevent this message, ensure that `SwcInstance`s correctly reference their `SwcBswMapping`s.

Variants:

F002150, E002150, W002150, I002150

2151:

RTA-RTE inferred the `BswImplementation` for a `SwcBswMappingRef`

Description:

This message indicates that a `SwcBswMapping` was not referenced by any relevant `BswImplementation`, but RTA-RTE was able to use the referenced `InternalBehavior` to infer the `BswImplementation` and continue processing.

How to Solve:

To prevent this message, ensure that `BswImplementation`s correctly reference their `SwcBswMapping`s.

Variants:

F002151, E002151, W002151, I002151

2152:

RTA-RTE could not determine the `SwcInstance` for a `SwcBswMappingRef`

Description:

This message indicates that a `SwcBswMapping` was not referenced by any relevant `SwcInstance`, and RTA-RTE was not able to infer any `SwcInstance` via the referenced `InternalBehavior`.

How to Solve:

To prevent this message, ensure that `SwcInstance`s correctly reference their `SwcBswMapping`s.

Variants:

F002152, E002152, W002152, I002152

2153:

RTA-RTE could not determine the `BswImplementation` for a `SwcBswMappingRef`

Description:

This message indicates that a `SwcBswMapping` was not referenced by any relevant `BswImplementation`, and RTA-RTE was not able to infer any `BswImplementation` via the referenced `InternalBehavior`.

How to Solve:

To prevent this message, ensure that `SwcInstance`s correctly reference their `SwcBswMapping`s.

Variants:

F002153, E002153, W002153, I002153

2154:

Multiple `TextTableMapping`s exist, however both of them are in the same direction

Description:

This message indicates that multiple `TextTableMapping`s with indentical directions were detected by the RTA-RTE.

How to Solve:

Please supply one forward and one reverse mapping direction or alternatively supply one bidirectional mapping.

Variants:

F002154, E002154, W002154, I002154

2155:

Multiple `TextTableMappings` exist, however one is Bidirectional

Description:

This message indicates that multiple `TextTableMappings` with incompatible directions were detected by the RTA-RTE.

How to Solve:

Please supply a forward and a reverse mapping directions or only one bidirectional mapping.

Variants:

F002155, E002155, W002155, I002155

2156:

A non-unique text table mapping has been detected

Description:

RTA-RTE has detected an `ValuePair` element which contains a value which is already mapped.

How to Solve:

Ensure the text table mappings are unique.

Variants:

F002156, E002156, W002156, I002156

2157:

Unsupported sub element mapping for data prototypes of identical types

Description:

RTA-RTE has detected a sub element mapping between Data Prototypes of the same type.

How to Solve:

Ensure that both first and second data prototypes are of differing types or remove the sub element mappings.

Variants:

F002157, E002157, W002157, I002157

2158:

A non-unique sub element mapping has been detected

Description:

RTA-RTE has detected an `subElementMapping` element which contains a value which is already mapped.

How to Solve:

Ensure the sub element mappings are unique.

Variants:

F002158, E002158, W002158, I002158

2159:

Invalid number of `TextTableMapping` elements has been detected

Description:

This message indicates that more `TextTableMapping`s were detected by the RTA-RTE than expected.

How to Solve:

Please supply a forward and a reverse mapping directions or only one bidirectional mapping.

Variants:

F002159, E002159, W002159, I002159

2160:

Missing `index` attribute

Description:

RTA-RTE failed to locate a required `index` attribute at the referenced element.

How to Solve:

A valid `index` attribute is required to resolve the array element reference and is therefore mandatory. Ensure schema validation is enabled and that the input configuration is correct.

Variants:

F002160, E002160, W002160, I002160

2161:

Invalid `index` attribute

Description:

RTA-RTE detected an invalid `index` attribute at the referenced element.

How to Solve:

A valid `index` attribute is required to resolve the array element reference and is therefore mandatory. Ensure schema validation is enabled and that the input configuration is correct.

Variants:

F002161, E002161, W002161, I002161

2162:

A non-unique mode value has been detected in a mode group

Description:

RTA-RTE has detected a `ModeDeclaration. value` that clashes with another `ModeDeclaration. value`.

How to Solve:

Ensure the `ModeDeclaration` values within a `ModeDeclarationGroup` are unique.

Variants:

F002162, E002162, W002162, I002162

2163:

Missing `ModeDeclaration.`value attribute

Description:

RTA-RTE failed to locate an expected `value` element for the `ModeDeclaration` at the referenced element.

How to Solve:

A valid unique `value` element is required for each `ModeDeclaration` in a `ModeDeclarationGroup` having `category` `EXPLICIT_ORDER`.

Variants:

F002163, E002163, W002163, I002163

2164:

Default value used for `onTransitionValue`

Description:

When a `ModeDeclarationGroup. category` is `EXPLICIT_ORDER` then there may be a `ModeDeclarationGroup. onTransitionValue` specifying a unique value to use during mode transition. RTA-RTE did not find a specified `onTransitionValue` so used a value one greater than the highest `ModeDeclaration.`value in the `ModeDeclarationGroup`.

How to Solve:

To avoid this info message, choose a unique value for `ModeDeclarationGroup. onTransitionValue`, or set the `ModeDeclarationGroup. category` to `ALPHABETIC_ORDER`

Variants:

F002164, E002164, W002164, I002164

2165:

`ModeDeclarationGroup. onTransitionValue` clashes with `ModeDeclaration.`value

Description:

If `ModeDeclarationGroup. onTransitionValue` is specified, it must be set to a value not used by any `ModeDeclaration` in the `ModeDeclarationGroup`.

How to Solve:

Choose a unique value for `ModeDeclarationGroup. onTransitionValue`. (Typically, add one to the highest `ModeDeclaration.`value specified within the `ModeDeclarationGroup`.)

Variants:

F002165, E002165, W002165, I002165

2166:

Unexpected `ModeDeclaration. value`

Description:

The `ModeDeclaration. .value` attribute is only valid when the `ModeDeclarationGroup` has `category EXPLICIT_ORDER`. RTA-RTE unexpectedly encountered a `ModeDeclaration. .value` attribute in a `ModeDeclarationGroup` of `category ALPHABETIC_ORDER`.

How to Solve:

Either remove the `VALUE` elements or change the `ModeDeclarationGroup. category` to `EXPLICIT_ORDER`.

Variants:

F002166, E002166, W002166, I002166

2167:

Unexpected `onTransitionValue`

Description:

The `ModeDeclarationGroup. onTransitionValue` attribute is only valid when the `ModeDeclarationGroup` has `category EXPLICIT_ORDER`. RTA-RTE unexpectedly encountered an `onTransitionValue` attribute on a `ModeDeclarationGroup` of `category ALPHABETIC_ORDER`.

How to Solve:

Either remove the `onTransitionValue` element or change the `ModedeclarationGroup. category` to `EXPLICIT_ORDER`.

Variants:

F002167, E002167, W002167, I002167

2168:

Mode `ImplementationDataType` range is too small for the `ModeDeclarationGroup`

Description:

RTA-RTE has detected that either one of the mode values or the on transition value exceeds the maximum supported range value of a `ImplementationDataType` mapped to the `ModeDeclarationGroup`.

How to Solve:

Increase the size of the `ImplementationDataType` or ensure the mode values/on transition value do not exceed the maximum range of the `ImplementationDataType`

Variants:

F002168, E002168, W002168, I002168

2169:

Missing `SwBaseType` encoding

Description:

RTA-RTE has been unable to locate a `baseTypeEncoding` element.

How to Solve:

Ensure that strict XML schema validation is enabled and that the input is correct.

Variants:

F002169, E002169, W002169, I002169

2170:

Invalid initial mode

Description:

RTA-RTE has detected that the referenced `ModeDeclaration` in `initialModeRef` element does not belong to the specified `ModeDeclarationGroup`.

How to Solve:

Ensure that the `ModeDeclarationGroup` specifies an initial `ModeDeclaration` that belongs to it.

Variants:

F002170, E002170, W002170, I002170

2171:

Conflicting `displayFormat` detected on types

Description:

RTA-RTE has detected that the `displayFormat` is a specified on both an Application and its mapped Implementation data type.

How to Solve:

Ensure that the `displayFormat` is specified on either the Application or Implementation data type.

Variants:

F002171, E002171, W002171, I002171

2172:

Conflicting `CompuMethod` detected on types

Description:

RTA-RTE has detected that the `CompuMethod` is a specified on both an Application and its mapped Implementation data type.

How to Solve:

Ensure that the `CompuMethod` is specified on either the Application or Implementation data type.

Variants:

F002172, E002172, W002172, I002172

2173:

Variable or parameter reference element contains multiple references

Description:

RTA-RTE has detected that the specified element contains multiple references in different styles. Only one reference is allowed.

How to Solve:

Ensure that exactly one reference to one variable or parameter is given per containing element.

Variants:

F002173, E002173, W002173, I002173

2174:

`InstantiationDataDefProps` contains multiple references

Description:

RTA-RTE has detected that the specified `InstantiationDataDefProps` element contains both a `VariableInstance` reference and a `ParameterInstance` reference. Only one reference is allowed.

How to Solve:

Ensure that exactly one reference to either a variable or a parameter is given per `InstantiationDataDefProps` element.

Variants:

F002174, E002174, W002174, I002174

2175:

Conflicting `InstantiationDataDefPropsS`

Description:

RTA-RTE has detected that the specified `InstantiationDataDefProps` element assigns a value for the given `SwDataDefProp` to the specified element of data that conflicts with the value set by one or more other `InstantiationDataDefProps` elements for the same element of data.

How to Solve:

Ensure that all `InstantiationDataDefProps` elements reference the intended elements of data and apply only the intended `SwDataDefProp` values.

Variants:

F002175, E002175, W002175, I002175

2176:

Conflicting shared input axis

Description:

RTA-RTE has encountered a shared axis input reference property on either a `ParameterAccess` or `InstantiationDataDefProps` that conflicts with that specified on the data type.

How to Solve:

A curve or map can only use a solitary shared input axis reference.

Variants:

F002176, E002176, W002176, I002176

2177:

Input axis conflicting calibration parameter

Description:

Multiple calibration parameters have been specified as the shared axis calibration parameter for the same input axis of a curve or map calibration parameter. When a curve or map uses shared axes then, for a given instance of the curve/map and a given axis index, all calibration parameter references in all associated `DataDefProps` must ultimately reference the same calibration parameter instance. Relevant `DataDefProps` might be found in a `ParameterAccess` or `InstantiationDataDefProps` and may be related to objects connected via `AssemblySwConnectors` or `DelegationSwConnectors`. RTA-RTE is unable to give a single location for where the error is injected because the problem relates to a conflict at the point of instantiation between values that are valid when considered in isolation.

How to Solve:

Ensure that the values of both the properties on source axis and destination axis are the same

Variants:

F002177, E002177, W002177, I002177

2178:

Multiple input axes

Description:

RTA-RTE has encountered an attempt to use multiple per-instance shared calibration input axes as a single input axis.

How to Solve:

Ensure that there is only one shared calibration input axis instance.

Variants:

F002178, E002178, W002178, I002178

2179:

Conflicting `SwAddrMethod`s

Description:

RTA-RTE has encountered two `SwAddrMethod`s of the same name supplied on the input that have conflicting properties.

How to Solve:

Ensure `SwAddrMethod`s with the same `shortName` do not have conflicting values for their `sectionType`, `sectionInitializationPolicy`, or `memoryAllocationKeyword`.

Variants:

F002179, E002179, W002179, I002179

2180:

Invalid `memoryAllocationKeywordPolicy` on a `SwAddrMethod`

Description:

RTA-RTE has encountered a `memoryAllocationKeywordPolicy` whose value is invalid.

How to Solve:

Set the `memoryAllocationKeywordPolicy` to either 'ADDR-METHOD-SHORT-NAME' or 'ADDR-METHOD-SHORT-NAME-AND-ALIGNMENT'.

Variants:

F002180, E002180, W002180, I002180

2181:

MemoryAllocationKeywordPolicy overridden

Description:

RTA-RTE has encountered two `SwAddrMethod`s with the same name and has merged the Memory Allocation Keyword Policy value from the `SwAddrMethod` of which said property was set.

How to Solve:

It is valid to have two `SwAddrMethod`s of the same name where one has a Memory Allocation Keyword Policy set and the other does not. Alternatively, the same value of Memory Allocation Keyword Policy can be set on both `SwAddrMethod`s of the same name.

Variants:

F002181, E002181, W002181, I002181

2182:

RTA-RTE will choose a default `SwAddrMethod. sectionType` based on the use case

Description:

A `SwAddrMethod` on the input has no `sectionType`. RTA-RTE has chosen a value for the property based on the use-case, for example, if the `SwAddrMethod` is used by port data then the `sectionType` is set to `VAR`. This may cause conflicts if the `SwAddrMethod` is used for a conflicting purpose, eg a calibration parameter.

How to Solve:

Consider explicitly specifying `SwAddrMethod. sectionType`, or check that `SwAddrMethod`s are used for a single purpose (eg do not mix calibration and port data on the same `SwAddrMethod`)

Variants:

F002182, E002182, W002182, I002182

2183:

Inconsistent `InstantiationDataDefProps SwAddrMethod`s for an instance of data

Description:

The RTE generator has found more than one `SwAddrMethod` associated with an instance of data via `InstantiationDataDefProps`s.

How to Solve:

Ensure that any `InstantiationDataDefProps`s that reference a single instance of a `DataPrototype` use `SwAddrMethod`s that are equivalent.

Variants:

F002183, E002183, W002183, I002183

2185:

RTA-RTE has encountered conflicting properties of two `SwAddrMethod`s of the same name

Description:

RTA-RTE has encountered two `SwAddrMethod`s of the same name that have conflicting properties. One of the `SwAddrMethod`s has been internally created by RTA-RTE and it conflicts with an input `SwAddrMethod` of the same name. Since the RTA-RTE version of this `SwAddrMethod` adheres to the AUTOSAR specification or documented RTA-RTE behavior, the properties used in the internal RTA-RTE `SwAddrMethod` will be used.

How to Solve:

Ensure the input `SwAddrMethod` with the same AUTOSAR or RTA-RTE defined `shortName` conforms to the appropriate specified property value.

Variants:

F002185, E002185, W002185, I002185

2186:

Could not choose a default `SwAddrMethod` property

Description:

RTA-RTE has encountered a `SwAddrMethod` where the given property is unset and has attempted to default the property based on the use case. This was not possible because the `SwAddrMethod` was used in different use cases.

How to Solve:

Either supply the property on the input or use different `SwAddrMethod`s for the different uses cases.

Variants:

F002186, E002186, W002186, I002186

2187:

RTA-RTE choosing `sectionInitializationPolicy`

Description:

A `SwAddrMethod` on the input has no `sectionInitializationPolicy`. RTA-RTE has chosen a value for the property based on the use-case, for example, if the `SwAddrMethod` is used by data with an intial value then the `sectionInitializationPolicy` is set to `INIT`. This may cause conflicts if the `SwAddrMethod` is used for other purposes.

How to Solve:

Consider explicitly specifying `SwAddrMethod. sectionInitializationPolicy`, or ensure that either all or none of the objects referencing this `SwAddrMethod` require initialization.

Variants:

F002187, E002187, W002187, I002187

2188:

Missing SectionNamePrefix.symbol

Description:

A `SectionNamePrefix` has a missing `symbol`. Although this passes the schema check, the AUTOSAR metamodel states that `SectionNamePrefix. symbol` is mandatory.

How to Solve:

Supply a `sectionNamePrefix. symbol`.

Variants:

F002188, E002188, W002188, I002188

2189:

Calibration Parameter has no Initial Value

Description:

RTA-RTE was unable to determine the initial value for the specified Calibration Parameter.

How to Solve:

Ensure an initial value is present.

Variants:

F002189, E002189, W002189, I002189

2190:

Conflicting memmap file name

Description:

When a `RequiredArtifacts.DependencyOnArtifact` has `category MEMMAP` then its `ArtifactDescriptor.shortLabel` is used for the filename of the Memory Mapping Header File. RTA-RTE found more than one such `DependencyOnArtifact` which conflicted over what the filename should be.

How to Solve:

Ensure that all `ArtifactDescriptor.shortLabel` of `DependencyOnArtifact` with `category MEMMAP` specify the same `shortLabel`.

Variants:

F002190, E002190, W002190, I002190

2191:

`ArtifactDescriptor` of `category MEMMAP`

Description:

To specify the filename of the Memory Mapping Header File, you should create a `RequiredArtifacts.DependencyOnArtifact` with `category MEMMAP`. This message indicates that RTA-RTE encountered a `RequiredArtifacts.DependencyOnArtifact` with a `category` other than `MEMMAP` but containing an `ArtifactDescriptor` with `category MEMMAP`. This does not rename the Memory Mapping Header File, but that was probably the intent.

How to Solve:

If the `DependencyOnArtifact` was intended to rename the Memory Mapping Header File, ensure the `DependencyOnArtifact` itself has `category MEMMAP` rather than its child ArtifactDescriptor.

Variants:

F002191, E002191, W002191, I002191

2192:

Missing `shortLabel`

Description:

RTA-RTE encountered a `RequiredArtifacts. DependencyOnArtifact` with `category MEMMAP` containing an `ArtifactDescriptor` which does not provide a `shortLabel`. Since, in this context, the `shortLabel` is used for the Memory Mapping Header File name this `ArtifactDescriptor` has been ignored.

How to Solve:

An `ArtifactDescriptor. shortLabel` must be provided.

Variants:

F002192, E002192, W002192, I002192

2193:

No `ArtifactDescriptor` found in a required `DependencyOnArtifact`

Description:

RTA-RTE found a `RequiredArtifacts. DependencyOnArtifact` having `category MEMMAP` but could not rename the Memory Mapping Header File without an `ArtifactDescriptor`.

How to Solve:

A `DependencyOnArtifact` with `category MEMMAP` must contain an `ArtifactDescriptor`, whose `shortLabel` is then used for the filename of the Memory Mapping Header File.

Variants:

F002193, E002193, W002193, I002193

2194:

MemorySection references foreign `ExecutableEntity`

Description:

A `MemorySection. executableEntityRef` references an `ExecutableEntity` in a `BswInternalBehavior` other than that referenced by the `BswImplementation. behaviorRef`.

How to Solve:

`ResourceConsumption. MemorySection. executableEntityRef`s should only reference `ExecutableEntity`s aggregated by the `InternalBehavior` implied by the `BswImplementation. behaviorRef`.

Variants:

F002194, E002194, W002194, I002194

2195:

Multiple MemorySections reference the same `ExecutableEntity`

Description:

RTA-RTE has encountered more than one `MemorySection` being used for an `ExecutableEntity`, which is forbidden.

How to Solve:

Remove the `executableEntityRef`s for the `MemorySection`s that reference the `ExecutableEntity`, except for the desired one.

Variants:

F002195, E002195, W002195, I002195

2196:

`System` constant value assignment without a system constant reference

Description:

RTA-RTE has encountered and ignored a `SwSystemconstValue` without a `SwSystemconstRef`.

How to Solve:

Add a `SwSystemconstRef` to the mentioned `SwSystemconstValue` to indicate which system constant the value should be applied to, or remove the `SwSystemconstValue` if it is not required.

Variants:

F002196, E002196, W002196, I002196

2197:

System constant value assignment without a value

Description:

RTA-RTE has encountered and ignored a `SwSystemconstValue` without a `Value`.

How to Solve:

Add a `Value` to the mentioned `SwSystemconstValue` or remove the `SwSystemconstValue` if it is not required.

Variants:

F002197, E002197, W002197, I002197

2198:

A `CompuMethod. compu` contains the wrong number or type of `CompuScale`s

Description:

RTA-RTE has encountered a `CompuMethod. compu` that contains the wrong number or type of valid `CompuScale`s for the `CompuMethod. category` and is ignoring it. Note that this message can arise if there `CompuScale`s were ignored. In that case, separate messages will have been written.

How to Solve:

Amend the referenced `Compu` to aggregate `CompuScale`s as described in the message or change the `category` of the `CompuMethod`. Also check for other warnings or errors relating to aggregated `CompuScale`s.

Variants:

F002198, E002198, W002198, I002198

2199:

Bitfield text table used on non-integer `ImplementationDataType`

Description:

RTA-RTE has encountered a `CompuMethod` of `category BITFIELD_TEXTTABLE` referenced from an `ImplementationDataType` that is not of integer type.

How to Solve:

Ensure that `CompuMethod`s of `category BITFIELD_TEXTTABLE` are only referenced from `ImplementationDataType`s of `category VALUE` with base types having an integer encoding, or from `ImplementationDataType`s of `category TYPE_REFERENCE` that alias such types, or from `ApplicationPrimitiveDataType`s that are mapped to such types.

Variants:

F002199, E002199, W002199, I002199

2200:

Bitfield Text Table mapped to non-integer `ImplementationDataType`

Description:

RTA-RTE has encountered a `CompuMethod` of `category BITFIELD_TEXTTABLE` referenced from an `ApplicationDataType` but that is mapped to an `ImplementationDataType` that is not of integer type.

How to Solve:

Ensure that `CompuMethod`s of `category BITFIELD_TEXTTABLE` are only referenced from `ImplementationDataType`s of `category VALUE` with base types having an integer encoding, or from `ImplementationDataType`s of `category TYPE_REFERENCE` that alias such types, or from `ApplicationPrimitiveDataType`s that are mapped to such types.

Variants:

F002200, E002200, W002200, I002200

2201:

A `Compu` in wrong direction for `category`

Description:

A `CompuMethod` of this `category` must be specified in a particular direction (Phys-to-Internal vs. Internal-to-Phys). RTA-RTE has encountered a `CompuMethod.compu` that is in the wrong direction for the `CompuMethod`.

How to Solve:

Ensure that the `CompuMethod.compu` is in the correct direction.

Variants:

F002201, E002201, W002201, I002201

2202:

No suitable C-identifier can be derived for a text table value

Description:

RTA-RTE has detected that there is no valid C-identifier for the generation of a text table value definition for a `CompuScale` in a text table `CompuMethod`.

How to Solve:

Ensure that all `CompuScale`s with point ranges within the `CompuMethod` contain a `shortLabel` or a `symbol` or a valid C-identifier (`[0-9]`, `[a-z]`, `[A-Z]` or `'_'`) in the `CompuConst.vt`.

Variants:

F002202, E002202, W002202, I002202

2203:

Two `CompuScales` with the same `shortLabel` contain conflicting bitmask properties

Description:

RTA-RTE has detected two `CompuScale`s with the same `shortLabel` within a `CompuMethod` of `category BITFIELD_TEXTTABLE` that contain conflicting `mask` properties. Within a bitfield text table `CompuScale`s with the same `shortLabel` form a single bitfield so must have the same `mask`.

How to Solve:

Ensure that all `CompuScale`s with the same `shortLabel` do not have conflicting properties.

Variants:

F002203, E002203, W002203, I002203

2204:

A `CompuScale` contains a value that is outside the range allowed by the mask

Description:

RTA-RTE has detected that a `CompuScale` contains a value that is outside the range allowed by the mask.

How to Solve:

Ensure that all the values defined by the `CompuScale`s within a bitfield are inside the range allowed by the mask.

Variants:

F002204, E002204, W002204, I002204

2205:

Unmatched text label

Description:

RTA-RTE was unable find a corresponding numerical value for a text label given in a `TextValueSpecification` or `ApplicationValueSpecification`, considering any relevant `CompuMethod`s.

How to Solve:

Check that the data type to which the `ValueSpecification` applies references a `CompuMethod` of a text table `category`, or is mapped to an `ImplementationDataType` that references such a `CompuMethod`. Ensure that the text label specified matches the `CompuConst. vt`, `shortLabel` or `symbol` of one of the `CompuScale`s of the `CompuMethod`.

Variants:

F002205, E002205, W002205, I002205

2206:

Ambiguous text label in enumeration value

Description:

RTA-RTE has detected multiple possible enumeration values for a text label given in a `TextValueSpecification` or `ApplicationValueSpecification`, considering the relevant `CompuMethod`s. For example, the specified text label might not match the `CompuConst. vt` of any `CompuScale` but match the `shortLabel` of multiple `CompuScale`s with different point ranges.

How to Solve:

Give a text label that matches the desired `CompuScale` uniquely, considering that `CompuConst. vt` is matched with highest precedence, then `shortLabel`, then `symbol`.

Variants:

F002206, E002206, W002206, I002206

2207:

Empty `CompuMethod` with non-`IDENTICAL` `category` but no computation specified

Description:

RTA-RTE has encountered a `CompuMethod` with a `category` that is not `IDENTICAL` but does not contain a `Compu` in either direction. Since no conversion is specified this has the effect of a `CompuMethod` of `category` `IDENTICAL`.

How to Solve:

Ensure that the referenced `CompuMethod` contains a `Compu` with valid `CompuScale`s in either the internal to physical direction, as applicable for the `category`, or change the `category` of the `CompuMethod`.

Variants:

F002207, E002207, W002207, I002207

2208:

A bitfield within a `BITFIELD_TEXTTABLE` `CompuMethod` is missing a `mask`

Description:

RTA-RTE has been unable to derive a bitmask for the bitfield formed by `CompuScale`s with the given `shortLabel` in the given `CompuMethod` of `category` `BITFIELD_TEXTTABLE`. None of the `CompuScale`s with the given `shortLabel` have a `mask`.

How to Solve:

Specify the `mask` for the bitfield on at least one `CompuScale` with the given `shortLabel`.

Variants:

F002208, E002208, W002208, I002208

2209:

Bitfield Text Table does not fit in mapped `ImplementationDataType`

Description:

RTA-RTE has encountered a `CompuMethod` of `category` `BITFIELD_TEXTTABLE` referenced from an `ImplementationDataType` that does not have sufficient size; at least one `CompuScale` has a mask that requires more bits than the `ImplementationDataType` has.

How to Solve:

Ensure that the `ImplementationDataType` is of a sufficient width to accommodate all `value`s and `mask`s specified in the referenced `CompuMethod` of `category` `BITFIELD_TEXTTABLE`.

Variants:

F002209, E002209, W002209, I002209

2210:

Bitfield Text Table does not fit in mapped `ImplementationDataType`

Description:

RTA-RTE has encountered a `CompuMethod` of `category` `BITFIELD_TEXTTABLE` referenced from an `ApplicationDataType` but that is mapped to an `ImplementationDataType` that does not have sufficient size; at least one `CompuScale` has a mask that requires more bits than the `ImplementationDataType` has.

How to Solve:

Ensure that the `ImplementationDataType` mapped to the `ApplicationDataType` is of a sufficient width to accommodate all `value`s and `mask`s specified in the referenced `CompuMethod` of `category` `BITFIELD_TEXTTABLE`.

Variants:

F002210, E002210, W002210, I002210

2211:

`Comp` conflicts with `category IDENTICAL`

Description:

RTA-RTE encountered a `CompuMethod` with `category IDENTICAL` but with a `Compu` specifying a non-identity transformation conversion. RTA-RTE is honouring the `category` and ignoring the `CompuScale`s.

How to Solve:

Correct the `category` of the `CompuMethod` or remove any `Compu`s.

Variants:

F002211, E002211, W002211, I002211

2212:

A `CompuScale` in a `CompuMethod` has incorrect contents

Description:

RTA-RTE has encountered a `CompuScale` that contains both a `CompuConst` and a `CompuRationalCoeffs` and is ignoring it. Each `CompuScale` must map its range to either a constant value or a rational formula.

How to Solve:

Amend the `CompuScale` to supply either a `CompuConst` or a `CompuRationalCoeffs`, depending on the `category` of the `CompuMethod`.

Variants:

F002212, E002212, W002212, I002212

2213:

Invalid `compuScale` for `CompuMethod`

Description:

RTA-RTE has encountered a `CompuScale` that contains the wrong type of value for the `CompuMethod`. `category` and is ignoring it. Either a numeric value is given when a textual value is required or vice versa.

How to Solve:

Amend the `CompuConst` or change the `category` of the `CompuMethod`.

Variants:

F002213, E002213, W002213, I002213

2214:

Wrong number of coefficients

Description:

RTA-RTE has encountered a `CompuScale` that contains the wrong number of coefficients that evaluate non-zero for the `CompuMethod. category` and is ignoring it.

How to Solve:

Amend the `CompuRationalCoeffs` or change the `category` of the `CompuMethod`.

Variants:

F002214, E002214, W002214, I002214

2215:

A `CompuScale` contains a `mask` that is not a valid integer value

Description:

RTA-RTE has encountered a `CompuScale. mask` that is not a valid integer value and has ignored it.

How to Solve:

Ensure that the `CompuScale. mask` is a valid integer.

Variants:

F002215, E002215, W002215, I002215

2216:

A `VaritionPointProxy`'s attributes do not match its `category`

Description:

RTA-RTE has encountered a `VariationPointProxy` that contravenes `constr_1253`.

How to Solve:

Ensure that the `VariationPointProxy`'s `category` is `CONDITION` or `VALUE`, and that it aggregates exactly one `conditionAccess` or `valueAccess`, respectively. Note that the `category` is mandatory.

Variants:

F002216, E002216, W002216, I002216

2217:

Invalid `DataTransformation`

Description:

RTA-RTE has encountered an invalid `DataTransformation`. The specific reason is in the error text.

How to Solve:

A `DataTransformation` must reference at least one valid `TransformationTechnology` and no `TransformerClass` can appear more than once within the `DataTransformation`.

Variants:

F002217, E002217, W002217, I002217

2219:

Invalid `TransformationTechnology`

Description:

Invalid `TransformationTechnology`. The specific reason is in the error text.

How to Solve:

The `TransformationTechnology` must aggregate a valid `BufferProperties` and a boolean value of `inPlace`. It must also have a `TransformerClass` of `SERIALIZER`, `SAFETY`, `SECURITY` or `CUSTOM`.

Variants:

F002219, E002219, W002219, I002219

2220:

Invalid `XfrmImplementationMapping`

Description:

RTA-RTE has encountered an invalid `XfrmImplementationMapping`. The specific reason is in the error text.

How to Solve:

The `XfrmImplementationMapping` must aggregate an `XfrmTransformationTechnologyRef` that references a `TransformationTechnology`, and an `XfrmTransformerBswModuleEntryRef` and/or `XfrmInvTransformerBswModuleEntryRef` that maps to a `BswModuleEntry`. It must aggregate either a `XfrmISignalRef` or a `XfrmVariableDataPrototypeInstanceRef` referencing an `ISignal` or `VariableDataPrototype` respectively. In the event that multiple mappings exist, ie multiple `XfrmImplementationMapping`s map the same `ISignal` and `TransformationTechnology`, then each of these mappings must contain an `XfrmVariableDataPrototypeInstanceRef`.

Variants:

F002220, E002220, W002220, I002220

2221:

Unsupported `XfrmImplementationMapping`

Description:

RTA-RTE encountered an unsupported `XfrmImplementationMapping`. The specific reason is in the error text.

How to Solve:

The `XfrmImplementationMapping` must reference an `ISignal` and is forbidden to reference a `VariableDataPrototype`.

Variants:

F002221, E002221, W002221, I002221

2223:

Unsupported `DataTransformation` configuration

Description:

Unsupported `DataTransformation`. The specific reason is in the error text.

How to Solve:

RTA-RTE only supports a `DataTransformation` that references exactly one `TransformationTechnology`.

Variants:

F002223, E002223, W002223, I002223

2225:

Invalid `BufferProperties`

Description:

RTA-RTE encountered an invalid `BufferProperties`. The specific reason is in the error text.

How to Solve:

The `BufferProperties` should express a linear computation in its `bufferComputation`, its `headerLength` should be between 0 and 65535 inclusive, and must have an `inPlace` attribute.

Variants:

F002225, E002225, W002225, I002225

2227:

Unused `XfrmImplementationMapping`

Description:

RTA-RTE encountered a `XfrmImplementationMapping` that has no effect.

How to Solve:

Either remove the unused `XfrmImplementationMapping` or check that it maps a relevant `ISignal` to a `TransformationTechnology` referenced by the `DataTransformation` referenced by the `ISignal`.

Variants:

F002227, E002227, W002227, I002227

2401:

Unable to resolve Port instance reference

Description:

The RTE generator is unable to resolve an `IRef` element to a port prototype within an SWC type.

How to Solve:

Validate the specified instance reference to ensure that if refers to a valid port prototype within the correct SWC.

Variants:

F002401, E002401, W002401, I002401

2403:

Unknown element in instance reference

Description:

RTA-RTE encountered an unexpected element within an instance reference.

How to Solve:

Ensure schema validation is enabled and that the input configuration is correct.

Variants:

F002403, E002403, W002403, I002403

2404:

Missing or invalid `dest` attribute

Description:

RTA-RTE failed to locate a required `dest` attribute at the referenced element.

How to Solve:

A `dest` attribute is required to resolve ambiguity and is therefore mandatory. Ensure schema validation is enabled and that the input configuration is correct.

Variants:

F002404, E002404, W002404, I002404

2410:

Instance reference resolution failed

Description:

RTA-RTE failed to resolve the specified instance reference.

How to Solve:

Ensure schema validation is enabled and that the input configuration is correct. In particular check that all references within the specified instance reference are valid.

Variants:

F002410, E002410, W002410, I002410

2411:

Reference resolution failed

Description:

RTA-RTE failed to resolve the specified reference.

How to Solve:

Ensure schema validation is enabled and that the input configuration is correct. In particular check that the reference is valid.

Variants:

F002411, E002411, W002411, I002411

2412:

Type reference resolution failed

Description:

RTA-RTE failed to resolve the specified type reference.

How to Solve:

Ensure schema validation is enabled and that the input configuration is correct. In particular check that the reference is valid.

Variants:

F002412, E002412, W002412, I002412

2413:

Invalid software component instance reference

Description:

RTA-RTE was unable to resolve the specified software component instance in the given instance reference.

How to Solve:

Check that the instance reference contains a valid sequence of references to component prototypes from the prototype in the root composition down through any nested compositions to a component instance that is instantiated on the ECU instance for which the RTE is being generated.

Variants:

F002413, E002413, W002413, I002413

2414:

Type reference loop detected

Description:

RTA-RTE failed to resolve the specified type reference because it (ultimately) resolves to itself.

How to Solve:

Change the referenced type definition or one of the type definitions to refer-

encea type of `category` `VALUE`.

Variants:

F002414, E002414, W002414, I002414

2415:

Target of local reference not local to the software component

Description:

The specified component-local reference resolved to something that is not within the local component.

How to Solve:

A local reference must be to something that is directly within the same software component type / BSW module type. It may be possible to refer to something in eg a port on the software component using a different style of reference instead, for example in an `AutosarVariable` element an `AutosarVariableInImplDatatype` instance reference can be used instead of a `LocalVariableRef` in order to reference a data prototype in an interface of one of the ports on the software component.

Variants:

F002415, E002415, W002415, I002415

2416:

Invalid reference in context of context-ref

Description:

The specified instance reference contains a reference element that is not applicable for the context in which the instance reference exists. For example an instance reference to a sub-element of data prototype that appears within a `SubElementMapping` cannot validly contain a `PortPrototypeRef`, since the port is implicit in the context of the `SubElementMapping`, but the same style of instance reference can validly contain a `PortPrototypeRef` when within an `InstantiationDataDefProps`.

How to Solve:

Ensure that the specified instance reference contains only those reference elements that are valid for the context in which the instance reference exists.

Variants:

F002416, E002416, W002416, I002416

2417:

Malformed instance reference

Description:

The specified instance reference is malformed. It does not contain exactly one target reference or the target reference is not the last reference in the

sequence.

How to Solve:

Ensure that the instance reference includes a target reference, that the target reference is the last reference and that there is only one target reference.

Variants:

F002417, E002417, W002417, I002417

2418:

Reference in an instance reference to the wrong type of object

Description:

The specified instance reference cannot be resolved because the specified reference is not to a type of object that is valid for that reference. For example it is not valid for a `PortPrototypeRef` to reference a `DataPrototype`.

How to Solve:

Ensure that every reference in the instance reference has a `dest` attribute that is permitted by the schema and that is consistent with the tag of the referenced element.

Variants:

F002418, E002418, W002418, I002418

2420:

Invalid mode instance referenced

Description:

RTA-RTE failed to resolve mode declaration group and/or mode within the specified instance reference.

How to Solve:

Ensure that the first reference is to a port, the second to a `ModeDeclarationGroupPrototype` within the interface categorizing the port and the third to a mode within the group.

Variants:

F002420, E002420, W002420, I002420

2421:

Invalid mode instance reference

Description:

RTA-RTE failed to resolve the port prototype within the specified instance reference.

How to Solve:

Ensure that the first reference within the instance reference element references a port, the second a `ModeDeclarationGroupPrototype` within the inter-

facecategorizing the port and the final reference a mode declaration within the group.

Variants:

F002421, E002421, W002421, I002421

2422:

Failed to resolve an instance reference

Description:

The specified instance reference cannot be resolved. The first referenced prototype was not in the resolution context, typically the root composition or the atomic component type containing the component-local instance reference. This error can also occur if the first reference is to a component prototype that is not mapped to the ECU for which the RTE is being generated.

How to Solve:

Ensure that the first reference within the specified instance reference is to a component prototype that is within the root composition and is (at least partially) mapped to the ECU for which the RTE is being generated, or for a component-local instance reference that the first reference is to a prototype within the same component type.

Variants:

F002422, E002422, W002422, I002422

2423:

Failed to resolve an instance reference

Description:

The specified instance reference cannot be resolved. The specified reference is to a prototype that is not in the type / interface of the previous prototype. This error can also occur if the reference is to a component prototype that is not mapped to the ECU for which the RTE is being generated.

How to Solve:

Ensure that the specified reference within the specified instance reference is to a prototype that is within the type / interface of the previously referenced prototype. If the reference is to a component prototype ensure that it is (at least partially) mapped to the ECU for which the RTE is being generated.

Variants:

F002423, E002423, W002423, I002423

2430:

Invalid SWC type reference

Description:

RTA-RTE detected a `typeTref` within a `ComponentPrototype` that references

an invalid element type.

How to Solve:

Ensure that the reference is to either an SWC type or a Composition.

Variants:

F002430, E002430, W002430, I002430

2440:

Invalid operation

Description:

The referenced operation parameter is not contained within the operation.

How to Solve:

Ensure that the operation and parameter references are consistent.

Variants:

F002440, E002440, W002440, I002440

2450:

No mode switch event source

Description:

RTA-RTE has been unable to resolve the `ModeGroup` within the specified instance reference.

How to Solve:

Check that the `EventSource` references a `ModeSwitchPoint`.

Variants:

F002450, E002450, W002450, I002450

2451:

Unable to resolve exclusive area reference

Description:

RTA-RTE has been unable to resolve the specified reference to an exclusive area.

How to Solve:

Ensure that XML schema validation is enabled and that the input is correct.

Variants:

F002451, E002451, W002451, I002451

2452:

Unexpected reference destination

Description:

RTA-RTE has detected a reference that resolved to an unexpected element

type.

How to Solve:

Ensure that the specified reference is correct.

Variants:

F002452, E002452, W002452, I002452

2453:

Reference out of scope

Description:

RTA-RTE has detected a reference that resolves to an element that is not part of the active configuration.

How to Solve:

Ensure that the specified reference is correct.

Variants:

F002453, E002453, W002453, I002453

2454:

No reference

Description:

RTA-RTE failed to find an expected reference within the specified element.

How to Solve:

Ensure that XML schema validation is enabled and that the input is correct.

Variants:

F002454, E002454, W002454, I002454

2455:

Invalid reference

Description:

RTA-RTE detected an invalid reference at the indicated node.

How to Solve:

Ensure that XML schema validation is enabled and that the input is correct.

Variants:

F002455, E002455, W002455, I002455

2456:

Reference failed filter

Description:

RTA-RTE resolved a reference at the indicated node but it failed to pass the internal filter.

How to Solve:

Ensure that XML schema validation is enabled and that the input is correct.

Variants:

F002456, E002456, W002456, I002456

2457:

Multiple application

Description:

RTA-RTE has detected that a task is mapped to multiple Os applications.

How to Solve:

Ensure that each task is mapped to at most one Os application.

Variants:

F002457, E002457, W002457, I002457

2458:

No application

Description:

RTA-RTE failed to detect an assignment for the specified task to an `OsApplication`

How to Solve:

Ensure that if `Partitions` and/or `OsApplications` are used then every task is assigned to an `OsApplication`.

Variants:

F002458, E002458, W002458, I002458

2459:

SWC instance spans OS application

Description:

RTA-RTE has detected that the runnables of the specified SWC Instance are mapped to tasks in more than one `OsApplication`.

How to Solve:

Ensure that no SWCI spans an `OsApplication`.

Variants:

F002459, E002459, W002459, I002459

2460:

Inconsistent partition and OS application mapping

Description:

RTA-RTE has detected that the runnables of the specified SWC Instance are

mapped to tasks in an `OsApplication` that is not marked as implementing the `Partition` (s) to which the SWC Instance is mapped.

How to Solve:

Ensure that the `OsApplication` references the same `Partition` as references the SWC Instance.

Variants:

F002460, E002460, W002460, I002460

2461:

SWC instance mapped to multiple partitions

Description:

RTA-RTE has detected that the specified SWC Instance is mapped to more than one `Partition`.

How to Solve:

Ensure that only one `Partition` references the SWC Instance.

Variants:

F002461, E002461, W002461, I002461

2462:

SWC instance mapped to a partition multiple times

Description:

RTA-RTE has detected that the specified SWC Instance is mapped to a `Partition` more than once.

How to Solve:

Ensure that only one reference to the SWC Instance exists in the desired `Partition`.

Variants:

F002462, E002462, W002462, I002462

2463:

Inconsistent partition and OS application mapping

Description:

RTA-RTE has detected that the runnables of the specified SWC Instance are mapped to tasks in an `OsApplication` that is marked as implementing a `Partition` to which the SWC Instance is not mapped.

How to Solve:

Ensure that the `OsApplication` references the same `Partition` as references the SWC Instance.

Variants:

F002463, E002463, W002463, I002463

2464:

No partition

Description:

RTA-RTE has detected that the specified software component instance does not belong to any partition.

How to Solve:

Ensure that if `Partitions` are used then every SWC Instance is referenced from some `Partition`.

Variants:

F002464, E002464, W002464, I002464

2465:

Inconsistent partition and OS application mapping

Description:

RTA-RTE has detected that the schedulable entities of the specified BSW module instance are mapped to tasks in an `OsApplication` that is not marked as implementing the BSW `Partition`.

How to Solve:

Ensure that the `OsApplication` references the `Partition` that is marked as containing the BSW for the ECU. Alternatively, specify the command-line option –deviate-bsw-any-partition.

Variants:

F002465, E002465, W002465, I002465

2466:

`AutosarDataPrototypes` require a reference to exactly one `ARType`

Description:

RTA-RTE encountered a Data Prototype that had no associated type.

How to Solve:

Add a `typeTref` to the Data Prototype, referencing either an `ApplicationDataType` or an `ImplementationDataType` (a `SwBaseType` alone is not sufficient)

Variants:

F002466, E002466, W002466, I002466

2467:

Non-local exclusive area reference

Description:

RTA-RTE has detected a reference to a non-local exclusive area (ie one defined in a different SWC). All exclusive areas must be used locally within the SWC within which they are defined.

How to Solve:

Ensure that local exclusive areas (defined within the same internal behavior) are used.

Variants:

F002467, E002467, W002467, I002467

2468:

Missing reference to `ComSignal` or `ComSignalGroup`

Description:

RTA-RTE needs each `ISignalToIPduMapping` element to reference a `ComSignal` or `ComSignalGroup` in the Ecuc, either to take its `shortName` or its `ComSignalHandleId`. It was unable to find and resolve a suitable reference.

How to Solve:

Make sure the whole ECU Extract is present, including the configuration of COM, or remove any inter-ECU commuication from the `System` view.

Variants:

F002468, E002468, W002468, I002468

2469:

Multiple-instantiated `StaticMemory`

Description:

AUTOSAR forbids `StaticMemory` in multiply instantiated SWCTs. This violates AUTOSAR constr_2028 which states that it is an error to have `supportsMultipleInstantiation` =true in a SWCT having `StaticMemory`s. This message can be degraded to a warning with `--deviate-allow-supportsmulti-sharedmemorys`.

How to Solve:

If you require multiple instances of the SWCT, use an `InterRunnableVariable` or Port-based communication to pass the value around. If there will be only one instance of this SWCT in the ECU, then set `supportsMultipleInstantiation` to false, or use `--deviate-allow-supportsmulti-sharedmemorys`.

Variants:

F002469, E002469, W002469, I002469

2470:

Multiply instantiated SWCT contains `StaticMemory`

Description:

AUTOSAR forbids `StaticMemory` in multiply instantiated SWCTs. Communication between Software Component Instances must be via the RTE API.

How to Solve:

Consider whether it is appropriate to instantiate this SWCT multiple times on one ECU. If you do require multiple instances of the SWCT, use an `InterRunnableVariable` or Port-based communication to pass the value around.

Variants:

F002470, E002470, W002470, I002470

2471:

A `ConstantMemory` within a SWCT cannot be referenced by a `FlatInstanceDescriptor`

Description:

In general, `FlatInstanceDescriptor`s are used to set the names of generated C data items for the purposes of measurement. RTA-RTE does not generate any API for `ConstantMemory`s or `StaticMemory`s so no renaming can occur.

How to Solve:

Remove the `FlatInstanceDescriptor`. If measurement of the `ConstantMemory` is required, consider the option `--mcsd-policy:emit-memorys`.

Variants:

F002471, E002471, W002471, I002471

2472:

Invalid context of a reference within an instance reference

Description:

RTA-RTE has detected a reference for either a `contextElementRef` or `targetRef` element which is invalid within the context of the previous `contextElementRef` element.

How to Solve:

Ensure that each of the references are valid within the context of the previous reference.

Variants:

F002472, E002472, W002472, I002472

2473:

Type reference resolution failed

Description:

RTA-RTE failed to resolve the type reference for the specified `contextElementRef` element.

How to Solve:

Ensure schema validation is enabled and that the input configuration is correct. In particular check that the reference is valid.

Variants:

F002473, E002473, W002473, I002473

2474:

Invalid OS application mapping to partition

Description:

RTA-RTE has detected that the runnables of the specified SWC Instance are mapped to tasks in an `OsApplication` which does not reference a `Partition`. When any `Partition` is used in an ECU extract, all software components in the ECU extract must use `Partition`s.

How to Solve:

Ensure that the `OsApplication` references a `Partition` which references the SWC Instance.

Variants:

F002474, E002474, W002474, I002474

2600:

Invalid communication models

Description:

RTA-RTE has detected an invalid connection of interface models, eg client-server and sender-receiver.

How to Solve:

Ensure that interfaces with the same model are connected. In AUTOSAR R4.0 `NvData` and Sender-receiver interfaces can be connected.

Variants:

F002600, E002600, W002600, I002600

2601:

Mismatched provide-require sender-receiver interfaces

Description:

The require interface contains the named data item and there is no data item in the provide interface with the same name. No port interface mapping is referenced by the connector (AUTOSAR R4.0 only) so data items are associated by matching `shortName`.

How to Solve:

For AUTOSAR R4.0 a port interface map can be used to connect dissimilar interfaces. Prior releases associate data items by name and therefore the names must match.

Variants:

F002601, E002601, W002601, I002601

2602:

Incomplete port interface mapping for provide-require sender-receiver interfaces

Description:

The require interface contains the named data item that is not mapped to any data item in the provide interface by the port interface mapping referenced from the given connector.

How to Solve:

Ensure that the port interface mapping associates all pairs of data items that are to be connected, including any pairs with matching `shortName`.

Variants:

F002602, E002602, W002602, I002602

2603:

Invalid event semantics

Description:

RTA-RTE has detected conflicting event semantics within the specified data item.

How to Solve:

Ensure that matching data items in connected interfaces have the same event (queued/unqueued) semantics.

Variants:

F002603, E002603, W002603, I002603

2604:

Mismatched provide-require client-server interfaces

Description:

The require interface contains the named operation and there is no operation in the provide interface with the same name. No port interface mapping is referenced by the connector (AUTOSAR R4.0 only) so operations are associated by matching `shortName`.

How to Solve:

Client-server connections must normally be made between interfaces containing `Operation`s with the same name and same-named arguments. For AUTOSAR R4.0 a port interface map can be used to connect dissimilar interfaces.

Variants:

F002604, E002604, W002604, I002604

2611:

Mismatched provide-require mode switch interfaces

Description:

The require interface contains the named mode group and there is no mode group in the provide interface with the same name. No port interface mapping is referenced by the connector (AUTOSAR R4.0 only) so mode groups are associated by matching `shortName`.

How to Solve:

For AUTOSAR R4.0 a port interface map can be used to connect dissimilar interfaces. Prior releases associate mode groups by name and therefore the names must match.

Variants:

F002611, E002611, W002611, I002611

2613:

Invalid server runnable

Description:

RTA-RTE has detected that a runnable is triggered by multiple `OperationInvokedEvent`s but the parameters in the associated operations are not compatible.

How to Solve:

Ensure that the operations are consistent.

Variants:

F002613, E002613, W002613, I002613

2617:

Mismatched provide-require trigger interfaces

Description:

The require interface contains the named trigger and there is no trigger in the provide interface with the same name. No port interface mapping is referenced by the connector so triggers are associated by matching `shortName`.

How to Solve:

A port interface map can be used to connect dissimilar interfaces.

Variants:

F002617, E002617, W002617, I002617

2618:

Incompatible application error lists

Description:

RTA-RTE has detected that the possible application errors of connected operations are incompatible either because no provide application error exists or similarly named errors have different values.

How to Solve:

Ensure that all connected operations declared compatible lists of possible application errors.

Variants:

F002618, E002618, W002618, I002618

2621:

Mismatched provide-require calibration interfaces

Description:

The require interface contains the named parameter and there is no parameter in the provide interface with the same name. No port interface mapping is referenced by the connector (AUTOSAR R4.0 only) so parameters are associated by matching `shortName`.

How to Solve:

For AUTOSAR R4.0 a port interface map can be used to connect dissimilar interfaces. Prior releases associate parameters by name and therefore the names must match.

Variants:

F002621, E002621, W002621, I002621

2622:

Port interface mapping not between connector interfaces

Description:

RTA-RTE has found a connector that references a port interface mapping that does not map between the interfaces of the connected ports.

How to Solve:

Ensure that the connector references a port interface mapping between the interfaces of the connected ports.

Variants:

F002622, E002622, W002622, I002622

2623:

Connector between service and application interfaces

Description:

The port at one end of the connector has a service interface whereas the port at the other end of the connector has an application (non-service) interface.

How to Solve:

Ensure that all ports involved connections to services are characterised by interfaces with the `isService` flag set, including any delegation ports on compositions that delegate service ports.

Variants:

F002623, E002623, W002623, I002623

2624:

Incomplete port interface mapping for provide-require calibration interfaces

Description:

The require interface contains the named parameter that is not mapped to any parameter in the provide interface by the port interface mapping referenced from the given connector.

How to Solve:

Ensure that the port interface mapping associates all pairs of parameters that are to be connected, including any pairs with matching `shortName`.

Variants:

F002624, E002624, W002624, I002624

2625:

Incomplete port interface mapping for provide-require client-server interfaces

Description:

The require interface contains the named operation that is not mapped to any operation in the provide interface by the port interface mapping referenced from the given connector.

How to Solve:

Ensure that the port interface mapping associates all pairs of operations that are to be connected, including any pairs with matching `shortName`.

Variants:

F002625, E002625, W002625, I002625

2626:

Incomplete port interface mapping for provide-require trigger interfaces

Description:

The require interface contains the named trigger that is not mapped to any trigger in the provide interface by the port interface mapping referenced from the given connector.

How to Solve:

Ensure that the port interface mapping associates all pairs of triggers that are to be connected, including any pairs with matching `shortName`.

Variants:

F002626, E002626, W002626, I002626

2627:

Incomplete port interface mapping for provide-require mode switch interfaces

Description:

The require interface contains the named mode group that is not mapped to any mode group in the provide interface by the port interface mapping referenced from the given connector.

How to Solve:

Ensure that the port interface mapping associates all pairs of mode groups that are to be connected, including any pairs with matching `shortName`.

Variants:

F002627, E002627, W002627, I002627

2628:

Incomplete sub-element mappings

Description:

The require interface contains the named data item that is mapped to a data item in the provide interface by the port interface mapping referenced from the given connector. Within the data prototype mapping that associates the pair of data prototypes there are sub-element mappings that associate pairs of sub-elements but no sub-element mapping associates the named sub-element in the require data prototype with any element in the provide data prototype.

How to Solve:

Ensure that the sub-element mappings within the data prototype mapping associate all pairs of sub-elements at all levels of nesting that are to be connected, including any pairs with matching `shortName` or index.

Variants:

F002628, E002628, W002628, I002628

2629:

A `PortInterfaceMapping` remaps an Interface onto itself

Description:

`PortInterfaceMapping`s are for connecting interfaces having different names, scaling, resolution, or `invalidationPolicy`. The indicated `PortInterfaceMapping` remaps an interface to itself. This is not valid since the `PortInterfaceMapping` could be applied in either direction for every connector referencing it. That is, the pair of interfaces at the start and end of connector could equally match the pair of mapped interfaces as (first, second) or as (second, first).

How to Solve:

Remove references to the `PortInterfaceMapping` from connectors where a direct one-to-one connection is desired. In the exceptional case that a

remapping really is necessary for a connection between ports characterised by the same interface consider wrapping one of the affected software components in a composition component, delegating the port to one characterised by a different but compatible interface, then referencing a port interface mapping between the two interfaces from the delegation connector.

Variants:

F002629, E002629, W002629, I002629

2630:

A `DataPrototypeMapping` references a `DataPrototype` that is not in an interface

Description:

A `DataPrototypeMapping` can only be used to map a `DataPrototype` in one interface to a `DataPrototype` in another interface. A `DataPrototypeMapping` in the indicated `PortInterfaceMapping` references the indicated `DataPrototype` that does not belong to an interface, for example it may be a `StaticMemory` of a software component.

How to Solve:

Correct the references in the `DataPrototypeMapping`s of the indicated `PortInterfaceMapping`.

Variants:

F002630, E002630, W002630, I002630

2631:

A `PortInterfaceMapping` contains inconsistent mappings between Interface elements

Description:

The indicated `PortInterfaceMapping` contains mappings that are not all between the same pair of Interfaces.

How to Solve:

All First*Ref references in the `PortInterfaceMapping` must reference prototypes in the Interface at one end of the associated Connector, and all Second*Ref references must reference prototypes in the Interface at the other end of the associated Connector; and the Interfaces cannot be the same interface.

Variants:

F002631, E002631, W002631, I002631

2632:

Runtime data conversion between floating-point values is not supported

Description:

Runtime data conversion is implemented using integer types, and data conversion computed between floating-point values is likely to be truncated. The generation continues for compatibilty with some legacy projects.

How to Solve:

Ensure that connected floating-point values do not require data conversion, or use integer equivalent (fixed point) representations.

Variants:

F002632, E002632, W002632, I002632

2633:

RTA-RTE does not currently support inter-ECU data conversion on complex types

Description:

RTA-RTE encountered an `ISignal` that references a subelement of a complex type and references a `CompuMethod`. RTA-RTE does not implement data conversion on complex types in inter-ECU communication and is unable to determine whether data conversion is actually requested. Since the presence of a `CompuMethod` indicates a risk that data conversion was intended, this warning is raised. You should review the model to check that data conversion is not expected of the inter-ECU complex data.

How to Solve:

Ensure that no data conversion is needed. Eiither remove the `NetworkRepresentationProps`. `CompuMethod` or suppress the warning by the `--disable-warnings` option.

Variants:

F002633, E002633, W002633, I002633

2636:

Unsupported Runtime data conversion use-case

Description:

Runtime data conversion for inter-ECU transmission or reception is only supported when the signal mappings reference an inner port on a software component instance and the relevant ISignal supplies NetworkRepresentationProps that include a `SwBaseType` and `CompuMethod`. The case of signal mappings referencing an outer, delegated, port is only supported when no runtime data conversion is needed across the delegation connection.

How to Solve:

Alter the signal mappings to reference the inner port on the software component instance, change the dataTypePolicy of the ISignal to LEGACY and supply NetworkRepresentationProps on that ISignal that correspond to the data type from the originally used outer port.

Variants:

F002636, E002636, W002636, I002636

2650:

Incompatible types mapped or connected

Description:

The referenced DataTypes are incompatible with one another

How to Solve:

Ensure that the two types are directly compatible with one another, or add an appropriate `PortInterfaceMapping` to `DataPrototype`s typed by those types, or remove the `SwConnector` or `DataTypeMap` linking the two types. Note that `PortInterfaceMapping`s are not supported with PRPorts.

Variants:

F002650, E002650, W002650, I002650

2681:

Mismatched `CompuMethod`s

Description:

When `DataPrototype`s are connected, eg via `AssemblySwConnector`s, and without `PortDataMapping`s applied, RTA-RTE applies the AUTOSAR "Compatibility of Data Types" rules. This error indicates that an incompatibility was found between the `CompuMethod`s associated with the participating `DataPrototype`s.

How to Solve:

If data conversion was intended, ensure that a `PortDataMapping` associates the two `DataPrototype`s. If data conversion is not intended, then examine the `CompuMethod`s associated with the `DataPrototype`s and correct the references or the `CompuMethod`s themselves.

Variants:

F002681, E002681, W002681, I002681

2682:

`CompuMethod`s reference incompatible `Unit`s

Description:

When `DataPrototype`s are connected, eg via `AssemblySwConnector`s, and without `PortDataMapping`s applied, RTA-RTE applies the AUTOSAR "Compatibility of Data Types" rules. This error indicates that an incompatibility was found between the `Unit`s referenced by the `CompuMethod`s associated with the participating `DataPrototype`s.

How to Solve:

If data conversion was intended, ensure that a `PortDataMapping` associates the two `DataPrototype`s. If data conversion is not intended, then examine the

`CompuMethod`s associated with the `DataPrototype`s and correct the references or the `CompuMethod`s themselves.

Variants:

F002682, E002682, W002682, I002682

2685:

Invalid unit

Description:

RTA-RTE has detected that the specified data type refers to `Unit` but also references a `CompuMethod` that refers to a different `Unit`.

How to Solve:

Ensure that the referenced `Unit`s are the same `Unit` or remove one of the references.

Variants:

F002685, E002685, W002685, I002685

2686:

Incompatible units

Description:

RTA-RTE has detected that connected data types reference (either directly or through their referenced `CompuMethod`s) incompatible, non-convertible units.

How to Solve:

Ensure that connected data types reference compatible or convertible units.

Variants:

F002686, E002686, W002686, I002686

2687:

Data conversion: Constant result

Description:

An expression was encountered during data conversion processing whose result does not depend on the input.

How to Solve:

Numerator1 (The second `V` within an `compuNumerator`) should never be zero. Note: such a Data Conversion might not be explicitly specified on the input, it may be derived from, eg the inverse of a `CompuMethod` defined in the opposite direction.

Variants:

F002687, E002687, W002687, I002687

2688:

Data conversion: Undefined result

Description:

An expression was encountered during data conversion processing which contained a division by zero.

How to Solve:

Denominator (The `V` within an `compuDenominator`) should never be zero. Note: such a Data Conversion might not be explicitly specified on the input, it may be derived from, eg the inverse of a `CompuMethod` defined in the opposite direction.

Variants:

F002688, E002688, W002688, I002688

2690:

Attempt to convert between unconvertible data types

Description:

No conversion is possible between the connected data types named in the error message. Connected data types must both have `CompuMethod`s of `category TEXTTABLE`, or must both have Linear `CompuMethod`s (`category LINEAR`, `IDENTICAL` or `RAT_FUNC` that specifies a linear type conversion).

How to Solve:

Only connect compatible or convertible data types with `AssemblyConnector`s.

Variants:

F002690, E002690, W002690, I002690

2692:

Data conversion: Intermediate Value too wide

Description:

During Data Conversion processing, the range of an intermediate value was such that no platform type could be found to contain it.

How to Solve:

Consider using a different `ApplicationDataType` to reduce the extreme values in the necessary data conversion.

Variants:

F002692, E002692, W002692, I002692

2693:

Data conversion: No width

Description:

Data Conversion tried to retrieve the `baseTypeWidth` of the type of its parameter and failed.

How to Solve:

Make sure that `ApplicationDataTypes` used in `VariableDataPrototypes` are mapped to `ImplementationDataTypes` that reference `SwBaseTypes` and that the referenced `SwBaseTypes` contain valid `baseTypeWidth` elements.

Variants:

F002693, E002693, W002693, I002693

2694:

Data conversion: No base type

Description:

Data Conversion tried to retrieve the `baseTypeWidth` of the type of its parameter and failed.

How to Solve:

Make sure that `ApplicationDataTypes` used in `VariableDataPrototypes` are mapped to `ImplementationDataTypes` that reference `SwBaseTypes` and that the referenced `SwBaseTypes` contain valid `baseTypeWidth` elements.

Variants:

F002694, E002694, W002694, I002694

2697:

Data conversion: Division by zero

Description:

An expression was encountered during data conversion processing which contained a division by zero.

How to Solve:

Denominator (The `V` within an `compuDenominator`) should never be zero. Note: such a Data Conversion might not be explicitly specified on the input, it may be derived from, eg the inverse of a `CompuMethod` defined in the opposite direction.

Variants:

F002697, E002697, W002697, I002697

2699:

Unsupported connection type

Description:

RTA-RTE has detected an unsupported connection type.

How to Solve:

Ensure that both first and second reference within the port interface mapping are valid.

Variants:

F002699, E002699, W002699, I002699

2807:

Complex value in `ApplicationValueSpecification`

Description:

`ApplicationValueSpecification`s only apply for values typed by `ApplicationPrimitiveDataType`s, not `ApplicationRecordDataType`s or `ApplicationArrayDataType`s. In a future release, it will become an error to use an `ApplicationValueSpecification` to express a value typed by a `ApplicationRecordDataType` or `ApplicationArrayDataType`.

How to Solve:

To express a value typed by an `ApplicationRecordDataType`, use a `RecordValueSpecification` which itself eventually aggregates `ApplicationValueSpecification`s in the primitive leaf nodes. To express a value typed by an `ApplicationArrayDataType`, use an `ArrayValueSpecification` in a similar way.

Variants:

F002807, E002807, W002807, I002807

2850:

Common published information error

Description:

The RTE's Common Published Information within the `RteGeneration` parameters specifies an invalid AUTOSAR revision. The error message indicates the range with which this release is compatible. All input files must be conformant to the same release.

How to Solve:

Ensure the common published information is within the supported range.

Variants:

F002850, E002850, W002850, I002850

2851:

Invalid tool version

Description:

The RTE's Common Published Information within the `RteGeneration` parameters specifies an invalid software tool version. The error message indicates the range with which this release is compatible.

How to Solve:

Ensure the tool version common published information is within the supported range. Newer versions of RTA-RTE are backwards compatible and therefore the tool version in the parameters can be less than the current version.

Variants:

F002851, E002851, W002851, I002851

2852:

Version check omitted

Description:

The RTE Configuration contains multiple Common Published Information containers. RTA-RTE has therefore not applied the version check.

How to Solve:

Ensure the RTE configuration parameters contain at most one common published information container.

Variants:

F002852, E002852, W002852, I002852

2880:

No `category`

Description:

RTA-RTE has been unable to locate a `category` element.

How to Solve:

Ensure that XML schema validation is enabled and that the input is correct.

Variants:

F002880, E002880, W002880, I002880

2881:

Unexpected `SwDataDefProp`

Description:

RTA-RTE found an unexpected `SwDataDefProp` element when processing the specified element.

How to Solve:

Ensure that XML schema validation is enabled and that the input is correct.

Variants:

F002881, E002881, W002881, I002881

2883:

Property multiplicity check failed

Description:

RTA-RTE encountered an invalid number of elements for the specified property according to the AUTOSAR specification.

How to Solve:

Review the datadefprops at the error location and ensure that the specific `SwDataDefProp` referenced in the error occurs the correct number of times.

Variants:

F002883, E002883, W002883, I002883

2884:

Invalid `category`

Description:

RTA-RTE encountered an invalid `category` within the specified context.

How to Solve:

Ensure that the element is correct for the specified context. See the *RTA-RTE Reference Manual* for more details.

Variants:

F002884, E002884, W002884, I002884

2885:

Invalid `category`

Description:

RTA-RTE encountered an unsupported `category`.

How to Solve:

Ensure that the element is correct for the specified context. See the *RTA-RTE Reference Manual* for more details.

Variants:

F002885, E002885, W002885, I002885

2886:

`SwFillChar` must be an Integer

Description:

RTA-RTE was not able to interpret `SwFillChar` as an integer.

How to Solve:

`SwFillChar` shall be an integer expressing the fill character encoded in the the relevant encoding.

Variants:

F002886, E002886, W002886, I002886

2887:

Invalid or missing `maxTextSize`

Description:

RTA-RTE was not able to locate and interpret `maxTextSize`.

How to Solve:

`maxTextSize` is mandatory within `SwTextProp`s, and must be a positive integer.

Variants:

F002887, E002887, W002887, I002887

2888:

Unsupported data type `category`

Description:

RTA-RTE does not currently support types with `category`s of `FUNCTION_REFERENCE`, `HOST` or `BIT`.

How to Solve:

Avoid using Types with `category`s of `FUNCTION_REFERENCE`, `HOST` or `BIT`.

Variants:

F002888, E002888, W002888, I002888

2889:

Unsupported mode order

Description:

RTA-RTE encountered an unsupported mode order `category`. AUTOSAR R4.0.3 introduced 'explicit' ordering for mode declarations as well as alphabetical sort based order.

How to Solve:

Ensure that the mode declaration group's `category` is set to either `ALPHABETIC_ORDER` or `EXPLICIT_ORDER`.

Variants:

F002889, E002889, W002889, I002889

2890:

Missing mode order

Description:

RTA-RTE has been unable to locate a mode order `category`.

How to Solve:

Ensure that the mode declaration group's `category` is set to either `ALPHABETIC_ORDER` or `EXPLICIT_ORDER`.

Variants:

F002890, E002890, W002890, I002890

2891:

Invalid `category`

Description:

RTA-RTE encountered an invalid `category` within the specified context. The `category` is not valid in the specified context in any use case.

How to Solve:

Ensure that the element is correct for the specified context. See the *RTA-RTE Reference Manual* for more details.

Variants:

F002891, E002891, W002891, I002891

2892:

`SwDataDefProp` used where no longer allowed

Description:

RTA-RTE encountered the specified property in a location where it is no longer allowed according to the AUTOSAR specification.

How to Solve:

Review the datadefprops at the warning location and ensure that the specific `SwDataDefProp` referenced in the warning occurs the correct number of times according to latest AUTOSAR revision supported by RTA-RTE.

Variants:

F002892, E002892, W002892, I002892

2893:

The Data Constraint `SwDataDefProp` values do not match among connected SWCs

Description:

RTA-RTE encountered a `DataConstr` that does not match (ie have different values) among different SWCs.

How to Solve:

The values of the connected `DataConstr` should not conflict.

Variants:

F002893, E002893, W002893, I002893

2894:

The Display Format `SwDataDefProp` values do not match among connected SWCs

Description:

RTA-RTE encountered a `displayFormat` that does not match (ie have different values) among different SWCs.

How to Solve:

The values of the connected `displayFormat` should not conflict.

Variants:

F002894, E002894, W002894, I002894

2895:

Mismatched Software Interpolation Methods among connected SWCs

Description:

RTA-RTE encountered a `SwInterpolationMethod` that does not match (ie have different values) among different SWCs.

How to Solve:

The values of the connected `SwInterpolationMethod` should not conflict.

Variants:

F002895, E002895, W002895, I002895

2900:

Missing object

Description:

The generated RTE requires a specific OS object but during the strict configuration check this object was not found on the input.

How to Solve:

Either modify the configuration so that the object is not required, eg associate periodic events with an `OsAlarm` to avoid the need to generate a schedule table, or supply the required OS object on the input.

Variants:

F002900, E002900, W002900, I002900

2901:

Missing expiry point

Description:

The RTE implementation requires that the specified schedule table declare an expiry point with offset the specified number ticks. This object is not declared in the input.

How to Solve:

Either modify the configuration so that the object is not required, eg associate periodic events with an `OsAlarm` to avoid the need to generate a schedule table, or supply the required OS object on the input.

Variants:

F002901, E002901, W002901, I002901

2903:

Missing OS Event referenced

Description:

The generated RTE implementation requires that the specified task is declared as accessing the specified `OsEvent`.

How to Solve:

Ensure that either the `OsEvent` is declared in the input or that the configuration of wait points and RTE event mappings mean the use of extended tasks is not required

Variants:

F002903, E002903, W002903, I002903

2904:

Missing alarm activation

Description:

RTA-RTE has detected that the specified alarm does not activate a task yet this is required based on RTE event mappings.

How to Solve:

Ensure that the configuration configures the alarm to activate the specified task.

Variants:

F002904, E002904, W002904, I002904

2905:

Invalid alarm activation

Description:

RTA-RTE has detected that the specified alarm activates an unexpected task. Based on RTE event mappings the configuration requires the alarm to activate the specified task

How to Solve:

Ensure that the alarm is configured to activate the correct task, or change the RTE event mappings.

Variants:

F002905, E002905, W002905, I002905

2906:

Unexpected alarm activation

Description:

RTA-RTE has detected that the specified alarm activate a task yet this is not required based on RTE event mappings

How to Solve:

Ensure that the alarm is configured to perform the correct action.

Variants:

F002906, E002906, W002906, I002906

2907:

Missing OS Event set by OS alarm

Description:

RTA-RTE has determined that the specified alarm is required by the configuration to dispatch the specified task by setting an OS event. No such event setting could be found.

How to Solve:

Ensure that the alarm is configured to perform the correct action.

Variants:

F002907, E002907, W002907, I002907

2908:

Invalid OS Event set by OS alarm

Description:

RTA-RTE has determined that the specified alarm is required by the configuration to dispatch the specified task by setting a particular OS event yet it is configured to set a different event.

How to Solve:

Ensure that the alarm is configured to perform the set the expected event.

Variants:

F002908, E002908, W002908, I002908

2909:

Unexpected OS Event set by OS alarm

Description:

RTA-RTE has determined that the specified alarm is dispatches the specified task by setting an OS event yet the configuration merely requires that the task be activated.

How to Solve:

Ensure that the alarm is configured to perform the correct action.

Variants:

F002909, E002909, W002909, I002909

2910:

>Missing callback by OS alarm

Description:

The specified OS Alarm is required by the configuration to invoke the specified callback but no such callback was detected.

How to Solve:

Ensure that the alarm is configured to perform the correct action.

Variants:

F002910, E002910, W002910, I002910

2911:

>Invalid callback by OS alarm

Description:

The specified OS Alarm is required by the configuration to invoke the specified callback but actually invokes a different callback.

How to Solve:

Ensure that the alarm is configured to perform the correct action.

Variants:

F002911, E002911, W002911, I002911

2912:

>Unexpected callback by OS alarm

Description:

The specified OS Alarm invokes the specified callback but no such callback is required by the configuration.

How to Solve:

Ensure that the alarm is configured to perform the correct action.

Variants:

F002912, E002912, W002912, I002912

2913:

Invalid schedule table activation

Description:

RTA-RTE has detected an invalid schedule table expiry point. The configuration requires an action that activates the specified task.

How to Solve:

Ensure that the specified schedule table entry activates the specified task.

Variants:

F002913, E002913, W002913, I002913

2914:

Invalid schedule table OS event set

Description:

RTA-RTE has detected an invalid schedule table expiry point. The configuration requires an action that dispatches the specified task by setting the specified OS event.

How to Solve:

Ensure that the specified schedule table entry dispatches the specified task by setting the expected event.

Variants:

F002914, E002914, W002914, I002914

2915:

Non-repeating schedule table

Description:

The specified schedule table is required by the configuration to be repeating.

How to Solve:

Ensure that the schedule table is defined as repeating.

Variants:

F002915, E002915, W002915, I002915

2916:

Invalid schedule table duration

Description:

The specified schedule table has an invalid duration (repeat schedule).

How to Solve:

Ensure that the schedule table has the expected duration or modify RTE event mappings and periods to suit.

Variants:

F002916, E002916, W002916, I002916

2917:

Unexpected counter

Description:

RTA-RTE has detected that schedule table is ticked by an unexpected.

How to Solve:

Ensure that the schedule table is ticked by the specified counter.

Variants:

F002917, E002917, W002917, I002917

2918:

Missing counter

Description:

RTA-RTE has could not determine the counter that ticks the specified schedule table.

How to Solve:

Ensure that the schedule table is ticked by the required counter.

Variants:

F002918, E002918, W002918, I002918

2919:

Expected AUTO OS Event mask

Description:

RTA-RTE could not detect the required OS event mask. The configuration requires that the specified OS Event have an automatically assigned mask.

How to Solve:

Ensure that the OS Event mask is assigned correctly.

Variants:

F002919, E002919, W002919, I002919

2920:

Invalid OS Event mask

Description:

RTA-RTE could detected an invalid or unexpected OS event mask.

How to Solve:

Ensure that the OS Event mask is assigned the specified mask.

Variants:

F002920, E002920, W002920, I002920

2921:

osNeeds.arxml does not contain `OsApplication` configuration

Description:

RTA-RTE has written osNeeds.arxml containing the OS objects it requires, but these objects have not been integrated into the `OsApplication` configuration.

How to Solve:

Manually update OS configuration.

Variants:

F002921, E002921, W002921, I002921

2950:

No symbol table

Description:

RTA-RTE has attempted to evaluate an system constant referenced without a symbol table.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F002950, E002950, W002950, I002950

2951:

No symbol table entry

Description:

RTA-RTE has failed to find an identifier within the system constant symbol table.

How to Solve:

Ensure that all referenced system constants are defined.

Variants:

F002951, E002951, W002951, I002951

2952:

Failed to resolve reference to a system constant

Description:

RTA-RTE has encountered an AUTOSAR formula language function that takes as an argument a reference to a system constant where the reference cannot be evaluated.

How to Solve:

Ensure that the reference is valid (eg in the correct form) and resolves to a system constant.

Variants:

F002952, E002952, W002952, I002952

2953:

Undefined function argument

Description:

RTA-RTE has encountered an AUTOSAR formula language function with an undefined argument.

How to Solve:

Ensure that all function arguments are present and valid.

Variants:

F002953, E002953, W002953, I002953

2954:

Unknown AUTOSAR Formula function

Description:

RTA-RTE has encountered an unknown function when processing an AUTOSAR formula language expression.

How to Solve:

Please see the *RTA-RTE Reference Manual* for details of supported functions.

Variants:

F002954, E002954, W002954, I002954

2955:

Unknown AUTOSAR Formula operator

Description:

RTA-RTE has encountered an unknown operator when processing an AUTOSAR formula language expression.

How to Solve:

Please see the *RTA-RTE Reference Manual* for details of supported operators.

Variants:

F002955, E002955, W002955, I002955

2956:

Division by zero

Description:

An AUTOSAR Formula expression has attempted to divide by zero.

How to Solve:

Ensure the denominator of any expression does not evaluate to zero.

Variants:

F002956, E002956, W002956, I002956

2957:

Unexpected real operand

Description:

RTA-RTE has detected an AUTOSAR Formula function passed an unexpected real argument.

How to Solve:

Ensure that function arguments are of the correct type.

Variants:

F002957, E002957, W002957, I002957

2958:

Unexpected string operand

Description:

RTA-RTE has detected an AUTOSAR Formula function passed an unexpected string argument.

How to Solve:

Ensure that function arguments are of the correct type.

Variants:

F002958, E002958, W002958, I002958

2959:

Invalid numeric operand

Description:

RTA-RTE has detected an AUTOSAR Formula function passed an invalid number argument.

How to Solve:

Ensure that numeric arguments are valid. For example, check that octal numbers do not include the digits '8' and '9'.

Variants:

F002959, E002959, W002959, I002959

2960:

Syntax error

Description:

RTA-RTE has detected a syntax error within an AUTOSAR Formula expression

How to Solve:

Ensure that all expressions are syntactically correct with respect to the AUTOSAR documentation.

Variants:

F002960, E002960, W002960, I002960

2961:

Expected string operand not found

Description:

RTA-RTE has detected an AUTOSAR Formula function that expected a string argument is actually passed an argument of an incorrect type.

How to Solve:

Ensure that function arguments are of the correct type.

Variants:

F002961, E002961, W002961, I002961

2962:

Undefined system constant value

Description:

RTA-RTE has encountered an AUTOSAR formula language expression containing a reference to a system constant and that system constant has not been assigned any value by the `PredefinedVariant` that is selected in the ECU configuration. Note that this can also occur if no `PredefinedVariant` is selected in the ECU configuration.

How to Solve:

Ensure that the correct `PredefinedVariant` is referenced in the ECU configuration, that the selected `PredefinedVariant` references the correct `SwSystemconstantValueSet`s and that one of those `SwSystemconstantValueSet`s defines a value for the given `SystemConstant`.

Variants:

F002962, E002962, W002962, I002962

2963:

Conflicting system constant values

Description:

RTA-RTE has encountered an AUTOSAR formula language expression containing a reference to a system constant and that system constant has been assigned multiple conflicting values by the `PredefinedVariant`(s) that is(are) selected in the ECU configuration.

How to Solve:

Ensure that only one `SwSystemconstantValue` that defines a value for the given `SystemConstant` is active, via references from the ECU configuration to a `PredefinedVariant` and hence to the containing `SwSystemconstantValueSet`, or that all active `SwSystemconstantValue`s for the given `SystemConstant` define the same value.

Variants:

F002963, E002963, W002963, I002963

2964:

Cyclical dependency in system constant value expressions

Description:

RTA-RTE has encountered an AUTOSAR formula language expression containing a reference to a system constant and that system constant is assigned a value expression by the `PredefinedVariant` that is selected in the ECU configuration that is ultimately in terms of the system constant itself.

How to Solve:

Ensure that where system constants are defined in terms of other system constants they are all ultimately defined in terms of system constants with simple values defined.

Variants:

F002964, E002964, W002964, I002964

2965:

An `ISignal` references more than one `DataTransformation`

Description:

RTA-RTE has encountered an `ISignal` that references more than one `DataTransformation`. At most one `DataTransformation` may be referenced by an `ISignal`.

How to Solve:

Ensure that there is at most one referenced `DataTransformation`.

Variants:

F002965, E002965, W002965, I002965

2966:

No `BswModuleEntry` mapped for transformation (sending side)

Description:

The sending `ISignal` references a `DataTransformation`, so it is necessary to map all the `TransformationTechnology`s referenced by that `DataTransformation` to `BswModuleEntry`s that implement the transformers. RTA-RTE could not find a valid mapping for the `TransformationTechnology` referenced in the error text.

How to Solve:

Ensure that the ECU Parameter Values contain an `XfrmImplementationMapping` that references the `ISignal` and the `TransformationTechnology`, and has a valid `XfrmTransformationTechnologyRef`.

Variants:

F002966, E002966, W002966, I002966

2967:

No `BswModuleEntry` mapped for transformation (receiving side)

Description:

The receiving `ISignal` references a `DataTransformation`, so it is necessary to map all the `TransformationTechnology`s referenced by that `DataTransformation` to `BswModuleEntry`s that implement the retransformers. RTA-RTE could not find a valid mapping for the `TransformationTechnology` referenced in the error text.

How to Solve:

Ensure that the ECU Parameter Values contain an `XfrmImplementationMapping` that references the `ISignal` and the `TransformationTechnology`, and has a valid `XfrmInvTransformationTechnologyRef`.

Variants:

F002967, E002967, W002967, I002967

2968:

Missing XfrmBufferSizeInBytes

Description:

An `XfrmImplementationMapping` does not contain an `XfrmBufferSizeInBytes`. This is a mandatory, vendor-specific parameter.

How to Solve:

Calculate the desired output buffer size and enter it in the `XfrmImplementationMapping/XfrmBufferSizeInBytes`.

Variants:

F002968, E002968, W002968, I002968

2970:

Unknown `SwAddrMethod. sectionType` in input

Description:

A `SwAddrMethod` on the input has a `sectionType` that was not recognised. RTA-RTE will act as if no `sectionType` has been specified.

How to Solve:

Supported values for `SwAddrMethod. sectionType` are `CALPRM`, `CODE`, `CONST`, and `VAR`.

Variants:

F002970, E002970, W002970, I002970

2971:

Missing, mandatory networkRepresentationProps

Description:

The `ISignal` has a `dataTypePolicy` of `OVERRIDE` or `LEGACY`, but does not have a `networkRepresentationProps`. This violates AUTOSAR 4.2.2 `[SWS_Rte_07842]`

How to Solve:

Either change the `dataTypePolicy` or add a `networkRepresentationProps`.

Variants:

F002971, E002971, W002971, I002971

2972:

Multiple signatures implied for Xfrm function prototype

Description:

RTA-RTE tried to generate multiple incompatible function prototypes for a transformer implementation function. This may indicate that the same function has been requested for transformation of both transmitting and receiving `ISignal`s, or for transformation of different data types that are not mapped to the same `ImplementationDataType`.

How to Solve:

Check the `BswModuleEntry`s referenced from all `XfrmImplementationMappings` and ensure that the same C function is not used in situations that result in the creation of incompatible function prototypes.

Variants:

F002972, E002972, W002972, I002972

2973:

MEMORY optimization strategy is not supported with data transformation

Description:

This error is raised when an input configuration contains any `XfrmImplementationMapping`s and the optimization strategy is set to MEMORY.

How to Solve:

Either change the optimization strategy to RUNTIME, or remove the `XfrmImplementationMapping`s from the input configuration if data transformation is not required.

Variants:

F002973, E002973, W002973, I002973

2974:

Unsupported `ImplementationDataTypeElement.category` for transformers

Description:

RTA-RTE does not support passing of arrays whose `ImplementationDataTypeElement` is of a `category` other than `TYPE_REFERENCE`. This is because, to conform with `[SWS_Rte_05107]`, arrays are passed as pointer-to-element-type. When the `ImplementationDataTypeElement` is of `category TYPE_REFERENCE`, the "element type" is the `shortName` of the referenced `ImplementationDataType`. For other `category`s, RTA-RTE would be

forced to write a type expression inline, which violates MISRA [AUTOSAR_SWS_Rte appendix C].

How to Solve:

Extract the data type definition from the `ImplementationDataTypeElement` and create a named `ImplementationDataType`. Change the `ImplementationDataTypeElement` to be `category TYPE_REFERENCE`, referencing the new `ImplementationDataType`.

Variants:

F002974, E002974, W002974, I002974

2975:

Invalid `DataTransformation` when first `TransformationTechnology` is not `SERIALIZER`

Description:

The first `TransformationTechnology` referenced from a `DataTransformation` must be of class `SERIALIZER`. RTA-RTE does not support `DataTransformation`s where the first referenced `TransformationTechnology` is not `SERIALIZER`, in accordance with AUTOSAR [constr_3123].

How to Solve:

Change the `DataTransformation` so that the first `TransformationTechnology` is of class `SERIALIZER`, or remove the references to that `DataTransformation` from all `ISignal`s.

Variants:

F002975, E002975, W002975, I002975

2976:

A command-line option is deprecated

Description:

This error is raised when a deprecated command-line option was presented to RTA-RTE, or it was presented in a deprecated form.

How to Solve:

Refer to the release notes. The issue reference is given in the error text.

Variants:

F002976, E002976, W002976, I002976

2977:

An argument given for a command-line option is deprecated

Description:

This error is raised when a command-line option is presented to RTA-RTE with a deprecated argument.

How to Solve:

Refer to the release notes. The issue reference is given in the error text.

Variants:

F002977, E002977, W002977, I002977

2978:

Port involved in inter-ecu dataTransformation must be a delegation port

Description:

Port involved in inter-ecu dataTransformation must be a delegation port.

How to Solve:

Ensure the ISignal references a SystemSignal that is mapped to an outer delegation port via a SenderReceiverToSignalMappings.

Variants:

F002978, E002978, W002978, I002978

2979:

Transformation not supported: multiple signals mapped to the same data element

Description:

RTA-RTE does not support the use of data transformation where multiple signals or signal groups are mapped to the same data element in a delegation port on the `RootSwComposition`, or any of its connected ports (ie with signal fan-in or fan-out).

How to Solve:

Map only a single transforming `SystemSignal` to the data element in the delegation port on the `RootSwComposition` and remove the `SenderReceiverToSignalMappings` or `SenderReceiverToSignalGroupMappings` for all other signals or signal groups that reference the same data element. Alternatively, remove the data transformation from all mapped signals by removing the `DataTransformation` reference from the associated `ISignals`.

Variants:

F002979, E002979, W002979, I002979

2980:

Transformation unsupported for delegation port connected to multiple inner ports

Description:

RTA-RTE does not support the use of data transformation where multiple inner ports are connected to the same delegation port on the `RootSwComposition`.

How to Solve:

Map only a single inner port to the outer delegation port by removing the `DelegationSwConnectors`s for the other inner ports, or remove the data transformation by removing the `DataTransformation` reference from the associated `ISignal`.

Variants:

F002980, E002980, W002980, I002980

2981:

IOC may not be used to transport transformed data from a reception callback

Description:

RTA-RTE does not support scenarios in which the Com or LdCom module is mapped to a different `EcucParition` to one (or more) receivers of the data (which would require the use of the IOC to transport the untransformed data from the incoming Com/LdCom callback).

How to Solve:

This situation can fundamentally be solved in two different ways. A user can either set up their integration to avoid the use of the IOC when transporting data between callbacks and receivers or the user can use parameters to force RTA-RTE to perform inverse transformation inside the callback instead. For the former, a user can either configure their system such that all receivers are located in the same `EcucPartition` as the Com/LdCom module or use the RTA-RTE option –inter-ecu-no-ioc to force RTA-RTE to not use the IOC (although bear in mind that this may have safety implications). For the latter, the user can use the command line option –deviate-cbk-xfrm or ECUC parameter `RteXfrmInvUnqueuedDataCbk`.

Variants:

F002981, E002981, W002981, I002981

2982:

Invalid ManagedModeGroup reference

Description:

The BswModuleEntity references as ManagedModeGroup a `ModeDeclarationGroupPrototype` that is not aggregated as a ProvidedModeGroup within the parent BswModuleDescription. This may be because the `ModeDeclarationGroupPrototype` is aggregated within a different BswModuleDescription, or because the `ModeDeclarationGroupPrototype` is aggregated by a ModeSwitchInterface.

How to Solve:

Modify the BswModuleEntity.ManagedModeGroup so that it references a `ModeDeclarationGroupPrototype` that is aggregated as a ProvidedModeGroup within the parent BswModuleDescription.

Variants:

F002982, E002982, W002982, I002982

2983:

Transformation only supported for AUTOSAR systems with `category` ECU_EXTRACT

Description:

RTA-RTE does not support the use of data transformation when the AUTOSAR `System` is not of `category ECU_EXTRACT`. Data transformation is deemed to be in use when the ECU value file contains one or more `XfrmImplementationMappings`.

How to Solve:

Provide RTA-RTE with a system of `category ECU_EXTRACT` or remove all `XfrmImplementationMappings`s from the ECU value file.

Variants:

F002983, E002983, W002983, I002983

2984:

Transformation unsupported: delegation port connected to incompatible inner port

Description:

RTA-RTE does not support data transformation for signals that are mapped to a delegation port, when that delegation port is connected to an incompatible inner port. Ports are considered to be incompatible whenever a `PortInterfaceMapping` is referenced by the `DelegationSwConnector` that connects the inner and outer ports (whether or not data conversion is in use).

How to Solve:

Connect the inner port to a compatible delegation port and remove the `PortInterfaceMapping` reference from the `DelegationSwConnector`, or remove the data transformation by removing the `DataTransformation` reference from the associated `ISignal`.

Variants:

F002984, E002984, W002984, I002984

2985:

Transformation unsupported: delegation port connected to inner delegation port

Description:

RTA-RTE does not support data transformation for signals that are mapped to a delegation port on the `RootSwComposition`, when that delegation port is connected to a delegation port on an inner `CompositionSWC`.

How to Solve:

Remove the inner composition by flattening, so that the delegation port (to which the transformed signal is mapped) is connected directly to an inner port on an atomic SWC, or remove the data transformation by removing the `DataTransformation` reference from the associated `ISignal`.

Variants:

F002985, E002985, W002985, I002985

2986:

Transformation not supported: multiple ISignals mapped to the same SystemSignal

Description:

RTA-RTE does not support the use of data transformation where multiple `ISignal`s are mapped to a `SystemSignal` (ie with signal fan-in or fan-out).

How to Solve:

Map only a single transforming `ISignal` to the `SystemSignal` and remove the references to that `SystemSignal` from all other `ISignal`s, or remove the data transformation from all `ISignal`s that reference the `SystemSignal` by removing their `DataTransformation` reference.

Variants:

F002986, E002986, W002986, I002986

2987:

SwcModeSwitchEvent does not reference a valid number of ModeDeclarations

Description:

A `SwcModeSwitchEvent` must reference as `mode` attribute(s) one or two `ModeDeclaration`s. The referenced `SwcModeSwitchEvent` references either zero or more than two `ModeDeclaration`s.

How to Solve:

Modify the `SwcModeSwitchEvent` so that it references one or two `ModeDeclaration`s as `mode` attribute(s).

Variants:

F002987, E002987, W002987, I002987

2989:

ISignal does not reference a SystemSignal

Description:

An `ISignal` must reference a single `SystemSignal`. The referenced `ISignal` does not reference a `SystemSignal`.

How to Solve:

Modify the `ISignal` so that it references single `SystemSignal`, or remove the `ISignal` from the configuration.

Variants:

F002989, E002989, W002989, I002989

2990:

Unrecognized INI file section

Description:

An unrecognized section was found in the RTEGen.ini configuration file.

How to Solve:

Correct the section name or remove the unrecognized section and all its contents from the RTEGen.ini file.

Variants:

F002990, E002990, W002990, I002990

2991:

Error parsing input file

Description:

An error occurred while parsing the supplied input file. The following info message 2992 shows the specific error that was returned by the XML parser.

How to Solve:

Ensure that all AUTOSAR XML description files use the AUTOSAR 4.0 namespace, contain the standard AUTOSAR document header `[TR_APRXML 6.2]`, and conform with the AUTOSAR R23-11 XML schema. The following info message 2992 contains the error returned by the XML parser which may help to identify the source of the problem.

Variants:

F002991, E002991, W002991, I002991

2992:

XML parser returned an error

Description:

The XML parser returned an error when attempting to process an input file. This information may help in diagnosing the problem with the input XML.

How to Solve:

(none)

Variants:

F002992, E002992, W002992, I002992

2993:

RteInitializationRunnableBatch is not supported

Description:

At least one `RteInitializationRunnableBatch` was found in the input configuration. RTA-RTE does not support the mapping of events to `RteInitializationRunnableBatch`es.

How to Solve:

Remove all `RteInitializationRunnableBatch` containers from the ECU value file.

Variants:

F002993, E002993, W002993, I002993

2996:

Overriding schema for XML validation

Description:

The XML schema used for validating ARXML input files was overridden in the `RTEGen.ini` file.

How to Solve:

Validate the input against the default schema by removing the `ValidateARXMLAgainstSchema` option from the `RTEGen.ini` file

Variants:

F002996, E002996, W002996, I002996

2997:

Invalid overriding schema location

Description:

The XML schema used for validating ARXML input files was overridden in the `RTEGen.ini` file, but a valid XML schema (XSD) file could not be found at the specified location.

How to Solve:

Ensure that the path supplied in the `RTEGen.ini` file is correct and references a valid XML schema (XSD) file.

Variants:

F002997, E002997, W002997, I002997

2999:

`InitEvent` instance is not referenced by an `RteEventToTaskMapping`

Description:

The input configuration contained an `InitEvent` instance that is not referenced by an `RteEventToTaskMapping`. An `RteEventToTaskMapping` is required to map the `InitEvent` instance to an `OsTask` and to determine the runnable

ordering within that task.

How to Solve:

Ensure that the ECU value file contains an `RteEventToTaskMapping` for each `InitEvent` instance. The `RteEventToTaskMapping` must contain an `RtePositionInTask` and an `RteMappedToTaskRef`.

Variants:

F002999, E002999, W002999, I002999

3000:

`InitEvent` instance is referenced by multiple `RteEventToTaskMappings`

Description:

The input configuration contained an `InitEvent` instance that is referenced by multiple `RteEventToTaskMappings`. A single `RteEventToTaskMapping` is required to map the `InitEvent` instance to an `OsTask` and to determine the runnable ordering within that task.

How to Solve:

Ensure that the ECU value file contains a single `RteEventToTaskMapping` for each `InitEvent` instance. The following info messages 3001 show the `RteEventToTaskMappings` that reference the `InitEvent` instance.

Variants:

F003000, E003000, W003000, I003000

3001:

`RteEventToTaskMapping` references instance of `InitEvent` in SWCI

Description:

The input configuration contained multiple `RteEventToTaskMappings` for an instance of an `InitEvent`. This message provides details of those `RteEventToTaskMappings`.

How to Solve:

See error 3000.

Variants:

F003001, E003001, W003001, I003001

3002:

No `rtePositionInTask` provided for `InitEvent`

Description:

The input configuration contained an `InitEvent` instance that was not associated with any `RtePositionInTask` value.

How to Solve:

Ensure that the `RteEventToTaskMapping` referencing the `InitEvent` contains an `RtePositionInTask` value.

Variants:

F003002, E003002, W003002, I003002

3003:

Invalid `rtePositionInTask` value provided for `InitEvent`

Description:

The input configuration contained an `InitEvent` instance that was associated with an invalid value for `RtePositionInTask`.

How to Solve:

Ensure that the `RteEventToTaskMapping` referencing the `InitEvent` contains an `RtePositionInTask` value in the range 0 to 65535.

Variants:

F003003, E003003, W003003, I003003

3004:

`InitEvent` does not start any `RunnableEntity`

Description:

The input configuration contained an `InitEvent` that does not reference any `RunnableEntity` to be started when the event is triggered.

How to Solve:

Ensure that the `InitEvent` references a `RunnableEntity` in the role `startOnEvent`.

Variants:

F003004, E003004, W003004, I003004

3005:

`PRPortPrototype` only supported for AUTOSAR systems with `category` ECU_EXTRACT

Description:

RTA-RTE does not support the use of `PRPortPrototype` when the AUTOSAR `System` is not of `category` `ECU_EXTRACT`.

How to Solve:

Either provide RTA-RTE with a system of `category` `ECU_EXTRACT`, or remove all `PRPortPrototypes`s from the input configuration. The following info messages 1001 show the `PRPortPrototype`s found in the input configuration.

Variants:

F003005, E003005, W003005, I003005

3006:

`DelegationSwConnector` referencing a `PRPortPrototype` is not supported

Description:

RTA-RTE does not support the use of `PRPortPrototypes` for inter-ECU communications. This means that it is not permitted to reference a `PRPortPrototype` from a `DelegationSwConnector`.

How to Solve:

Remove all `DelegationSwConnectors` that reference a `PRPortPrototype` on a `SwComponentPrototype`, and replace any `PRPortPrototypes` on the `RootSwComposition` with either `PPortPrototype` or `RPortPrototype` as appropriate.

Variants:

F003006, E003006, W003006, I003006

3008:

`PRPortPrototype` typed by unsupported interface class

Description:

The configuration contained a `PRPort` that was typed by an unsupported class of interface. RTA-RTE only supports the use of `PRPortPrototypes` that are typed by `ModeSwitchInterfaces`, `SenderReceiverInterfaces`, or `NvDataInterfaces`

How to Solve:

Ensure that the configuration only contains `PRPortPrototypes` that are typed by `ModeSwitchInterfaces`, `SenderReceiverInterfaces`, or `NvDataInterfaces`

Variants:

F003008, E003008, W003008, I003008

3010:

`PRPortPrototype` connected to N:1 communication graph is not supported

Description:

The configuration contained an N:1 communication graph involving a `PRPortPrototype` via the referenced `AssemblySwConnector`. RTA-RTE does not support the use of PRPorts in N:1 communications.

How to Solve:

Remove the `PRPortPrototypes` from the N:1 communication graph. The creation of an adapter SWC may allow the same connections without directly involving the `PRPortPrototype` in an N:1 communication.

Variants:

F003010, E003010, W003010, I003010

3012:

Aggregation of `PRPortPrototype` in unsupported class of SoftwareComponentType

Description:

The configuration contained a `PRPort` that was aggregated in an class of `SoftwareComponentType` that does not support the use of PRPorts. RTA-RTE only supports the use of `PRPortPrototype`s in `ApplicationSoftwareComponentType`s, `ComplexDeviceDriverSoftwareComponentType`s, `ServiceSwComponentType`s, `SensorActuatorSwComponentType`s, `NvBlockSwComponentType`s or `EcuAbstractionSwComponentType`s.

How to Solve:

Ensure that the configuration only contains `PRPortPrototype`s that are aggregated in `ApplicationSoftwareComponentType`s, `ComplexDeviceDriverSoftwareComponentType`s, `ServiceSwComponentType`s, `SensorActuatorSwComponentType`s, `NvBlockSwComponentType`s or `EcuAbstractionSwComponentType`s.

Variants:

F003012, E003012, W003012, I003012

3013:

`InitEvent` instance not mapped to `OsTask` is unsupported

Description:

At least one `InitEvent` instance was not mapped to an `OsTask` in the input configuration. RTA-RTE requires `InitEvent`s instances to be mapped to `OsTask`s.

How to Solve:

Ensure that all `RteEventToTaskMapping`s in the ECU value file that reference `InitEvent`s include an `RteMappedToTaskRef`s referencing an appropriate `OsTask`.

Variants:

F003013, E003013, W003013, I003013

3014:

Incompatible event mappings for `OsTask` (`InitEvent` with other event classes)

Description:

The input configuration contained an `OsTask` with a mixture of `InitEvent`s and other event classes mapped to it. When an `InitEvent` is mapped to an `OsTask`, RTA-RTE does not permit any other class of `RteEvent` or `BswEvent` to be mapped to the same task.

How to Solve:

Ensure that if an `InitEvent` is mapped to a task, no other event classes are mapped to that task. The accompanying info messages 1001 show the `RteEventToTaskMappings` and `RteBswEventToTaskMappings` for the task in question.

Variants:

F003014, E003014, W003014, I003014

3015:

VFB tracing for COM TX for array elements is not supported

Description:

The input configuration contained inter-ECU transmission of an array with COM transmit hooks enabled for that array and VFB tracing enabled. RTA-RTE does not generate the hook for the array.

How to Solve:

(none)

Variants:

F003015, E003015, W003015, I003015

3016:

Event has no EventSourceRef

Description:

The specified Event has no EventSourceRef, and an EventSourceRef is mandatory for this Event type.

How to Solve:

Add an EventSourceRef to the Event.

Variants:

F003016, E003016, W003016, I003016

3019:

Invalidation not supported for data element in the context of `PRPortPrototype`

Description:

The `InvalidationPolicy` aggregated by the referenced `SenderReceiverInterface` enables acknowledgement for a `VariableDataPrototype` that exists in the context of a `PRPortPrototype`. The use of data element invalidation with PRPorts is not supported.

How to Solve:

Disable invalidation for the data element by removing the `InvalidationPolicy` aggregated by the referenced `SenderReceiverInterface` or setting its value to `dontInvalidate`.

Variants:

F003019, E003019, W003019, I003019

3020:

Status not supported for data element in the context of `PRPortPrototype`

Description:

A `NonQueuedReceiverComSpec` aggregated by the referenced `PRPortPrototype` enables status for a `VariableDataPrototype` that exists in the context of a `PRPortPrototype`. The use of status with PRPorts is not supported.

How to Solve:

Disable status for the data element by ensuring that the `NonqueuedReceiverComSpec` aggregated by the referenced `PRPortPrototype` meets all the following criteria: `aliveTimeout` value set to 0; `handleNeverReceived` set to `FALSE`; `handleDataStatus` set to `FALSE`; `handleOutOfRange` set to any value other than `none`.

Variants:

F003020, E003020, W003020, I003020

3021:

`PRPort` connected to `RPort` of incompatible interface type not supported

Description:

The referenced `AssemblySwConnector` connects a `PRPortPrototype` to an `RPortPrototype` typed by an incompatible `SenderReceiverInterface`. Note that, for `PRPortPrototype`s, RTA-RTE requires connected interfaces to be directly compatible according to [constr_1069 part 1a], without the use of a `PortInterfaceMapping`.

How to Solve:

Ensure that all receivers connected to a `PRPortPrototype` use a directly compatible interface type, and remove the `PortInterfaceMapping` from all `AssemblySwConnector`s that reference a `PRPortPrototype`.

Variants:

F003021, E003021, W003021, I003021

3022:

`PortInterfaceMapping` not supported with `PRPortPrototype`

Description:

The referenced `AssemblySwConnector` references a `PRPortPrototype` and a `PortInterfaceMapping`. For `PRPortPrototype`s, RTA-RTE requires connected interfaces to be directly compatible according to [constr_1069 part 1a], without the use of a `PortInterfaceMapping`.

How to Solve:

Ensure that all receivers connected to a `PRPortPrototype` use a directly compatible interface type, and remove the `PortInterfaceMapping` from all `AssemblySwConnector`s that reference a `PRPortPrototype`.

Variants:

F003022, E003022, W003022, I003022

3023:

InvalidationPolicy of externalReplacement not supported

Description:

The `InvalidationPolicy` aggregated by the referenced `SenderReceiverInterface` requests an `InvalidationPolicy` of `externalReplacement`, which is not currently supported.

How to Solve:

Use a different `InvalidationPolicy` for the data element - either `keep`, `replace` or `dontInvalidate`.

Variants:

F003023, E003023, W003023, I003023

3024:

handleDataStatus not supported for data element

Description:

A `NonqueuedReceiverComSpec` aggregated by the referenced `PortPrototype` has the `handleDataStatus` attribute set to `TRUE`. The use of the `handleDataStatus` attribute is not supported.

How to Solve:

Remove the `handleDataStatus` attribute from the `NonqueuedReceiverComSpec` aggregated by the referenced `PortPrototype`, or set its value to `FALSE`. The `NonqueuedReceiverComSpec` that requires modification will also reference the data element mentioned in the error message text.

Variants:

F003024, E003024, W003024, I003024

3025:

handleOutOfRange not supported for data element

Description:

A `ComSpec` aggregated by the referenced `PortPrototype` has the `handleOutOfRange` attribute set to a value other than `none`. The use of the `handleOutOfRange` attribute is not supported.

How to Solve:

Remove the `handleOutOfRange` attribute from the `ComSpec` aggregated by the referenced `PortPrototype`, or set its value to `none`. The `ComSpec` that requires modification will reference the data element mentioned in the error message text.

Variants:

F003025, E003025, W003025, I003025

3027:

Queued data element in `PRPort` violates [constr_1200]

Description:

The referenced `VariableDataPrototype` has a `swImplPolicy` of `queued`, and is used in the context of a `PRPortPrototype`. This violates AUTOSAR [constr_1200].

How to Solve:

Set the `swImplPolicy` of all data elements used in the context of PRPorts to a value other than `queued`.

Variants:

F003027, E003027, W003027, I003027

3028:

Aggregation of `PRPortPrototype` by multiply-instantiable SWC is not supported

Description:

The configuration contained an `AtomicSwComponentType` that is multiply-instantiable (`supportsMultipleInstantiation` set to `TRUE`) and that also aggregates a `PRPortPrototype`. RTA-RTE does not support the use of `PRPortPrototype`s with multiply-instantiable SWCs.

How to Solve:

Modify the configuration so that the SWC is not multiply-instantiable (set `supportsMultipleInstantiation` to `FALSE`) or remove the `PRPortPrototype` from the SWC.

Variants:

F003028, E003028, W003028, I003028

3029:

DataFilter not supported for data element in the context of `PRPortPrototype`

Description:

A `NonqueuedReceiverComSpec` aggregated by the referenced `PRPortPrototype` enables data filtering for a `VariableDataPrototype` that exists in the context of a `PRPortPrototype`. The use of filtering with PRPorts is not supported.

How to Solve:

Disable data filtering for the data element by removing the `dataFilter` attribute from the `NonqueuedReceiverComSpec` aggregated by the referenced `PRPortPrototype`.

Variants:

F003029, E003029, W003029, I003029

3030:

LdCom TP API is not supported

Description:

RTA-RTE does not support the use of the LdCom `TP` API.

How to Solve:

Use the LdCom `IF` API, by setting the LdComApiType to `LDCOM_IF` in the referenced `LdComIPdu`.

Variants:

F003030, E003030, W003030, I003030

3031:

LdCom is not supported with non-transformed signals

Description:

The supplied input configuration contained a non-transformed `ISignal` mapped to an `LdComIPdu`. RTA-RTE only supports the use of LdCom with signals that use data transformation.

How to Solve:

Enable data transformation for the `ISignal` by adding a reference to a `DataTransformation`, or map the `ISignal` to regular AUTOSAR COM.

Variants:

F003031, E003031, W003031, I003031

3032:

Multiple `ISignalToIPduMappings` reference the same `ISignal`

Description:

Having more than one `ISignalToIPduMapping` reference the same `ISignal` violates AUTOSAR constr_3514.

How to Solve:

Remove `ISignalRefs` from `ISignalToIPduMappings` which reference the same `ISignal` until only one `ISignalToIPduMapping` references the `ISignal`. The following info messages 1001 show the `ISignalToIPduMappings` referencing the same `ISignal`

Variants:

F003032, E003032, W003032, I003032

3034:

Did not expect locatable to be split

Description:

The configuration object was defined in multiple XML fragments but RTA-RTE did not expect the element to participate in an `atpSplitable` pattern. Some child elements might have been missed.

How to Solve:

RTEGen's validation here is too weak to tell whether it is a problem with the configuration or not. Check whether the named element is `atpSplitable` and consider raising a support ticket.

Variants:

F003034, E003034, W003034, I003034

3036:

`ISignalToIPduMapping` maps `ISignal` to multiple `LdComIPdu`s or `ComSignal`s

Description:

The input configuration mapped an `ISignal` to an `LdComIPdu` and a `ComSignal`, or to multiple `LdComIPdu`s. LdCom is only supported for configurations where an `ISignal` is mapped to a single `LdComIPdu`. The following Info 1001 messages give details of the `LdComIPdu`s and `ComSignal`s that reference the `ISignalToIPduMapping`.

How to Solve:

Map the `ISignal` to a single `LdComIPdu` by removing all other `LdComIPdu`s and `ComSignal`s that reference the `ISignalToIPduMapping`. Alternatively, map the signal to Com by removing all `LdComIPdu`s that reference the `ISignalToIPduMapping`.

Variants:

F003036, E003036, W003036, I003036

3037:

COMBased `DataTransformation` not supported when using LdCom

Description:

The supplied input configuration contains an `ISignal` mapped to an `LdComIPdu` which also uses a `DataTransformation` where the serializing transformer has `protocol COMBased`. This configuration is not supported by RTEGen.

How to Solve:

Remove the usage of the COMBased transformation from the referenced `ISignal`, or map the `ISignal` to COM instead of LdCom.

Variants:

F003037, E003037, W003037, I003037

3038:

`RunnableEntity` with implicit access started by multiple `OsTasks`

Description:

The supplied input configuration contains an implicitly accessing `RunnableEntity` started by multiple `RteEvent`s that are mapped to different `OsTask`s. RTA-RTE does not support this configuration as it may result in the corruption of implicitly-accessed data elements. The following Info 1001 messages give details of the `RteEventToTaskMapping`s that map the activating `RteEvent`s to `OsTask`s.

How to Solve:

Either ensure that all of the `RteEvent`s starting the referenced `RunnableEntity` are mapped to the same `OsTask`, or remove all `VariableAccess`s with the role `DataWriteAccess` or `DataReadAccess` from the referenced `RunnableEntity`.

Variants:

F003038, E003038, W003038, I003038

3041:

Evaluation of `CompuMethod` requires division by zero

Description:

The referenced `CompuMethod` aggregates either a `CompuPhysToInternal` with a first denominator ('D0') of 0, or a `CompuInternalToPhys` with a second numerator ('N1') of 0. When RTA-RTE attempts to evaluate the referenced `CompuMethod` in order to convert a value from its `INTERNAL` to `PHYSICAL` representation, it encounters a division by zero.

How to Solve:

If the `CompuMethod` aggregates a `CompuPhysToInternal`, ensure that the first denominator ('D0') is non-zero. If the `CompuMethod` aggregates a `CompuInternalToPhys`, ensure that the second numerator ('N1') is non-zero.

Variants:

F003041, E003041, W003041, I003041

3042:

`CompuMethod` evaluates to zero

Description:

The referenced `CompuMethod` aggregates either a `CompuPhysToInternal` with a second numerator ('N1') of 0, or a `CompuInternalToPhys` with a first denominator ('D0') of 0. When the `CompuMethod` is evaluated, its scale will be zero. If it is used in a scaling multiplication for data conversion, the resulting value from that will be zero. If it is used in a scaling division for data conversion, a division by zero will occur in the generated RTE code.

How to Solve:

If the `CompuMethod` aggregates a `CompuPhysToInternal`, ensure that the second numerator ('N1') is non-zero. If the `CompuMethod` aggregates a `CompuInternalToPhys`, ensure that the first denominator ('D0') is non-zero.

Variants:

F003042, E003042, W003042, I003042

3043:

`EcucValueCollection` contains multiple `EcuExtractRef`s

Description:

RTA-RTE has detected multiple `EcuExtractRef` instances for a single `EcucValueCollection`. An `EcucValueCollection` must only have a single unambiguous `EcuExtractRef`.

How to Solve:

Ensure that the `EcuExtractRef` contains a single `EcuExtractRef`. If the `EcucValueCollection` is split, ensure that all fragments that contain an `EcuExtractRef` have the same `EcuExtractRef` value.

Variants:

F003043, E003043, W003043, I003043

3044:

`CompuMethod` used for floating-point data conversion is not of `category` `IDENTICAL` or `LINEAR`

Description:

The referenced `CompuMethod` is referenced by the `NetworkRepresentationProps` of an `ISignal` which undergoes floating-point data conversion. Floating-point data conversion is only supported with `CompuMethod`s of `category` `IDENTICAL` or `LINEAR`.

How to Solve:

Either change the `category` of the referenced `CompuMethod` to `IDENTICAL` or `LINEAR`, or remove any references to the `CompuMethod` from any `NetworkRepresentationProps`.

Variants:

F003044, E003044, W003044, I003044

3045:

`----mcore-spinlocks-always` ignored on single core configuration

Description:

If the option `--mcore-spinlocks-always` is enabled on a single core configuration, RTA-RTE will continue as if the option had not been set.

How to Solve:

(none)

Variants:

F003045, E003045, W003045, I003045

3046:

Incorrect number of `RunnableEntityArguments`

Description:

When a `RunnableEntity` implements a service to an `OperationInvokedEvent` the number of `RunnableEntityArgument` must be equal to the number of associated `ArgumentDataPrototypes` plus the number of associated `PortDefinedArgumentValues`, in accordance with constr_1164.

How to Solve:

Ensure that the number of RunnableEntityArguments is equal to the number of associated ClientServerOperation's ArgumentDataPrototypes plus the number of associated PortDefinedArgumentValues

Variants:

F003046, E003046, W003046, I003046

3047:

Multiple instantiation of Service SWC, ECU Abstraction SWC, and Complex Device Driver SWC not supported

Description:

The configuration contained a multiply instantiable `ServiceSwComponentType`, `EcuAbstractionSwComponentType`, or `ComplexDeviceDriverSwComponentType` which is not supported

How to Solve:

Ensure that `SwcInternalBehavior. supportsMultipleInstantiation` is set to `FALSE` for `ServiceSwComponentTypes`, `EcuAbstractionSwComponentTypes`, and `ComplexDeviceDriverSwComponentTypes`.

Variants:

F003047, E003047, W003047, I003047

3048:

Invalid `LdComApiType` value

Description:

An `LdComApiType` can only be set to either `LDCOM_IF` or `LDCOM_TP`

How to Solve:

Ensure that the `LdComApiType` is set to either `LDCOM_IF` or `LDCOM_TP`

Variants:

F003048, E003048, W003048, I003048

3049:

IRV access does not reference an IRV that exists in the context of the same SWC as the IRV access

Description:

A `RunnableEntity` contains a `VariableAccess` defined in the role `writtenLocalVariable` or `readLocalVariable`. This `VariableAccess` contains an `AutosarVariableRef` which references a `VariableDataPrototype` defined in either of the roles `ImplicitInterRunnableVariable` or `ExplicitInterRunnableVariable` where that `VariableDataPrototype` is in a different `SwcInternalBehavior` to the `RunnableEntity`. This contravenes AUTOSAR constraint `constr_2026 @4.4.0`.

How to Solve:

Either remove the `VariableAccess` or change it to reference a `VariableDataPrototype` defined in the role `ImplicitInterRunnableVariable` or `ExplicitInterRunnableVariable` in the same `SwcInternalBehavior` as the `VariableAccess`.

Variants:

F003049, E003049, W003049, I003049

3050:

Unsupported interface type for `PRPortPrototype` in `NvBlockSwComponent`

Description:

RTA-RTE does not support a configuration containing an `NvBlockSwComponent` that aggregates a `PRPortPrototype` that is not typed by an `NvDataInterface`.

How to Solve:

Ensure that all `PRPortPrototype`s aggregated by an `NvBlockSwComponent` are typed by an `NvDataInterface`.

Variants:

F003050, E003050, W003050, I003050

3052:

`ModeDeclarationGroup` referenced from multiple `IncludedModeDeclarationGroupSet`s

Description:

A `ModeDeclarationGroup` is referenced from multiple `IncludedModeDeclarationGroupSet`s. The following info 1001 messages give information on each `SwcInternalBehavior` that contains an `IncludedModeDeclarationGroupSet` which references the `ModeDeclarationGroup`.

How to Solve:

Remove all but one of the references to the `ModeDeclarationGroup` by `IncludedModeDeclarationGroupSet`s.

Variants:

F003052, E003052, W003052, I003052

3053:

Runtime range check is not supported for `ISignals`

Description:

RTA-RTE does not support `handleOutOfRange` values other than `NONE` for `ISignalProps`

How to Solve:

Ensure that any `ISignalProps` aggregated by an `ISignal` has its `handleOutOfRange` set to `NONE`

Variants:

F003053, E003053, W003053, I003053

3056:

`RteEvent` references `RunnableEntity` in a different `SwcInternalBehavior`

Description:

An `RteEvent` references a `RunnableEntity` that is aggregated by a different `SwcInternalBehavior` to the `RteEvent`.

How to Solve:

Ensure that the `RteEvent` and the `RunnableEntity` that it references are aggregated by the same `SwcInternalBehavior`.

Variants:

F003056, E003056, W003056, I003056

3063:

Unsupported interface type for `PRPortPrototype` referenced as `requester`

Description:

The configuration contained a `PRPortPrototype` that was referenced as `requester` from an `AssemblySwConnector`, but that `PRPortPrototype` was not typed by an `NvDataInterface`.

How to Solve:

Ensure that any `PRPortPrototype` referenced by an `AssemblySwConnector` as `requester` is typed by an `NvDataInterface`.

Variants:

F003063, E003063, W003063, I003063

3064:

Instance of event is not referenced by an `EventToTaskMapping`

Description:

An instance of an event is not referenced by an `EventToTaskMapping`. This could be because the event should not be mapped to a `OsTask` or because the `EventToTaskMapping` is missing.

How to Solve:

Add the event to an `EventToTaskMapping` for the instance. Note `EventToTaskMapping` is not required to reference an `OsTask`.

Variants:

F003064, E003064, W003064, I003064

3065:

Unsupported connection to `PRPortPrototype`

Description:

The configuration contains a connection where the `requester` is a `PRPortPrototype`, but the `provider` port uses an unsupported interface type. RTA-RTE only supports `PRPortPrototype`s as `requester` where the connected `provider` is typed by an `NvDataInterface`.

How to Solve:

Ensure that when an `AssemblySwConnector` references a `PRPortPrototype` as `requester` then the port referenced by that `AssemblySwConnector` as `provider` is typed by an `NvDataInterface`.

Variants:

F003065, E003065, W003065, I003065

3066:

Implicit data access in a runnable that may not be activated

Description:

The `RunnableEntity` is not activated by an `RteEvent` but contains a `dataReadAccess` or a `dataWriteAccess`. At generation phase this could result in error 7223 unless the runnable is started by some other means, eg by synchronizing with a BSW `ExecutableEntity`.

How to Solve:

If BSW synchronization is intended for this runnable then the warning can be ignored. Otherwise, add an activating event.

Variants:

F003066, E003066, W003066, I003066

5016:

Too few signals

Description:

RTA-RTE has encountered a data mapping for a complex type with insufficient signals defined.

How to Solve:

Ensure that each element of the complex type (record or array) is mapped to a signal.

Variants:

F005016, E005016, W005016, I005016

5017:

Too many signals

Description:

RTA-RTE has encountered a data mapping for a complex type with too many signals defined.

How to Solve:

Ensure that each element of the complex type (record or array) is mapped to a signal is mapped exactly once.

Variants:

F005017, E005017, W005017, I005017

5019:

No signal group

Description:

RTA-RTE has encountered data mappings for the specified complex type but the signals are not mapped into a signal group.

How to Solve:

Ensure that signals for a complex type are mapped into a signal group to ensure atomic transmission.

Variants:

F005019, E005019, W005019, I005019

5026:

Ignored trailing characters

Description:

RTA-RTE has ignored trailing characters in the specified timeout string.

How to Solve:

Ensure that all timeouts are specified in seconds with no suffix.

Variants:

F005026, E005026, W005026, I005026

5028:

A required Platform Type is missing

Description:

RTA-RTE requires Platform Types for internal purposes. This error indicates that a particular Platform Type was required, but RTA-RTE could not find it on

the input.

How to Solve:

Provide the missing type on the input. To solve this in general, provide a full set of Platform Types in the input model. Note: these should reference a set of `SwBaseType`s tailored to your target platform

Variants:

F005028, E005028, W005028, I005028

5029:

Incompatible variables

Description:

RTA-RTE has detected incompatible types for variable declarations within the generated RTE.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F005029, E005029, W005029, I005029

5030:

Incompatible initial values

Description:

RTA-RTE has detected incompatible initial values for variable declarations within the generated RTE.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F005030, E005030, W005030, I005030

5031:

Incompatible array arity

Description:

RTA-RTE has detected incompatible arity for array variable declarations within the generated RTE.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F005031, E005031, W005031, I005031

5032:

Signal reused

Description:

RTA-RTE has detected that the specified system signal has been used for more than one Client-Server operation

How to Solve:

Ensure system signals are not reused.

Variants:

F005032, E005032, W005032, I005032

5033:

Incompatible return types

Description:

RTA-RTE has detected incompatible return type declarations for the specified function.

How to Solve:

Please contact ETAS GmbH for support. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F005033, E005033, W005033, I005033

5035:

Invalid implementation node

Description:

RTA-RTE has detected an implementation description node with an invalid type.

How to Solve:

Please contact ETAS GmbH for support. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F005035, E005035, W005035, I005035

5036:

Invalid function call parameter

Description:

RTA-RTE has detected an function call parameter implementation description object with no value.

How to Solve:

Please contact ETAS GmbH for support. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F005036, E005036, W005036, I005036

5037:

Unconnected mode user

Description:

Required mode port is unconnected. The mode instance will remain in the initial mode.

How to Solve:

None. This message can be suppressed if required.

Variants:

F005037, E005037, W005037, I005037

5038:

No receive buffer

Description:

RTA-RTE has been unable to locate a previously allocated receive buffer.

How to Solve:

Please contact ETAS GmbH for support. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F005038, E005038, W005038, I005038

5039:

Cannot find type

Description:

RTA-RTE has been unable to locate type for the specified item.

How to Solve:

Ensure all types are declared and correctly referenced. For application types ensure a mapping to an implementation type is present.

Variants:

F005039, E005039, W005039, I005039

5041:

Invalid operation parameter direction

Description:

RTA-RTE failed to find a valid operation parameter direction when testing parameter mapping.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F005041, E005041, W005041, I005041

5042:

Inter-ECU transmission of typedef

Description:

RTA-RTE has attempted inter-ECU transmission of a typedef.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F005042, E005042, W005042, I005042

5043:

Inter-ECU transmission of a function pointer

Description:

RTA-RTE has attempted inter-ECU transmission of a function-point.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F005043, E005043, W005043, I005043

5044:

Inter-ECU transmission of a invalid node type

Description:

RTA-RTE has attempted inter-ECU transmission of an unrecognized node type.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F005044, E005044, W005044, I005044

5045:

No server

Description:

RTA-RTE could not identify a server for the specified client operation. Client API will behave as if unconnected.

How to Solve:

Verify that a connected server operation exists and that it has an associated `OperationInvokedEvent` and runnable.

Variants:

F005045, E005045, W005045, I005045

5046:

Partial initialization of calibration parameter group

Description:

RTA-RTE failed to locate complete initialization for a calibration parameter group. Values were found for some but not all of the calibration parameters in the group. The values that were found have been ignored, since a full initializer could not be formed.

How to Solve:

Ensure that all elements are initialized – it is not possible to partially initialize a calibration parameter group.

Variants:

F005046, E005046, W005046, I005046

5047:

Calibration Parameter Group not (fully) initialized

Description:

RTA-RTE failed to locate complete initialization for a calibration parameter group. Some or all of the calibration parameters in the group were missing initial values. Full initialization is required because the group is for calibration parameters within a require port that is unconnected for at least one instance of the software component.

How to Solve:

Ensure that all elements are initialized.

Variants:

F005047, E005047, W005047, I005047

5049:

Partial initialization

Description:

RTA-RTE failed to locate complete initialization for InitRAM calibration support.

How to Solve:

Ensure that all elements are initialized – it is not possible to partially initialize a calibration group.

Variants:

F005049, E005049, W005049, I005049

5060:

No SWCI functions container

Description:

RTA-RTE has failed to find the SWC instance functions container

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F005060, E005060, W005060, I005060

5070:

No runnable wrapper

Description:

RTA-RTE has been unable to find an wrapper for a runnable entity while processing the specified SWC instance.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F005070, E005070, W005070, I005070

5071:

No instance wrapper

Description:

RTA-RTE has been unable to find an instance wrapper for a runnable entity

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F005071, E005071, W005071, I005071

5090:

Unconnected BSW mode user

Description:

BSW Required mode group prototype is unconnected. The mode instance will remain in the initial mode.

How to Solve:

None. This message can be suppressed if required.

Variants:

F005090, E005090, W005090, I005090

5100:

Identifier too long

Description:

RTA-RTE has detected identifiers that are not unique within the specified number of characters.

How to Solve:

Shorten names used within the configuration to enable RTA-RTE to create identifiers are that unique within the required length.

Variants:

F005100, E005100, W005100, I005100

5110:

Aliased server has more than one client

Description:

RTA-RTE has detected that a directly invoked server runnable that uses the implicit S/R API. In this situation RTA-RTE must use the client as an alias for the a server to determine its execution context however this is not possible when there are multiple clients.

How to Solve:

Either ensure that there is exactly one client or map the server to a task to disable direction invocation.

Variants:

F005110, E005110, W005110, I005110

5111:

Aliased server invoked by a servermore than one client

Description:

RTA-RTE has detected that a directly invoked server runnable that uses the implicit S/R API. In this situation RTA-RTE must use the client as an alias for the a server to determine its execution context however this is not possible when the client is itself a server.

How to Solve:

Either ensure that the clients for directly invoked servers that access the

implicit API are not themselves servers.

Variants:

F005111, E005111, W005111, I005111

5205:

Cannot connect NV `PRPorts` where both ports are aggregated by `ApplicationSwComponentTypes`

Description:

RTA-RTE does not support the connection of two `PRPortPrototypes` typed by `NvDataInterface` via an `AssemblySwConnector` where both ports are aggregated by `ApplicationSwComponentTypes`. An NV `PRPort` on an `ApplicationSwComponentType` must be connected to an NV `PRPort` on an `NvBlockSwComponentType`.

How to Solve:

Ensure that the `AssemblySwConnector` connects an NV `PRPort` aggregated by an `ApplicationSwComponentType` to an NV `PRPort` on an `NvBlockSwComponentType`.

Variants:

F005205, E005205, W005205, I005205

5206:

Cannot connect NV `PRPorts` where both ports are aggregated by `NvBlockSwComponentTypes`

Description:

RTA-RTE does not support the connection of two `PRPortPrototypes` typed by `NvDataInterface` via an `AssemblySwConnector` where both ports are aggregated by `NvBlockSwComponentTypes`. An NV `PRPort` on an `NvBlockSwComponentType` must be connected to an NV `PRPort` on an `ApplicationSwComponentType`.

How to Solve:

Ensure that the `AssemblySwConnector` connects an NV `PRPort` aggregated by an `NvBlockSwComponentType` to an NV `PRPort` on an `ApplicationSwComponentType`.

Variants:

F005206, E005206, W005206, I005206

5207:

All NV `PRPort` instances should be referenced by an `AssemblySwConnector`

Description:

An `ApplicationSwComponentType` instance aggregates an NV `PRPort`, but this NV `PRPort` is not referenced by any `AssemblySwConnector`. In such a case, the NV `PRPort` is not connected to NVRAM, thus writing and reading NV data

through this NV `PRPort` will not work as expected.

How to Solve:

Ensure that all NV `PRPort` instances are referenced by an `AssemblySwConnector`.

Variants:

F005207, E005207, W005207, I005207

6000:

Require SpecReq attribute missing

Description:

Buffering has encountered a SpecReq node without all required attributes set. The error message parameters indicate the missing attribute name.

How to Solve:

Internal error; contact ETAS GmbH for support.

Variants:

F006000, E006000, W006000, I006000

6022:

Unknown receive mode

Description:

Unrecognized value for `RECEIVE_MODE` for specified data item instance.

How to Solve:

Internal error; contact ETAS GmbH for support.

Variants:

F006022, E006022, W006022, I006022

6028:

No timeout

Description:

RTA-RTE has not detected a timeout for the specified data item instance. The generated API will wait indefinitely.

How to Solve:

None. If not required this message can be suppressed.

Variants:

F006028, E006028, W006028, I006028

6029:

Invalid SpecReq node

Description:

RTA-RTE has detected an invalid SpecReq node when generating the RTE.

How to Solve:

Internal error; contact ETAS GmbH for support.

Variants:

F006029, E006029, W006029, I006029

6030:

Unknown interface for port

Description:

RTA-RTE is unable to determine the interface for the specified port

How to Solve:

Ensure a value interface is referenced from the port prototype.

Variants:

F006030, E006030, W006030, I006030

6031:

Invalid buffer read node

Description:

RTA-RTE has detected an invalid node type within a buffer read node.

How to Solve:

Internal error; contact ETAS GmbH for support.

Variants:

F006031, E006031, W006031, I006031

6032:

No receiver found

Description:

RTA-RTE has attempted to configure transmission but cannot locate the receiver.

How to Solve:

Internal error; contact ETAS GmbH for support.

Variants:

F006032, E006032, W006032, I006032

6033:

No receiver data item found

Description:

RTA-RTE has attempted to configure transmission but cannot locate the

receiver's data item.

How to Solve:

Internal error; contact ETAS GmbH for support.

Variants:

F006033, E006033, W006033, I006033

6034:

Invalid filter for queued data item

Description:

Invalid filter specification for queued communication. The filters `NEW-IS-EQUAL`, `NEW-IS-DIFFERENT`, `MASKED-NEW-EQUALS-MASKED-OLD`, `MASKED-NEW-DIFFERS-MASKED-OLD`, `NEW-IS-GREATER`, `NEW-IS-LESS-OR-EQUAL`, `NEW-IS-LESS` and `NEW-IS-GREATER-OR-EQUAL` cannot be used with queued communication.

How to Solve:

Ensure correct filter configuration is used for queued communication.

Variants:

F006034, E006034, W006034, I006034

6051:

Mixed IOC communications

Description:

Detected an N:1 communication where some senders are in the same OS Application as the receiver and some senders are in a different OS Application.

How to Solve:

This configuration is not supported by RTA-RTE.

Variants:

F006051, E006051, W006051, I006051

6100:

Non-local Calibration

Description:

RTA-RTE has detected a calibration SWC instance that is not mapped to the same ECU as a requirer. Inter-ECU communication of calibration values is not permitted.

How to Solve:

Ensure calibration components and their requiring SWCs are mapped to the same ECU instance.

Variants:

F006100, E006100, W006100, I006100

6101:

Missing calibration SWC mapping

Description:

RTA-RTE cannot find an SWC mapping for calibration SWC instance in the same ECU as a requirer

How to Solve:

Ensure all calibration instances are mapped to the correct ECU instances.

Variants:

F006101, E006101, W006101, I006101

6113:

Multiple calibration providers

Description:

RTA-RTE has detected that a calibration `RPort` is connected to more than one provider. A require port categorized by a `CalPrmInterface` must have at most one provider.

How to Solve:

Ensure that each require port categorized by a calibration interface has exactly one provider.

Variants:

F006113, E006113, W006113, I006113

6114:

Unconnected item in an `RPort`

Description:

RTA-RTE has detected an unconnected item in a require port.

How to Solve:

Either enable support for unconnected require ports or ensure that the port instance is connected and a corresponding item exists in the connected provide port instance.

Variants:

F006114, E006114, W006114, I006114

6115:

Invalid calibration `RPort`

Description:

RTA-RTE has detected a calibration SWC with an `RPort`. `RPort`s must not be attached to Calibration SWCs.

How to Solve:

Ensure that calibration SWCs do not have require ports.

Variants:

F006115, E006115, W006115, I006115

6123:

Couldn't calculate `PhysConstr` for this `SwBaseType`

Description:

When creating `McSupportData` with `--mcsd-policy:phys-constr-always`, RTA-RTE forces all `McDataInstance`s to have a `PhysConstr`. The indicated `McDataInstance` did not have a `PhysConstr` attached to any `ApplicationDataType` or `ImplementationDataType`, nor was RTA-RTE able to derive a `PhysConstr` from the `SwBaseType` (must be `category NONE`, `2C` or `BOOLEAN`).

How to Solve:

Provide a `DataConstr` containing `PhysConstr`s, or provide a `DataConstr` containing `InternalConstr`s and a `CompuMethod` for RTA-RTE to calculate the `PhysConstr`s.

Variants:

F006123, E006123, W006123, I006123

6124:

Invalid mcsd-policy switch

Description:

The –mcsd-policy option was passed an invalid switch. See the *Reference Manual* for permitted values.

How to Solve:

Check the spelling and validity of the –mcsd-policy option argument.

Variants:

F006124, E006124, W006124, I006124

6125:

Couldn't calculate `PhysConstr` using this `CompuMethod`

Description:

When creating `McSupportData` with `--mcsd-policy:phys-constr-always`, RTA-RTE found a `CompuMethod` of an unsupported `category`. RTA-RTE treated the `CompuMethod` as `IDENTICAL`.

How to Solve:

Provide `PhysConstr`s directly in the `DataConstr`, or provide a `DataConstr` containing `InternalConstr`s and a `CompuMethod` of `category LINEAR` or `IDENTICAL` for RTA-RTE to calculate the `PhysConstr`s.

Variants:

F006125, E006125, W006125, I006125

6126:

Conflicting `SwCalprmAxis.category`

Description:

RTA-RTE encountered a `SwCalprmAxis` within a `ParameterAccess` or `InstantiationDataDefProps` that has a different `category` to the corresponding axis in the data type of the data prototype referenced by the `ParameterAccess` or `InstantiationDataDefProps`, or an inconsistent use of `SwAxisIndividual` / `SwAxisGrouped`.

How to Solve:

`SwCalprmAxis` elements within `InstantiationDataDefPropss` or `ParameterAccesss` must have the same `category` (and use of `SwAxisIndividual` / `SwAxisGrouped`) as the corresponding axis in the data type of the referenced data prototype.

Variants:

F006126, E006126, W006126, I006126

6127:

The input variable type is not a scalar type

Description:

RTA-RTE found that the input variable type for the input axis of a `SwCalprmAxis.SwAxisIndividual` does not have `ApplicationDataType.category` set to `VALUE` or `BOOLEAN`.

How to Solve:

The input variable should be a scalar value; ie `ApplicationDataType.category` equal to either `VALUE` or `BOOLEAN`.

Variants:

F006127, E006127, W006127, I006127

6129:

Axis index does match existing axis indexes

Description:

RTA-RTE found a `SwCalprmAxis` within an `InstantiationDataDefProps` or `ParameterAccess` whose index did not match the any of the axes in the curve/map/axis data type.

How to Solve:

Ensure that all axes in the `InstantiationDataDefProps` or `ParameterAccess` have an index that matches of one of the axes in the `ApplicationDataType` for the curve/map.

Variants:

F006129, E006129, W006129, I006129

6130:

Missing or invalid axis index

Description:

RTA-RTE needed a `SwAxisIndex` because there were multiple `SwCalprmAxis` elements aggregated in the `SwCalprmAxisSet`, but failed to find a `SwAxisIndex` containing a unique non-negative integer on one or more of the aggregated `SwCalprmAxis` elements.

How to Solve:

Make sure each `SwCalprmAxis` within the `SwCalprmAxisSet` for the `InstantiationDataDefProps` or `ParameterAccess` contains a `SwAxisIndex` that matches of one of the `SwCalprmAxis` indexes in the `ApplicationDataType` for the curve/map.

Variants:

F006130, E006130, W006130, I006130

6131:

Duplicate axis index within an `InstantiationDataDefProps` or `ParameterAccess`

Description:

RTA-RTE found `SwCalprmAxis` elements within an `InstantiationDataDefProps` or `ParameterAccess` with the same `SwAxisIndex`. This is not necessarily an error, because RTA-RTE can use the `inputVariable` to distinguish which `SwCalprmAxis` is in use.

How to Solve:

This warning indicates a condition that is not invalid, but might not be what you intended. Usually, we expect `SwAxisIndex` elements within a `SwCalprmAxisSet` to be unique, non-negative integers, but in this case, RTA-RTE can use the input variable to distinguish which `SwCalprmAxis` is in use. The warning can be suppressed with the `--disable-warning` option.

Variants:

F006131, E006131, W006131, I006131

6132:

Missing or invalid axis index in data type

Description:

RTA-RTE needed a `SwAxisIndex` because there were multiple `SwCalprmAxis` elements aggregated in the `SwCalprmAxisSet`, but failed to find a `SwAxisIndex` containing a unique non-negative integer on one or more of the aggregated `SwCalprmAxis` elements.

How to Solve:

Make sure each `SwCalprmAxis` within the `SwCalprmAxisSet` contains a unique non-negative `SwAxisIndex`.

Variants:

F006132, E006132, W006132, I006132

6133:

Duplicate axis index in data type

Description:

RTA-RTE found `SwCalprmAxis` elements within an `ApplicationDataType`. `SwCalprmAxisSet` with the same `SwAxisIndex`.

How to Solve:

When there is more than one `SwCalprmAxis` in a `SwCalprmAxisSet`, each must have a unique non-negative integer `SwAxisIndex`.

Variants:

F006133, E006133, W006133, I006133

6134:

Unexpected number of `SwCalprmAxis` elements found

Description:

RTA-RTE found an `ApplicationDataType`. `swCalprmAxisSet` with an unexpected number of `SwCalprmAxis` elements considering the `category` of the `ApplicationDataType`. For `category`s `COM_AXIS` and `RES_AXIS` there must be exactly one `SwCalprmAxis` element since the data type describes an axis. For `category`s `CURVE` and `MAP` the number of `SwCalprmAxis` elements is not important for RTE generation and they will all be reported to the `McSupportData`. This warning message merely indicates that the configuration may be incorrect and may cause problems with downstream measurement and calibration tooling.

How to Solve:

Review the number of `SwCalprmAxis` elements within the `ApplicationDataType`. `SwCalprmAxisSet`. A `COM_AXIS` or `RES_AXIS` data type must have exactly one `SwCalprmAxis`. For a `CURVE` or `MAP` data type this warning may be suppressed if the number of `SwCalprmAxis` elements is intentional.

Variants:

F006134, E006134, W006134, I006134

6135:

Curve/map input axis references non-axis data type

Description:

RTA-RTE found an input axis of a curve or map that references an Application Data Type as the shared axis type but the referenced data type is not an axis

data type.

How to Solve:

The `SwCalprmAxis`. `SwAxisGrouped`. `sharedAxisTypeRef` reference must point to an `ApplicationPrimitiveDataType` of `category` `COM_AXIS` or `RES_AXIS`.

Variants:

F006135, E006135, W006135, I006135

6136:

`SwCalprmAxis` incompatible with corresponding axis of data type

Description:

RTA-RTE encountered a `SwCalprmAxis` that has a different `category` to the corresponding axis in the data type or an inconsistent use of `SwAxisIndividual` / `SwAxisGrouped`.

How to Solve:

`SwCalprmAxis` elements within a `ApplicationDataType`. `SwCalprmAxisSet` must have the same `category` (`SwAxisIndividual` / `SwAxisGrouped`).

Variants:

F006136, E006136, W006136, I006136

6137:

Incompatible `category` of data item

Description:

RTA-RTE was applying the *Special Treatment of Arrays of Curves and Maps* (See the *Reference Manual*) because an `InstantiationDataDefProps` specified a `SwCalPrmAxisSet` to be applied to a complex `ApplicationDataType`. This error indicates that the `SwCalprmAxisSet` references an item of an incompatible `category`.

How to Solve:

The `SwCalprmAxisSet` should be compatible with the member elements of the referenced complex data type, for example it should be a primitive or boolean application data type when used as an input variable for an array of curves/maps, or it should be an axis type if it is to be used as a shared axis to an array of axes.

Variants:

F006137, E006137, W006137, I006137

6138:

An incompatible type of element cannot be assigned to a curve/map/axis

Description:

RTA-RTE was applying the *Special Treatment of Arrays of Curves and Maps* (See the *Reference Manual*) because an `InstantiationDataDefProps` specified a `SwCalPrmAxisSet` to be applied to a complex `ApplicationDataType`. This

error indicates that a member of the type referenced in the `SwCalprmAxisSet` was not suitable for use with the corresponding element of the object referenced by the `InstantiationDataDefProps`.

How to Solve:

Each element of the array of input variables or axes should be compatible with the corresponding element of the array of curves/maps/axes; If an element is an array or record type then the corresponding element should be an array or record type. Otherwise an input variable or axes must be primitive or of axes type, respectively.

Variants:

F006138, E006138, W006138, I006138

6139:

`InstantiationDataDefProps` on complex subelement

Description:

RTA-RTE was applying the *Special Treatment of Arrays of Curves and Maps* (See the *Reference Manual*) because an `InstantiationDataDefProps` specified a `SwCalPrmAxisSet` to be applied to a complex `ApplicationDataType`. This error indicates that the `SwCalprmAxisSet` referenced part of a complex data item that was neither a leaf nor the root. This is not permitted.

How to Solve:

Point the `InstantiationDataDefProps` at the root of the array or record of input variables or axes ie point at the whole data item.

Variants:

F006139, E006139, W006139, I006139

6140:

The size or shape of two corresponding arrays or records are different

Description:

RTA-RTE was applying the *Special Treatment of Arrays of Curves and Maps* (See the *Reference Manual*) because an `InstantiationDataDefProps` specified a `SwCalPrmAxisSet` to be applied to a complex `ApplicationDataType`. This error indicates that the two complex data items did not have the same shape. Complex data items used in this way must have the same `category` and recursively the same number and `category`s of child elements.

How to Solve:

The array or record of input variables or axes should have the same size and shape as the corresponding array of curves/maps/axes.

Variants:

F006140, E006140, W006140, I006140

6141:

`RES_AXIS` types are currently not supported

Description:

An instance of data is typed by an `ApplicationDataType` of `Category RES_AXIS`. `RES_AXIS` is currently not supported by RTA-RTE.

How to Solve:

Avoid using `ApplicationDataTypes` of `category RES_AXIS` as they are not currently supported. Consider using `ApplicationDataTypes` of `category COM_AXIS` instead.

Variants:

F006141, E006141, W006141, I006141

6142:

Shared axis references shared axis

Description:

A data instance is typed by an `ApplicationDataType` of `Category COM_AXIS` which has a `SwDataDefProps.swCalprmAxisSet` that contains a `SwCalprmAxis.SwAxisGrouped` which is forbidden; a shared axis should not reference another shared axis.

How to Solve:

All `ApplicationDataTypes` of `Category COM_AXIS` should contain a `SwDataDefProps.swCalprmAxisSet` consisting of a single `SwCalprmAxis.SwAxisIndividual`.

Variants:

F006142, E006142, W006142, I006142

6143:

Conflicting `dataConstr` between curve/map and axis

Description:

RTA-RTE has encountered a data prototype that is typed by an `ApplicationDataType` of `category CURVE` or `MAP`. This `ApplicationDataType` contains a `SwDataDefProps.dataConstr` and a `SwDataDefProps.valueAxisDataType`. `SwDataDefProps. dataConstr` which do not allow the same range of values and are consequently incompatible.

How to Solve:

`ApplicationDataType. SwDataDefProps. dataConstr` should allow the same range of values as `ApplicationDataType. SwDataDefProps. valueAxisDataType. SwDataDefProps. dataConstr` to be considered compatible.

Variants:

F006143, E006143, W006143, I006143

6144:

Conflicting `Unit` between curve/map and axis

Description:

RTA-RTE has encountered a data prototype that is typed by an `ApplicationDataType` of `category CURVE` or `MAP`. The `ApplicationDataType` references a `SwDataDefProps.unit` that has either been derived from the `SwDataDefProps.unit` or a `SwDataDefProps.compuMethod.unit`. Additionally, The `ApplicationDataType` contains a `SwDataDefProps.valueAxisDataType` that contains a `SwDataDefProps.unit` that has also been derived from either the `SwDataDefProps.unit` or a `SwDataDefProps.compuMethod.unit`. These two resulting `SwDataDefProps.units` are incompatible.

How to Solve:

All `SwDataDefProps.units` should be compatible on either the `ApplicationDataType` or the `ApplicationDataType` of the `ValueAxisDataType`.

Variants:

F006144, E006144, W006144, I006144

6145:

No type for `SwCalprmAxis. SwAxisGrouped. sharedAxisType` found

Description:

RTA-RTE has encountered a shared axis that does not contain a type for its `SwCalprmAxis. SwAxisGrouped. sharedAxisType`.

How to Solve:

Ensure that the `SwCalprmAxis. SwAxisGrouped. sharedAxisTypeRef` reference points to an `ApplicationPrimitiveDataType` of `category COM_AXIS` or `RES_AXIS`.

Variants:

F006145, E006145, W006145, I006145

6146:

Missing `SwCalprmAxis. SwAxisIndividual. inputVariableType`

Description:

RTA-RTE has encountered an embedded axis that contains input variables but does not contain any `SwCalprmAxis.SwAxisIndividual. inputVariableType`.

How to Solve:

Ensure that `SwCalprmAxis.SwAxisIndividual. inputVariableType` references an `ApplicationDataType` of `category VALUE` or `BOOLEAN`.

Variants:

F006146, E006146, W006146, I006146

6147:

`Filter`s with IOC are currently not supported

Description:

The communication path between the identified `DataPrototype`s crosses an `OsApplication` boundary and had filtering configured. This feature is not supported by RTA-RTE. Note that suppressing the error might lead to code that compiles but behaves differently to code generated for a non-IOC channel.

How to Solve:

Remove the filter specification from the `ComSpec`s for the `VariableDataPrototype`s.

Variants:

F006147, E006147, W006147, I006147

6220:

No client port interface

Description:

RTA-RTE has been unable to locate an interface from the specified port.

How to Solve:

Ensure that each client-server port has a correctly referenced interface.

Variants:

F006220, E006220, W006220, I006220

6221:

Invalid client-server communication model

Description:

RTA-RTE has detected an invalid client-server communication model.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F006221, E006221, W006221, I006221

6222:

Invalid receive mode for client

Description:

RTA-RTE has detected an invalid receive mode (ie not activation of runnable entity, blocking nor polled) for a client.

How to Solve:

Please contact support for assistance. In your error report please include a

sample configuration and command line that demonstrates the error.

Variants:

F006222, E006222, W006222, I006222

6223:

No runnable wrapper

Description:

RTA-RTE has been unable to find a wrapper for a for runnable entity

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F006223, E006223, W006223, I006223

6230:

Unnamed operation parameter

Description:

RTA-RTE failed to find a `shortName` for the specified operation parameter.

How to Solve:

Ensure all operation parameters are named.

Variants:

F006230, E006230, W006230, I006230

6231:

No data type for operation parameter

Description:

RTA-RTE failed to find a data type for the specified operation parameter.

How to Solve:

Ensure that all operation parameters have a referenced data type. For application data types ensure that a data type mapping exists.

Variants:

F006231, E006231, W006231, I006231

6233:

Multiple servers

Description:

RTA-RTE has detected multiple servers (provided client-server ports) for the specified instance of the client (`RPort`).

How to Solve:

Ensure that each required port (client) of client-server communication has at most one connected server.

Variants:

F006233, E006233, W006233, I006233

6234:

Multiple `OperationInvokedEvent`s

Description:

RTA-RTE has detected multiple `OperationInvokedEvent`s for the same server (provided client-server port/operation). This forms invalid 1:n client-server communication and is therefore forbidden by AUTOSAR.

How to Solve:

Ensure that each server port/operation pair has at most one connected `OperationInvokedEvent`.

Variants:

F006234, E006234, W006234, I006234

6235:

No client parameter signal

Description:

RTA-RTE could not determine the signal mapping for the specified client-server operation parameter instance.

How to Solve:

Ensure that all parameters are mapped.

Variants:

F006235, E006235, W006235, I006235

6236:

No application error signal

Description:

RTA-RTE could not determine the signal mapping for the application error for the specified client-server operation parameter instance.

How to Solve:

Ensure that the application error is mapped.

Variants:

F006236, E006236, W006236, I006236

6237:

Unable to locate runnable

Description:

RTA-RTE has been unable to locate the runnable started by an Asynchronous Server Call Returns Event.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F006237, E006237, W006237, I006237

6238:

Unconnected client

Description:

RTA-RTE has detected an unconnected required client-server port (a client).

How to Solve:

Either enable support for unconnected require ports or ensure that the port instance is connected.

Variants:

F006238, E006238, W006238, I006238

6250:

Incomplete signal mapping for operation parameter

Description:

RTA-RTE has failed to detect sufficient signal mappings for the specified operation parameter.

How to Solve:

Ensure that each sub-element of the operation parameter is mapped.

Variants:

F006250, E006250, W006250, I006250

6251:

Too many signal mappings for operation parameter

Description:

RTA-RTE has detected too many signal mappings for the specified operation parameter.

How to Solve:

Ensure that each sub-element of the operation parameter is mapped.

Variants:

F006251, E006251, W006251, I006251

6254:

Missing signal group for client

Description:

RTA-RTE cannot determine the signal group to initiate transmission of the request from the specified client instance.

How to Solve:

Ensure that the signal group is correctly mapped.

Variants:

F006254, E006254, W006254, I006254

6256:

Missing signal group for server

Description:

RTA-RTE cannot determine the signal group to initiate transmission of the reply from the specified server instance.

How to Solve:

Ensure that the signal group is correctly mapped.

Variants:

F006256, E006256, W006256, I006256

6257:

No mapped operation

Description:

RTA-RTE cannot locate an operation in the server's interface that corresponds to the client's operation.

How to Solve:

Either supply an operation interface mapping to connect dissimilar interfaces or ensure that operations are matched by name.

Variants:

F006257, E006257, W006257, I006257

6258:

No empty signal

Description:

RTA-RTE is unable to find a response signal mapping and/or unable to find the `EmptySignal` for the reply in mapping of the specified operation instance.

How to Solve:

Ensure that the signals are correctly mapped.

Variants:

F006258, E006258, W006258, I006258

6400:

Invalid attribute

Description:

RTA-RTE has detected an invalid attribute for implicit communication.

How to Solve:

Ensure that implicit access does not use a blocking API and is not configured for queued communication.

Variants:

F006400, E006400, W006400, I006400

6411:

Unable to locate function definition

Description:

RTA-RTE has been unable to locate the specified function definition.

How to Solve:

Please contact ETAS GmbH for support. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F006411, E006411, W006411, I006411

6413:

Unable to determine bit-pack type

Description:

RTA-RTE has been unable to determine a type for the specified bit-pack type.

How to Solve:

Ensure a valid type is referenced by the `bit-pack-type` type

Variants:

F006413, E006413, W006413, I006413

6414:

Runnable having implicit data access has no task

Description:

A runnable entity not mapped to a task contains implicit communication. RTA-RTE needs to know the priority of implicit readers or writers to guarantee data protection.

How to Solve:

Map the runnable entity to a task.

Variants:

F006414, E006414, W006414, I006414

6415:

Multiple loopbacks with different receive buffers

Description:

There are multiple provider-requirer loopback assembly connectors from the given provide port instance to require ports on the same sofware component instance with the same sender-receiver interface and the deviation of 'write as modify' semantics for providers with such a loopback is enabled on the command-line. There are multiple global receive buffers from which to pre-fill the implicit shadow write buffer for the different loopback require ports with different settings. RTA-RTE has choosen one arbitrarily.

How to Solve:

Use a different but compatible interface for the require ports that should not be seen as loopback require ports for the purposes of the implicit 'write as modify' deviation.

Variants:

F006415, E006415, W006415, I006415

6502:

Unknown type for Inter-runnable Variable

Description:

RTA-RTE cannot determine the type for an inter-runnable variable.

How to Solve:

Ensure Inter-runnable Variables use only known primitive types.

Variants:

F006502, E006502, W006502, I006502

6503:

Invalid runnable category

Description:

RTA-RTE has detected an inter-runnable variable accessed by category 2 runnable entity using implicit access.

How to Solve:

Only Category 1A and 1B runnable entities can access an IRV implicitly.

Variants:

F006503, E006503, W006503, I006503

6611:

No Call API available for client-server argument measurement copy point

Description:

RTA-RTE could not find any Call API in which to insert a copy point for measurement of a given client-server argument. The operation is not used in the given instance.

How to Solve:

Ensure that a ServerCallPoint exists for the given operation, or do not request measurement of the operation argument if the operation is intentionally unused.

Variants:

F006611, E006611, W006611, I006611

6613:

Invalid ARIANE report `category`

Description:

RTA-RTE has detected an invalid `category` for the ARIANE report.

How to Solve:

Ensure that only `category`s listed in the error message are used in items included in the ARIANE report.

Variants:

F006613, E006613, W006613, I006613

6614:

Missing ARIANE report `category`

Description:

RTA-RTE could not detect the `category` of a `CompuMethod` used for the ARIANE report.

How to Solve:

Ensure that all `CompuMethod`s used in the ARIANE report include a valid `category`.

Variants:

F006614, E006614, W006614, I006614

6615:

Measured nested record type encountered

Description:

When processing the ARIANE report a measured item has a nested record (record within record) data types. This is invalid for the ARIANE report.

How to Solve:

Nested record types are not supported.

Variants:

F006615, E006615, W006615, I006615

6616:

Invalid `category` encountered when processing ARIANE report

Description:

When processing the ARIANE report a compu-method with an invalid `category` has been encountered.

How to Solve:

Use only the defined `category`s for measured items within input XML when using the ARIANE report

Variants:

F006616, E006616, W006616, I006616

6617:

Unknown (to AUTOSAR) `category` encountered

Description:

When processing the ARIANE report a compu-method with a `category` not defined by AUTOSAR has been encountered.

How to Solve:

Use only AUTOSAR defined `category`s within input XML when using the ARIANE report

Variants:

F006617, E006617, W006617, I006617

6621:

Measurement for queued data

Description:

RTA-RTE has detected an attempt to measure queued sender-receiver communication. This configuration is not valid.

How to Solve:

Ensure that only unqueued communication is measured.

Variants:

F006621, E006621, W006621, I006621

6623:

Missing flatmap (data item)

Description:

RTA-RTE has not detected a `FlatInstanceDescriptor` for the specified data item. No `McSupportData` was generated for this item.

How to Solve:

If `McSupportData` is required then provide a suitable `FlatInstanceDescriptor`. If not then the message can be suppressed.

Variants:

F006623, E006623, W006623, I006623

6624:

Missing flatmap (operation parameter)

Description:

RTA-RTE has not detected a `FlatInstanceDescriptor` for the specified operation parameter. No `McSupportData` was generated for this item.

How to Solve:

If `McSupportData` is required then provide a suitable `FlatInstanceDescriptor`. If not then the message can be suppressed.

Variants:

F006624, E006624, W006624, I006624

6625:

Missing flatmap (IRV)

Description:

RTA-RTE has not detected a `FlatInstanceDescriptor` for the specified inter-runnable variable. No `McSupportData` was generated for this item.

How to Solve:

If `McSupportData` is required then provide a suitable `FlatInstanceDescriptor`. If not then the message can be suppressed.

Variants:

F006625, E006625, W006625, I006625

6626:

Missing flatmap (per-instance memory)

Description:

RTA-RTE has not detected a `FlatInstanceDescriptor` for the specified per-instance memory. No `McSupportData` was generated for this item.

How to Solve:

If `McSupportData` is required then provide a suitable `FlatInstanceDescriptor`. If not then the message can be suppressed.

Variants:

F006626, E006626, W006626, I006626

6627:

Missing flatmap (`RamBlock`)

Description:

RTA-RTE has not detected a `FlatInstanceDescriptor` for the specified NVRAM `RamBlock`. No `McSupportData` was generated for this item.

How to Solve:

If `McSupportData` is required then provide a suitable `FlatInstanceDescriptor`. If not then the message can be suppressed.

Variants:

F006627, E006627, W006627, I006627

6700:

Initializer assignment removed

Description:

RTA-RTE has optimized the first assignment of `RTE_E_OK` in a function.

How to Solve:

This message is for information only. If not required it can be suppressed.

Variants:

F006700, E006700, W006700, I006700

6701:

Latch removed

Description:

RTA-RTE has optimized the first latch in a function to an assignment.

How to Solve:

This message is for information only. If not required it can be suppressed.

Variants:

F006701, E006701, W006701, I006701

6702:

Optimized assignment data consistency

Description:

RTA-RTE has optimized the data consistency mechanism for an assignment operation as indicated in the error message.

How to Solve:

This message is for information only. If not required it can be suppressed.

Variants:

F006702, E006702, W006702, I006702

6703:

Optimized memcpy data consistency

Description:

RTA-RTE has optimized the data consistency mechanism for a memcpy operation as indicated in the error message.

How to Solve:

This message is for information only. If not required it can be suppressed.

Variants:

F006703, E006703, W006703, I006703

6704:

Node replacement failed

Description:

RTA-RTE failed to replace implementation description node

How to Solve:

Please contact ETAS GmbH for support. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F006704, E006704, W006704, I006704

6705:

Initial values reset

Description:

RTA-RTE has reset the initial values for the specified element.

How to Solve:

This message is for information only. If not required it can be suppressed.

Variants:

F006705, E006705, W006705, I006705

6708:

Invalid data consistency container

Description:

RTA-RTE cannot determine the accessing tasks and thus cannot determine the required data consistency mechanism for the specified assignment. Data consistency is not required if no tasks can access the variables in an assignment yet if this were the case no assignment would be necessary – therefore

an error has occurred.

How to Solve:

Check that the mapping from runnable entities to tasks is complete for the specified SWC instance. In particular check that all RTE events that can start a runnable are mapped – if direct function invocation is required then omit the `taskRef` from the event mapping rather than omit the mapping entirely

Variants:

F006708, E006708, W006708, I006708

6800:

User-configured resource ignored

Description:

The specified `OsResource` has a declared `ResourceProperty` (internal, standard) that conflicts with other aspects of the configuration. RTA-RTE will therefore create an alternative `OsResource` and ignore the existing declaration.

How to Solve:

Ensure the input configuration is consistent.

Variants:

F006800, E006800, W006800, I006800

6801:

Schedule table contains over 10,000 expiry points

Description:

RTA-RTE has calculated that a schedule table containing over 10,000 expiry points is required for this configuration. This may indicate an unintended explosion of expiry points due to the use of non-harmonic periods for periodic tasks or events. A schedule table containing a large number of expiry points may cause OS errors or result in high RAM usage.

How to Solve:

Use harmonic periods in the system schedule to reduce the number of generated expiry points. Alternatively, verify that the generated number of expiry points is not an issue with your operating system and disable the warning using the –disable-warning option.

Variants:

F006801, E006801, W006801, I006801

6802:

Integer overflow

Description:

RTA-RTE has detected an integer overflow when calculating least common multiple of two integers.

How to Solve:

This error occurs when non-harmonic runnable periods cause excessively long schedule tables. Ensure that runnable entities have harmonic periods (eg 1, 5, 10) or disable the RTE generators creation of a schedule table using the `--task-recurrence`.

Variants:

F006802, E006802, W006802, I006802

6900:

Runnable with wait-point in exclusive area

Description:

RTA-RTE has detected that the specified runnable instance is configured to run inside an Exclusive Area but contains a (possibly implicit) `WaitPoint`. This is an invalid configuration.

How to Solve:

Ensure a runnable that runs 'inside' an exclusive area cannot wait. This is required since the OS does not support suspension of a task with resources locked.

Variants:

F006900, E006900, W006900, I006900

6910:

Counter tick rate

Description:

Indicates the required rate at which the specified counter must be ticked to ensure correct scheduling of periodic activities and/or timeouts.

How to Solve:

None. This message is for information only.

Variants:

F006910, E006910, W006910, I006910

6927:

Zero tick stimulus

Description:

RTA-RTE has configured a periodic stimulus with zero tick period.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F006927, E006927, W006927, I006927

6930:

Activation limit mismatch

Description:

The configuration for the specified task defines a specific activation limit however RTA-RTE has calculated that it requires a different limit. The calculated figure has not been output.

How to Solve:

If not required this message can be suppressed. Alternatively select appropriate activation limit within the input configuration.

Variants:

F006930, E006930, W006930, I006930

6993:

Schedule table overflow

Description:

RTA-RTE has detected an integer overflow when determining the period of a schedule table.

How to Solve:

This error occurs when non-harmonic runnable periods cause excessively long schedule tables. Ensure that periodic runnable entities have harmonic periods (eg 1, 5, 10) or disable the RTE generators creation of a schedule table using the `--task-recurrence`.

Variants:

F006993, E006993, W006993, I006993

7001:

Invalid array element type

Description:

RTA-RTE has encountered an array element with an invalid type.

How to Solve:

Internal error; please contact ETAS GmbH for support. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007001, E007001, W007001, I007001

7002:

Invalid array element type

Description:

RTA-RTE has encountered an API object with an invalid type.

How to Solve:

Internal error; please contact ETAS GmbH for support. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007002, E007002, W007002, I007002

7009:

Forward declaration of non-struct

Description:

While processing the internal types database RTA-RTE has encountered a forward declaration of a non-struct type. This is invalid with respect to internal constraints.

How to Solve:

Internal error; please contact ETAS GmbH for support. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007009, E007009, W007009, I007009

7010:

Invalid type identifier

Description:

RTA-RTE has encountered an entry on the internal database of types with an invalid identifier.

How to Solve:

Internal error; please contact ETAS GmbH for support. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007010, E007010, W007010, I007010

7011:

No `NativeDeclaration` available when `typeEmitter` is RTE

Description:

An implementation data type has been configured with `TypeEmitter` set to 'RTE' but no `NativeDeclaration` is available in the base type. This configuration is invalid.

How to Solve:

Ensure that all implementation data types with type emitter set to "RTE" have a base type that specifies a `NativeDeclaration` or specify that the type is

emitted by another AUTOSAR module by specifying `typeEmitter` as "`OTHER`".

Variants:

F007011, E007011, W007011, I007011

7012:

Partially initialized array

Description:

RTA-RTE has encountered an array entry on the internal database of types with an incomplete initialization.

How to Solve:

Internal error; please contact ETAS GmbH for support. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007012, E007012, W007012, I007012

7013:

Invalid global type

Description:

RTA-RTE has encountered an entry on the internal database of global data with an invalid type.

How to Solve:

Internal error; please contact ETAS GmbH for support. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007013, E007013, W007013, I007013

7014:

Global has invalid type information

Description:

RTA-RTE has encountered an entry on the internal database of global data with invalid type information.

How to Solve:

Internal error; please contact ETAS GmbH for support. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007014, E007014, W007014, I007014

7020:

Multiply defined per-instance memories

Description:

RTA-RTE has detected multiple per-instance memories with the same name. Second and subsequent definitions will be ignored.

How to Solve:

Ensure that within an SWC all per-instance memory names are unique.

Variants:

F007020, E007020, W007020, I007020

7023:

Incompatible per-instance memory types

Description:

RTA-RTE has detected that the type of the specified per-instance memory clashes with an existing type.

How to Solve:

Ensure that all per-instance memories with the same Type have the same `TypeDefinition`.

Variants:

F007023, E007023, W007023, I007023

7024:

Unconnected R-port in DSMQ

Description:

RTA-RTE has encountered an unconnected R-port in a Distributed Shared Mode Queue (DSMQ).

How to Solve:

Either: (a) Connect the R-Port to a P-Port; or (b) Map the `ModeSwitchEvent` to a task that is not participating in the DSMQ.

Variants:

F007024, E007024, W007024, I007024

7061:

Inconsistent type definition

Description:

Type definitions in different packages are inconsistent.

How to Solve:

Where the same type is declared in multiple packages ensure that the type definitions result on the same definition.

Variants:

F007061, E007061, W007061, I007061

7062:

Duplicate enumerated value detected within context of SWC

Description:

The RTE generator has detected conflicting values for an enumeration. The error message indicates the enumeration label and the conflicting values.

How to Solve:

Ensure that all enumerations used by a SWC either contain unique labels or where duplicate labels are unavoidable that the values are also the same.

Variants:

F007062, E007062, W007062, I007062

7063:

Invalid runnable entity name detected within context of SWC

Description:

The RTE generator has detected an invalid runnable entity name (either empty or containing a forbidden character).

How to Solve:

Ensure that all runnable entities used by a SWC have non-empty, valid names.

Variants:

F007063, E007063, W007063, I007063

7101:

Unable to determine type

Description:

RTA-RTE is unable to determine an implementation data type for a per-instance memory typed by an application data type.

How to Solve:

Ensure the SWC includes a data type mapping for the application data type

Variants:

F007101, E007101, W007101, I007101

7200:

No runnable for RTE event

Description:

No runnable found corresponding to specified RTE Event.

How to Solve:

For activation of runnable entity, insert a `startOnEventRef`; for wake-up if wait-point, check that the `WaitPoint` and `DataReceivedEvent` refer to the correct data item.

Variants:

F007200, E007200, W007200, I007200

7201:

Unable to locate runnable container

Description:

RTA-RTE failed to locate a runnable entity container for the specified event.

How to Solve:

Ensure all wait-points have a corresponding read point

Variants:

F007201, E007201, W007201, I007201

7202:

No instance wrapper

Description:

RTA-RTE has been unable to find an instance wrapper for a for runnable entity

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007202, E007202, W007202, I007202

7210:

Unconnected receiver warning

Description:

RTA-RTE has detected an unconnected receiver.

How to Solve:

This message occurs when RTE generator's check for unconnected `RPorts` has been disabled. The message can be suppressed if not required.

Variants:

F007210, E007210, W007210, I007210

7216:

Blocking API for unqueued data

Description:

RTA-RTE has detected that the specified data item instance is configured without a queue, yet is also configured with a `WaitPoint`. This configuration is invalid.

How to Solve:

Ensure that only queued data if configured with a blocking receive.

Variants:

F007216, E007216, W007216, I007216

7218:

No queue configured

Description:

RTA-RTE has detected that the specified data item instance declared as queued but no queue is configured.

How to Solve:

Ensure that a queue is configured for all queued data items.

Variants:

F007218, E007218, W007218, I007218

7219:

Queue for unqueued data item

Description:

RTA-RTE has encountered a queue configuration for the specified data item yet it is declared as unqueued within the interface.

How to Solve:

Ensure that the com-spec element and the interface declaration are consistent.

Variants:

F007219, E007219, W007219, I007219

7220:

Unconnected receiver error

Description:

RTA-RTE has detected an unconnected receiver but such ports are not permitted.

How to Solve:

Either connect the specified port instance or disable the RTE generator's check for unconnected `RPorts`.

Variants:

F007220, E007220, W007220, I007220

7221:

Queued implicit read access

Description:

RTA-RTE has detected an attempt to configure implicit read access (`DataReadAccess`) to a queued data item. This configuration is invalid.

How to Solve:

Ensure only unqueued data is read implicitly.

Variants:

F007221, E007221, W007221, I007221

7222:

Read access to measurement point

Description:

RTA-RTE has detected an attempt to configure read access (`DataReadAccess`, `DataReceivePointByValue` or `DataReceivePointByArgument`) for a data item defined as a measurement point. This configuration is invalid.

How to Solve:

Ensure only non-measurement points have read access.

Variants:

F007222, E007222, W007222, I007222

7223:

Implicit port data access in a runnable that can never execute

Description:

RTA-RTE has detected an implicit read or write access to a data element in a `RunnableEntity` that has no activating `RTEEvent`s, or that is activated only by `RTEEvent`s that are not mapped to a task for invocation by direct function call but with no connected callers that can execute. Therefore there is no task whose shadow buffer the implicit APIs could access and so they cannot be implemented.

How to Solve:

Ensure that every `RunnableEntity` with implicit read or write accesses is directly or indirectly triggered by some `RTEEvent` that is mapped to a task.

Variants:

F007223, E007223, W007223, I007223

7224:

Implicit port data access in a directly invoked runnable

Description:

RTA-RTE has detected implicit read or write access to one or more data elements in a `RunnableEntity` that is invoked by direct function call within

another RTE API for which implicit shadow buffer fill and flush generation is not currently supported. At the time of writing, RTA-RTE only supports implicit accesses in runnables whose activating `RTEEvent`s are mapped to tasks or that are activated by unmapped `OperationInvokedEvent`s with a calling `RunnableEntity` that is mapped to a task.

How to Solve:

Ensure that every `RunnableEntity` with implicit read or write accesses is triggered only by `RTEEvent`s that are mapped to tasks or unmapped `OperationInvokedEvent`s with callers that are triggered by `RTEEvent`s that are mapped to tasks.

Variants:

F007224, E007224, W007224, I007224

7225:

Implicit port data access in a runnable that can execute in interrupt context

Description:

RTA-RTE has detected an implicit read or write access to a data element in a `RunnableEntity` that is synchronized with a `BswInterruptEntity`, or can be invoked by direct function call from (or ultimately from) a `BswInterruptEntity` or a `RunnableEntity` that is synchronized with a `BswInterruptEntity`.

How to Solve:

Ensure that every `RunnableEntity` with implicit read or write accesses is directly or indirectly triggered by some `RTEEvent` that is mapped to a task and cannot execute in interrupt context.

Variants:

F007225, E007225, W007225, I007225

7230:

Complex data read by value

Description:

The RTE generator has detected an invalid configuration containing a data element with a complex type (ie a record or an array) that is read by value.

How to Solve:

Ensure that the only primitive data types are read by value

Variants:

F007230, E007230, W007230, I007230

7231:

Queued data read by value

Description:

The RTE generator has detected an invalid configuration containing a queued data element that is read by value.

How to Solve:

Ensure that the data element is unqueued (ie has data semantics)

Variants:

F007231, E007231, W007231, I007231

7240:

Unconnected BSW receiver

Description:

RTA-RTE has detected that a required mode declaration group in the specified BSW instance is not connected to a provided mode declaration group either within the BSW or via an `SwcBswMapping`.

How to Solve:

Ensure that the required mode declaration group is connected (eg using a `RteBswRequiredModeGroupConnection` in Rte Module Configuration) or disable the RTE generator's check for unconnected `RPorts`.

Variants:

F007240, E007240, W007240, I007240

7250:

Wake-up-of-wait point detected for `NvData`

Description:

The RTE generator as detected a blocking API (wake-up-of-wait-point) configuration for the specified data item. This configuration is invalid.

How to Solve:

Remove the wait-point for the specified data item.

Variants:

F007250, E007250, W007250, I007250

7251:

Invalid port type referenced from Role based port assignment

Description:

The RTE generator has detected an invalid port reference within a Role based port assignment.

How to Solve:

Ensure that all role based port assignment elements reference client-server ports.

Variants:

F007251, E007251, W007251, I007251

7400:

No error check attribute

Description:

RTA-RTE was unable to locate a required attribute on an internal SpecReq node.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007400, E007400, W007400, I007400

7401:

Invalid node type

Description:

RTA-RTE encountered and invalid or unexpected node type.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007401, E007401, W007401, I007401

7410:

Unnamed task

Description:

RTA-RTE encountered an unnamed task.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007410, E007410, W007410, I007410

7412:

Same signal TX/RX

Description:

RTA-RTE attempted to both transmit and receive the same signal.

How to Solve:

Ensure that the data/operation mappings are correct.

Variants:

F007412, E007412, W007412, I007412

7413:

No source signal

Description:

RTA-RTE failed to find a source variable for a transmission node.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007413, E007413, W007413, I007413

7414:

No COM data type

Description:

RTA-RTE failed to find a data type for a transmission node.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007414, E007414, W007414, I007414

7415:

Missing pass type attribute

Description:

RTA-RTE failed to find a pass type attribute for a transmission node.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007415, E007415, W007415, I007415

7416:

Invalid pass type attribute

Description:

RTA-RTE failed to find a valid pass type attribute for a transmission node.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007416, E007416, W007416, I007416

7417:

No signal for COM callback

Description:

RTA-RTE failed to find a network signal specified on a ComCallback SpecReq node.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007417, E007417, W007417, I007417

7418:

No signal name for COM callback

Description:

RTA-RTE failed to find a signal name on ComCallback SpecReq node.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007418, E007418, W007418, I007418

7419:

No trigger attribute

Description:

RTA-RTE failed to find a trigger attribute set on SendSignal SpecReq node.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007419, E007419, W007419, I007419

7421:

Anonymous signal

Description:

RTA-RTE failed to locate a name for a network signal.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007421, E007421, W007421, I007421

7424:

No receive signal

Description:

RTA-RTE failed to determine the signal for a Receive SpecReq node.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007424, E007424, W007424, I007424

7425:

No signal

Description:

RTA-RTE failed to find a signal for a SpecReq node.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007425, E007425, W007425, I007425

7428:

No data destination

Description:

RTA-RTE could not find the data destination variable for a Receive SpecReq node

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007428, E007428, W007428, I007428

7429:

No data destination type

Description:

RTA-RTE failed to find the data type for a Receive SpecReq node.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007429, E007429, W007429, I007429

7434:

Mismatched data type

Description:

RTA-RTE has detected a mismatch in mapped AUTOSAR data type.

How to Solve:

Check that signal mappings for different AUTOSAR signals with different data types do not reference the same system signal.

Variants:

F007434, E007434, W007434, I007434

7438:

Signal group mismatch

Description:

RTA-RTE has detected that a signal is used in multiple signal groups.

How to Solve:

Ensure that data mappings are correct. In particular, check if the same `FrameType` is used for multiple `FrameInstances`.

Variants:

F007438, E007438, W007438, I007438

7440:

Signal initial value mismatch

Description:

RTA-RTE has detected conflicting initial values for the specified signal.

How to Solve:

Check if multiple data items (with different initial values specified) are mapped to a single signal.

Variants:

F007440, E007440, W007440, I007440

7442:

Unknown signal

Description:

RTA-RTE has detected that the signal is invalidated but could not found it in the signal database.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007442, E007442, W007442, I007442

7445:

Unexpected COM version

Description:

RTA-RTE has detected an unexpected COM version.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007445, E007445, W007445, I007445

7511:

IOC used from and to the same OS application

Description:

RTA-RTE needs to configure an IOC channel with both endpoints in the same `OsApplication`. This typically happens if you have N:1 communication where some senders are in the same `OsApplication` as the receiver and some senders are in a different `OsApplication` to the receiver. If you use RTA-OS this is okay and you can supress the warning, however, it does deviate from the AUTOSAR specification.

How to Solve:

Either supress this warning (if your OS will accept same-`OsApplication` IOC channels) or avoid combining cross-`OsApplication` and intra-`OsApplication` communication on a single N:1 `RPort`.

Variants:

F007511, E007511, W007511, I007511

7604:

Cannot identify task

Description:

RTA-RTE has been unable to identify the task associated with an RTE event instance.

How to Solve:

Check that the `RTEEventMapping` is correct and that the task name is not ambiguous.

Variants:

F007604, E007604, W007604, I007604

7606:

Task zero period

Description:

RTA-RTE has calculated a period of zero ticks for the specified task.

How to Solve:

Check that all `TimingEvent`s have an positive, non-zero, period.

Variants:

F007606, E007606, W007606, I007606

7607:

Invalid SpecReq node

Description:

RTA-RTE has detected an invalid SpecReq node in the context of scheduling.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007607, E007607, W007607, I007607

7608:

Invalid event mask

Description:

RTA-RTE has detected no OS events on which to wait when combining periodic/non-periodic event masks for the specified task. This indicates that no runnable mapped to the task can be triggered. Check the configuration of all RTE Events mapped to the task (eg connections, presence of call points for requirers of provided servers mapped to the task, etc.). The task will contain no generated code and no runnable invocations.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007608, E007608, W007608, I007608

7609:

No MDD disable node

Description:

RTA-RTE has failed to locate the expected mode disabling dependency node for the specified mode instance.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007609, E007609, W007609, I007609

7610:

Unconnected mode user

Description:

RTA-RTE has detected an unconnected required mode port.

How to Solve:

Either enable support for unconnected require ports or ensure that the port instance is connected.

Variants:

F007610, E007610, W007610, I007610

7611:

Unexpected RTE event

Description:

RTA-RTE encountered an unexpected RTE Event when processing a periodic task.

How to Solve:

Check that all mapped RTE Events are correctly configured (eg a mapped `DataSendCompletedEvent` must also have transmission acknowledge enabled)

Variants:

F007611, E007611, W007611, I007611

7612:

Could not resolve instance reference

Description:

Could not resolve the Instance Reference at the specified location.

How to Solve:

Check other error messages for more information.

Variants:

F007612, E007612, W007612, I007612

7620:

No activation condition

Description:

RTA-RTE could not determine an activation condition (eg time triggered, event triggered, etc.) for the identified runnable entity.

How to Solve:

Ensure that RTE event mappings are complete.

Variants:

F007620, E007620, W007620, I007620

7621:

`ModeSwitchEvent` for Cat.2 runnable

Description:

RTA-RTE has detected a `ModeSwitchEvent` that starts a category 2 runnable. This is an invalid configuration.

How to Solve:

Ensure that mode switch events only start category 1 runnables.

Variants:

F007621, E007621, W007621, I007621

7624:

Mode disabling dependency for unconnected port

Description:

RTA-RTE has detected a mode disabling dependency for the `RTEEvent` instance that specifies a mode that cannot be entered (the port instance is unconnected and therefore the mode remains in the initial mode). The disabling dependency has therefore been ignored.

How to Solve:

This message is for information only. If not required the message can be disabled.

Variants:

F007624, E007624, W007624, I007624

7625:

Mode disabling dependency for Cat.2 runnable

Description:

RTA-RTE has detected an `RTEEvent` that specifies a `ModeDisablingDependency` but cannot do so since it starts a category 2 runnable entity.

How to Solve:

RTA-RTE can optionally allow mode disabling dependencies for Cat.2

runnables using the `--deviate-implicit-cat2-mdd` option.

Variants:

F007625, E007625, W007625, I007625

7640:

Unmapped RTE event

Description:

RTA-RTE has detected an RTE Event instance that is not mapped to a task.

How to Solve:

Ensure that an RTE Event mapping exists within the input.

Variants:

F007640, E007640, W007640, I007640

7641:

No initial mode

Description:

RTA-RTE is unable to resolve the reference to the initial mode for the specified mode group.

How to Solve:

Ensure that the mode group specifies an initial mode.

Variants:

F007641, E007641, W007641, I007641

7643:

Mode disabling task priority

Description:

RTA-RTE has detected that the priority of a task containing mode disabling dependencies cannot be lower than the task containing the entry/exit runnables for the relevant mode instance. The task contains the specified RTE Event with a mode disabling dependency but the specified Task containing the entry/exit runnables has a greater priority.

How to Solve:

Ensure that task priorities are such that the mode disabling task has a higher priority than the entry/exit task.

Variants:

F007643, E007643, W007643, I007643

7644:

ECC tasks required

Description:

RTA-RTE requires OS events for the specified tasks scheduling however generation of extended (category 2) tasks has been disabled.

How to Solve:

Consult the user documentation on how to map runnables to avoid the need for extended tasks.

Variants:

F007644, E007644, W007644, I007644

7646:

Mixed fastinit activation

Description:

RTA-RTE has detected that runnable entities with `FastInit` activation mapped to the same task as runnables with AUTOSAR-compliant activation. This is a non-supported configuration.

How to Solve:

Ensure that task activation methods are not mixed within a task.

Variants:

F007646, E007646, W007646, I007646

7647:

Fastinit activation for invalid RTE event

Description:

RTA-RTE has detected `FastInit` activation applied to a non-`ModeSwitchEvent`.

How to Solve:

Ensure that fast init activation is only applied to `ModeSwitch` events.

Variants:

F007647, E007647, W007647, I007647

7648:

Force basic for task with wait-point

Description:

The specified task is configured as "force basic" but contains a runnable with a wait-point. This is an invalid configuration since force basic only affects the RTE generated scheduling code.

How to Solve:

Ensure that force basic tasks do not contain runnables with a wait-point.

Variants:

F007648, E007648, W007648, I007648

7649:

Force basic task with no timing event

Description:

RTA-RTE has detected that the specified task (which uses forced-basic semantics) does not have any Timing Events mapped. This is invalid because it means the task will never be activated.

How to Solve:

Either map at least one Timing Event to the task, or remove the forced-basic restriction for this task.

Variants:

F007649, E007649, W007649, I007649

7650:

Force basic in use

Description:

The specified task uses forced-basic semantics which can introduce some lag in sporadic `RTEEvents`. If your Operating System supports Os Events, consider removing this restriction.

How to Solve:

None. If not required the message can be suppressed.

Variants:

F007650, E007650, W007650, I007650

7651:

Force basic initialization

Description:

RTA-RTE has detected that the specified task (which uses forced-basic semantics) has initialization and periodic runnable(s) mapped to it. Activating the task for the initialization runnable(s) would cause periodic runnable(s) to activate too soon.

How to Solve:

Ensure that periodic and initialization runnables are not mapped to the same force basic task.

Variants:

F007651, E007651, W007651, I007651

7652:

Invalid background mapping

Description:

The task mapping of the specified background event is invalid. No `OsAlarm` or `OsScheduleTable` is configured for the event mapping and therefore it must

be mapped to the lowest priority task.

How to Solve:

Ensure the event is mapped to the lowest priority task or configure an activating OS object.

Variants:

F007652, E007652, W007652, I007652

7653:

Mask and no mask

Description:

Invalid RTE Event mapping detected for the specified task. `OsEvent`s are required to schedule task yet one or more mappings for the task have an explicit `OsAlarm` or `OsScheduleTable` specification without also specifying a `UsedOsEvent`.

How to Solve:

Either ensure all mappings specify an `OsEvent` or ensure that all mappings are triggered by the same `OsAlarm` or `OsScheduleTable` in which case no `OsEvent` is required.

Variants:

F007653, E007653, W007653, I007653

7654:

`minStartInterval` for background event

Description:

RTA-RTE has detected a background event that starts a runnable with a Minimum start interval. This is an invalid configuration.

How to Solve:

Remove the minimum start interval specification from the runnable.

Variants:

F007654, E007654, W007654, I007654

7655:

`ModeSwitch` event with `minStartInterval`

Description:

RTA-RTE has detected that the specified `ModeSwitchEvent` starts a runnable with a `minStartInterval` but is mapped to an extended task. This configuration is not supported.

How to Solve:

Modify the RTE event mappings to ensure that an extended task is not required. See the RTA-RTE User Guide for more details on the required task mappings.

Variants:

F007655, E007655, W007655, I007655

7656:

Force basic task with external triggering

Description:

RTA-RTE has detected that the specified task is forced to be a basic task by the configuration but contains a mapped RTE event that is triggered by an external `OsAlarm` or `OsScheduleTable`. This configuration is not supported.

How to Solve:

Either remove the force basic configuration or remove the external trigger.

Variants:

F007656, E007656, W007656, I007656

7657:

Invalid task period

Description:

RTA-RTE has detected an invalid user-specified recurrence rate for a task.

How to Solve:

The user specified rate must be an integral divider of the RTA-RTE calculated rate. For example, if RTA-RTE calculates 1.0s then both 0.5s and 0.25s are valid user rates.

Variants:

F007657, E007657, W007657, I007657

7658:

Missing task period

Description:

No user specified recurrence rate has been specified for a task. When a recurrence is specified for one task then a recurrence must also be specified for all tasks containing periodic events.

How to Solve:

Ensure a recurrence rate is specified for either all tasks or for none.

Variants:

F007658, E007658, W007658, I007658

7659:

Invalid event recurrence

Description:

An RTE Event has been mapped to a task with an explicit recurrence rate (command-line option –task-recurrence). The specified recurrence rate for the task is not suitable for the mapped RTE Event.

How to Solve:

Either change the specified recurrence rate to the rate specified in the error message (may require changes to the configuration external to the RTE) or map the event to a task having a suitable recurrence rate.

Variants:

F007659, E007659, W007659, I007659

7660:

Missing event recurrence

Description:

No user specified recurrence rate has been specified for an event. When a recurrence is specified for one event then a recurrence must also be specified for all periodic events.

How to Solve:

Ensure a recurrence rate is specified for either all periodic events or for none.

Variants:

F007660, E007660, W007660, I007660

7680:

Empty task

Description:

All runnables in the named task have been optimized away; the task will be generated with an empty body and will never be activated.

How to Solve:

This message is for information only. Consider not mapping the runnables to a task (or mapping them to a different task) to optimize away the task completely.

Variants:

F007680, E007680, W007680, I007680

7710:

No container

Description:

RTA-RTE has been unable to find container for a particular RTE event

How to Solve:

Please contact support for assistance. In your error report please include a ample configuration and command line that demonstrates the error.

Variants:

F007710, E007710, W007710, I007710

7711:

No instance wrapper

Description:

RTA-RTE has been unable to find an instance wrapper for a runnable entity

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007711, E007711, W007711, I007711

7722:

No feedback timeout

Description:

RTA-RTE could not locate a timeout specification for mode switch or transmission feedback.

How to Solve:

Ensure that XML schema validation is enabled and that the input is correct.

Variants:

F007722, E007722, W007722, I007722

7730:

Queuing mismatch

Description:

RTA-RTE has detected conflicting queue semantics within the specified data items.

How to Solve:

Ensure that matching data items have the same event (queued/unqueued) semantics.

Variants:

F007730, E007730, W007730, I007730

7731:

NULL transmit API generated

Description:

RTA-RTE has forced NULL transmission API generation for the specified port. A null API is required to support the indirect API. The generated API functions

take no action but enable the relevant entries in the generated component data structure to be filled

How to Solve:

No action required. The information message can be disabled if required.

Variants:

F007731, E007731, W007731, I007731

7732:

NULL transmission acknowledge API generated

Description:

RTA-RTE has forced NULL transmission acknowledge API generation for the specified port. A null API is required to support the indirect API. The generated API functions take no action but enable the relevant entries in the generated component data structure to be filled

How to Solve:

No action required. The information message can be disabled if required.

Variants:

F007732, E007732, W007732, I007732

7733:

Unconnected measurable sender

Description:

RTA-RTE has detected that the specified provided port instance is not connected to a requiring port but all provided data elements are measurable. Measurement points and null RTE API calls will be generated.

How to Solve:

No action required. The information message can be disabled if required.

Variants:

F007733, E007733, W007733, I007733

7740:

Invalid transmission acknowledgment

Description:

RTA-RTE has detected an attempt to enable transmission acknowledgment for 1:n communication. This configuration is invalid.

How to Solve:

Ensure that transmission acknowledgment is only enabled for 1:1 communication.

Variants:

F007740, E007740, W007740, I007740

7741:

No transmission acknowledge signal

Description:

RTA-RTE has detected that transmission acknowledgment was enabled for the specified data instance but could not locate a valid mapping into signals.

How to Solve:

Ensure that a valid mapping is provided or remove the configuration of transmit acknowledge.

Variants:

F007741, E007741, W007741, I007741

7742:

Multiple transmission acknowledge signals

Description:

RTA-RTE has detected that transmission acknowledgment was enabled for the specified data instance but multiple mappings into signals, or multiple signal instances, were found for this data item. A unique mapping is required.

How to Solve:

Ensure that exactly one valid mapping is provided.

Variants:

F007742, E007742, W007742, I007742

7750:

Unknown type

Description:

RTA-RTE was unable to location information about the specified type.

How to Solve:

Please contact support for assistance. In your error report please include a ample configuration and command line that demonstrates the error.

Variants:

F007750, E007750, W007750, I007750

7760:

Invalid or missing mode queue length in port (ASW)

Description:

RTA-RTE was unable to derive a valid queue length for a `ModeMachineInstance`

How to Solve:

The queue length for a mode machine instance in ASW must be specified on the Provide Port that initiates mode switches in that mode machine instance. The queue length must be 1 or greater.

Variants:

F007760, E007760, W007760, I007760

7761:

Elided mode queue (instantaneous mode switch)

Description:

RTA-RTE has detected that no task-mapped `ModeSwitchEvent`s are configured for the specified mode instance. Therefore no queue of mode requests is required and hence queue support has been omitted for the mode instance.

How to Solve:

No action required. The information message can be disabled if required.

Variants:

F007761, E007761, W007761, I007761

7762:

Elided mode queue (max one pending mode switch)

Description:

When only one mode switch can be pending or in progress, there is no need for generated code to implement a queue. This message indicates that this optimization has occurred. This optimization occurs `queueLength` is 1.

How to Solve:

This message is informational only. No action required. The message can be disabled if desired.

Variants:

F007762, E007762, W007762, I007762

7770:

No mode switch task

Description:

RTA-RTE could not locate a `taskRef` for the specified RTE event instance.

How to Solve:

Ensure that all `ModeSwitchEvent`s that start a runnable entity with a `MinimumStartInterval` have a mapping to an `OsTask`.

Variants:

F007770, E007770, W007770, I007770

7771:

Invalid or missing mode queue length in provided mode group (BSW)

Description:

RTA-RTE was unable to derive a valid queue length for a `ModeMachineInstance`

How to Solve:

The queue length for a mode machine instance in BSW must be specified on the provide mode group that initiates mode switches in that mode machine instance. The the queue length must be 1 or greater.

Variants:

F007771, E007771, W007771, I007771

7800:

Optimized server

Description:

RTA-RTE has optimized a client-server call to direct function call.

How to Solve:

This message is for information only. It can be suppressed if required.

Variants:

F007800, E007800, W007800, I007800

7801:

Unconnected server

Description:

RTA-RTE has detected that a provided client-server port is not connected to a requiring port or there are no `OperationInvokedEvent`s that reference the port. The servers for operations in the categorizing interface will never be activated.

How to Solve:

This message is for information only. It can be suppressed if required.

Variants:

F007801, E007801, W007801, I007801

7810:

No server port interface

Description:

RTA-RTE has been unable to locate an interface from the specified port.

How to Solve:

Ensure that each client-server port has a correctly referenced interface.

Variants:

F007810, E007810, W007810, I007810

7811:

No container

Description:

RTA-RTE has been unable to find container for a particular RTE event

How to Solve:

Please contact support for assistance. In your error report please include a ample configuration and command line that demonstrates the error.

Variants:

F007811, E007811, W007811, I007811

7812:

No instance wrapper

Description:

RTA-RTE has been unable to find an instance wrapper for a runnable entity

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007812, E007812, W007812, I007812

7813:

Unexpected client location

Description:

RTA-RTE has detected an unexpected client location when processing a server operation.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007813, E007813, W007813, I007813

7821:

No client operation

Description:

RTA-RTE is unable to locate the operation in the interface categorizing the client's port that corresponds to the server's operation.

How to Solve:

For AUTOSAR R4.0 a port interface map can be used to connect dissimilar interfaces. Prior releases associate operations by name and therefore the names must match.

Variants:

F007821, E007821, W007821, I007821

7822:

Duplicate parameter name

Description:

RTA-RTE has detected a client-server operation with duplicated parameter names.

How to Solve:

Ensure all parameter names are unique.

Variants:

F007822, E007822, W007822, I007822

7823:

Parameter mismatch

Description:

RTA-RTE has detected a mismatch in parameter types between calls in different SWC instances. This is invalid.

How to Solve:

Please contact support for assistance. In your error report please include a sample configuration and command line that demonstrates the error.

Variants:

F007823, E007823, W007823, I007823

7827:

Invalid implementation policy

Description:

RTA-RTE has detected an invalid configuration for an operation parameter. The parameter has an implementation policy of 'use array base type' but the parameter is not itself an array type.

How to Solve:

Implementation policy 'use array base type' can only be applied to array types.

Variants:

F007827, E007827, W007827, I007827

7828:

Invalid policy for primitive type

Description:

RTA-RTE has detected an invalid configuration for an operation parameter. The parameter has an implementation policy of 'use void' but the parameter is not itself a complex type.

How to Solve:

Implementation policy 'use void' can only be applied to complex types.

Variants:

F007828, E007828, W007828, I007828

7850:

No initiation signal

Description:

RTA-RTE could not locate the initiation signal (signal group) for the specified client-server.

How to Solve:

Ensure that the signal mapping for the communication includes the signal group.

Variants:

F007850, E007850, W007850, I007850

7854:

No server parameter signal

Description:

RTA-RTE could not determine the signal mapping for the specified client-server operation parameter instance.

How to Solve:

Ensure that all parameters are mapped both for the client's call and the server's reply.

Variants:

F007854, E007854, W007854, I007854

7900:

No invalid value

Description:

RTA-RTE was unable to determine the invalid value for the specified data type.

How to Solve:

Ensure an invalid value is defined for the specified data type.

Variants:

F007900, E007900, W007900, I007900

7901:

No initial value

Description:

RTA-RTE was unable to determine the initial value for the specified data item. An initial value is needed for items which handle invalidation by replacement.

How to Solve:

Ensure an initial value is present.

Variants:

F007901, E007901, W007901, I007901

7902:

Invalidation of queued data

Description:

RTA-RTE has detected an invalidation request for queued data.

How to Solve:

Invalidation is only supported for non-queued data.

Variants:

F007902, E007902, W007902, I007902

7903:

No receiver invalidation handling

Description:

RTA-RTE was unable to determine the reception handling for the specified data item which is marked as invalidatable in the transmitting port. By default this item will use 'keep' handling.

How to Solve:

Ensure specification of invalidation is consistent for provider and receiver.

Variants:

F007903, E007903, W007903, I007903

8100:

Internal trigger connected to non-local ECU

Description:

An invalid external trigger configuration between SWC instances on different ECU instances was detected when processing the port specified. Non-local connections are not permitted.

How to Solve:

Ensure both provider and requirer are located on the same ECU instance.

Variants:

F008100, E008100, W008100, I008100

8101:

Multiple providers detected for external trigger

Description:

RTA-RTE has detected that the specified port instance has multiple providers. External triggering with multiple providers (N:1 communication) is not valid.

How to Solve:

Ensure each required external trigger has at most one provider

Variants:

F008101, E008101, W008101, I008101

8102:

Minimum start interval specified for unmapped RTE event

Description:

RTA-RTE has detected that an RTE Event is not mapped to a task however the event starts a runnable with a Minimum Start Interval and therefore must be mapped.

How to Solve:

Map the specified event to a task.

Variants:

F008102, E008102, W008102, I008102

8103:

Unable to locate trigger

Description:

RTA-RTE has failed to find a matching trigger in the specified port. When synchronizing released BSW triggers and required SWC triggers the triggers must have the same name.

How to Solve:

Ensure matching names are used

Variants:

F008103, E008103, W008103, I008103

8110:

NULL API generated for port

Description:

RTA-RTE has forced NULL API generation for the specified port. A null API is required to support the indirect API. The generated API functions take no action but enable the relevant entries in the generated component data

structure to be filled

How to Solve:

No action required. The information message can be disabled if required.

Variants:

F008110, E008110, W008110, I008110

8300:

Invalid option value for ConfData report

Description:

The ConfData report requires the `--os-task-as-function` option to be enabled.

How to Solve:

Enable the option (parameter value = 1) to permit the report to be generated.

Variants:

F008300, E008300, W008300, I008300

8500:

Invalid option value for PaVaSt report

Description:

The PaVaSt report requires the `--os-task-as-function` option to be enabled.

How to Solve:

Enable the option (parameter value = 1) to permit the report to be generated.

Variants:

F008500, E008500, W008500, I008500

9100:

Invalid error

Description:

RTA-RTE has detected an invalid enumeration value for `RteRipsSupport`.

How to Solve:

Ensure that the '`RteRipsSupport`' enum is defined with a valid value.

Variants:

F009100, E009100, W009100, I009100

9101:

Multiplicity check failure for `RteRips`

Description:

RTA-RTE has encountered an invalid number of `RteRips` containers according to the defined multiplicity rules which state that there must be at most one

such container.

How to Solve:

Ensure that no more than one `RteRips` container exists in the RTE module configuration processed by RTA-RTE.

Variants:

F009101, E009101, W009101, I009101

9102:

Multiple instance SWC with RIPS Implicit S/R API

Description:

RTA-RTE has encountered an SWC with multiple instances on the target ECU instance that configures RIPS implementation for the Implicit API. This configuration is not supported.

How to Solve:

Ensure that at most one instance exists for any SWC that uses the RIPS Implicit API. If this is not possible the AUTOSAR implementation can be used by removing the RIPS-relevant `FlatInstanceDescriptor`.

Variants:

F009102, E009102, W009102, I009102

9104:

No SWCI for RTE Event

Description:

RTA-RTE has failed to locate a SWC instance based on a referenced RTE Event. RTA-RTE first resolves the SWC type and then assumes that the type is instantiated exactly once. A failure to instantiate the SWC type or multiple instantiation will raise this error.

How to Solve:

Ensure that exactly one such SWC instance exists and it is mapped to the ECU instance.

Variants:

F009104, E009104, W009104, I009104

9105:

No SWCI for SWC

Description:

RTA-RTE has failed to locate a SWC instance based on a referenced SWC type. RTA-RTE assumes that the type is instantiated exactly once. A failure to instantiate the SWC type or multiple instantiation will raise this error.

How to Solve:

Ensure that exactly one such SWC instance exists and it is mapped to the

ECU instance.

Variants:

F009105, E009105, W009105, I009105

9106:

No OriginalEvent ref

Description:

RTA-RTE has failed to locate a single OriginalEvent reference within a RIPS fill/flush event. RTA-RTE requires that exactly one such reference exists.

How to Solve:

Ensure that exactly one such reference exists and that it has GID `'RIPS-ORIGINAL-OPERATION'`.

Variants:

F009106, E009106, W009106, I009106

9107:

`RteRips` container does not contain exactly one `RteRipsSupport`

Description:

The input configuration contained an `EcuParamConfContainerDef` named `RteRips` that did not contain exactly one `EcucEnumerationParamDef`s named `RteRipsSupport`.

How to Solve:

Ensure that the `RteRips EcuParamConfContainerDef` contains exactly one `EcucEnumerationParamDef`s named `RteRipsSupport`.

Variants:

F009107, E009107, W009107, I009107

9108:

Application SwComponentPrototype has multiple SwcToImplMappings and RIPS enabled

Description:

The input configuration contained `ApplicationSwc SwComponentPrototype` with multiple `SwcToImplMapping`s, whilst `RteRipsSupport` had value `RTE_RIPS_ON`.

How to Solve:

Ensure that the `ApplicationSwc SwComponentPrototype` contains exactly one `SwcToImplMapping`, or change `RteRipsSupport` to `RTE_RIPS_OFF`.

Variants:

F009108, E009108, W009108, I009108

9109:

RIPS-relevant data element requires Data Element With Status

Description:

A RIPS-relevant data element requires a status [SWS_Rte_02608]. This is not supported.

How to Solve:

Either switch off RIPS globally, ensure that the data element is not RIPS-relevant or remove the need for Data Element With Status (see [SWS_Rte_02608]).

Variants:

F009109, E009109, W009109, I009109

9110:

Application SwComponentPrototype supports multiple instances and RIPS enabled

Description:

The input configuration contained an `ApplicationSwc` which supports multiple instances, whilst `RteRipsSupport` had value `RTE_RIPS_ON`.

How to Solve:

Ensure that the `ApplicationSwc` does not support multiple instantiation or change `RteRipsSupport` to `RTE_RIPS_OFF`.

Variants:

F009110, E009110, W009110, I009110

9111:

Use of RIPS with –deviate-split-swci-support option not supported

Description:

The supplied input configuration enables use of RIPS, and the `--deviate-split-swci-support` option is enabled on the command line. The use of RIPS at the same time as the `--deviate-split-swci-support=on` option is not supported.

How to Solve:

Either disable RIPS, or remove the `--deviate-split-swci-support=on` option from the command line. When RIPS is enabled, RTA-RTE allows runnables belonging to the same SWC instance to be mapped to different `OsApplications` without the use of `--deviate-split-swci-support=on`.

Variants:

F009111, E009111, W009111, I009111

100001:

RIPS mode is enabled

Description:

The supplied input configuration enables use of RIPS

How to Solve:

(none)

Variants:

F100001, E100001, W100001, I100001

100002:

AUTOSAR RIPS mode is enabled

Description:

The supplied input configuration enables use of standard AUTOSAR RIPS.

How to Solve:

(none)

Variants:

F100002, E100002, W100002, I100002

100003:

RIPS plugin detected

Description:

The supplied input configuration contains a RIPS plugin.

How to Solve:

(none)

Variants:

F100003, E100003, W100003, I100003

100004:

Configuration contains AUTOSAR and legacy RIPS FIDs

Description:

The supplied input configuration contains a mixture of AUTOSAR RIPS `FlatInstanceDescriptor`s and legacy RIPS `FlatInstanceDescriptor`s. RTA-RTE does not support the use of AUTOSAR and legacy RIPS simultaneously.

How to Solve:

Ensure that all RIPS `FlatInstanceDescriptor`s used by the configuration are of the appropriate type, depending on whether AUTOSAR or legacy RIPS is in use.

Variants:

F100004, E100004, W100004, I100004

100005:

No RIPS `FlatInstanceDescriptor`s when RIPS mode is enabled

Description:

The supplied input configuration enables RIPS mode but does not contain any RIPS `FlatInstanceDescriptor`s. At least one RIPS `FlatInstanceDescriptor` must be present so that RTA-RTE can determine whether RIPS is being used in AUTOSAR or legacy mode. A RIPS `FlatInstanceDescriptor` is a `FlatInstanceDescriptor` that aggregates an `RtePluginProps`.

How to Solve:

Either disable RIPS or ensure that at least one RIPS `FlatInstanceDescriptor` is present.

Variants:

F100005, E100005, W100005, I100005

100006:

RIPS plugin detected but RIPS mode is disabled

Description:

The supplied input configuration contains a RIPS plugin but RIPS mode is disabled. Any RIPS plugins are ignored.

How to Solve:

Either remove the RIPS plugin or enable RIPS mode

Variants:

F100006, E100006, W100006, I100006

100007:

RIPS mode is not enabled but the configuration contains RIPS FIDs

Description:

The supplied input configuration contains RIPS `FlatInstanceDescriptor`s but does not enable RIPS mode.

How to Solve:

Either enable RIPS mode or remove all the RIPS `FlatInstanceDescriptor`s from the configuration.

Variants:

F100007, E100007, W100007, I100007

100009:

Runnable activated by cross partition event

Description:

The supplied input configuration contains an event that is mapped to a task in a partition that is different from the partition of its parent SWC.

How to Solve:

This configuration is valid if RIPS mode is enabled and the command line option `--allow-cross-partition-event-mapping` is enabled.

Variants:

F100009, E100009, W100009, I100009

100012:

RIPS-relevant client server communication graph detected

Description:

The supplied input configuration contains a RIPS-relevant client server communication graph.

How to Solve:

(none)

Variants:

F100012, E100012, W100012, I100012

100014:

Legacy RIPS mode is forced

Description:

The supplied input configuration forces the use of Legacy RIPS.

How to Solve:

(none)

Variants:

F100014, E100014, W100014, I100014

100015:

Legacy RIPS configuration contains AUTOSAR RIPS FIDs

Description:

The supplied input configuration forces legacy RIPS but contains AUTOSAR RIPS `FlatInstanceDescriptor`s.

How to Solve:

Remove all instances of AUTOSAR RIPS `FlatInstanceDescriptor`s.

Variants:

F100015, E100015, W100015, I100015

100017:

`SoftwareComponentPrototype` containing cross partition mapped events is referenced by multiple `FlatInstanceDescriptors`

Description:

The supplied input configuration contains multiple `FlatInstanceDescriptors` referencing the same `SoftwareComponentPrototype` which contains events that are mapped cross partition. The trusted functions generated for activating cross partition events use the name of any `FlatInstanceDescriptors` referencing their owning `SoftwareComponentPrototype` as part of their symbol, which cannot be disambiguated if multiple `FlatInstanceDescriptors` reference the same `SoftwareComponentPrototype`.

How to Solve:

Ensure that any `SoftwareComponentPrototype`s participating in cross partition event activations only appear as the `TARGET-REF` in a single `FlatInstanceDescriptor`.

Variants:

F100017, E100017, W100017, I100017

100018:

`ImplementationDataType` supports optional elements but is not of `category` `STRUCTURE`

Description:

The supplied input configuration contains an `ImplementationDataType` that is not a structure type but is configured to support optional elements (ie `isStructWithOptionalElement` is `TRUE`). Only structure types can be configured to support optional elements.

How to Solve:

Ensure that all `ImplementationDataType`s configured to support optional elements have the `category` set to STRUCTURE.

Variants:

F100018, E100018, W100018, I100018

100019:

Violation of `[SWS_Rte_Constr_80005]`

Description:

A `FlatInstanceDescriptor` that enables RIPS for a client/server communication graph references the `ClientServerOperation` in the context of an `RPortPrototype`, but the server for that communication is present in the configuration. This contradicts `[SWS_Rte_Constr_80005]` (AUTOSAR 4.4.0).

How to Solve:

Change the `FlatInstanceDescriptor` so that it references the `ClientServerOperation` in the context of the `PPortPrototype`.

Variants:

F100019, E100019, W100019, I100019

100020:

`ImplementationDataType` does not support optional elements but contains optional `ImplementationDataTypeElements`

Description:

The supplied input configuration contains an `ImplementationDataType` that is not configured to support optional elements (ie `isStructWithOptionalElement` is `TRUE`) but contains optional `ImplementationDataTypeElements` (ie `isOptional` is `TRUE`). The following Info 1001 messages give details of the optional `ImplementationDataTypeElements`.

How to Solve:

Ensure that all `ImplementationDataTypes` that contain optional elements are configured to support optional elements or that no optional `ImplementationDataTypeElements` exist in non-optional `ImplementationDataTypes`.

Variants:

F100020, E100020, W100020, I100020

100021:

Optional `ImplementationDataTypeElement` cannot be of `category STRUCTURE`

Description:

The supplied input configuration contains an `ImplementationDataTypeElement` of `category STRUCTURE` that is also optional (ie `isOptional` is `TRUE`). This violates `[constr_1639]` (AUTOSAR 4.4.0).

How to Solve:

Ensure that `ImplementationDataTypeElements` that have `category STRUCTURE` do not have the attribute `isOptional` set to `TRUE`.

Variants:

F100021, E100021, W100021, I100021

100022:

First `ImplementationDataTypeElement` of a structure `ImplementationDataType` is optional

Description:

The supplied input configuration contains a structure `ImplementationDataType` in which the first element is optional. This contravenes `[constr_1638]` (AUTOSAR 4.4.0).

How to Solve:

Ensure that the first `ImplementationDataTypeElement` of a structure `ImplementationDataTypes` is not optional.

Variants:

F100022, E100022, W100022, I100022

100024:

RIPS `FlatInstanceDescriptor` references a data element via a SWC that does not access it

Description:

The supplied input configuration contains a RIPS `FlatInstanceDescriptor` that references a data element via a `PortPrototype` on an instance of a `SwComponentType`, and that `SwComponentType` does not contain any `VariableAccess`s referencing that same data element.

How to Solve:

Ensure that all RIPS `FlatInstanceDescriptor`s reference a data element via a SWC which has a `VariableAccess` to that data element.

Variants:

F100024, E100024, W100024, I100024

100026:

Deprecated core ID assignment

Description:

The supplied input configuration contains an `OsApplication` that has a core ID assigned using an `OsApplicationCoreAssignment`. This method of assigning a core ID was superseded in AUTOSAR 4.1 and is deprecated in RTA-RTE.

How to Solve:

Use a reference to an `EcucCoreDefinition` to assign a core ID to the `OsApplication`.

Variants:

F100026, E100026, W100026, I100026

100027:

Conflicting values for core ID

Description:

The supplied input configuration contains an `OsApplication` that has an `OsApplicationCoreAssignment` value different from the value of the referenced `EcucCoreDefinition`.

How to Solve:

Remove the deprecated `OsApplicationCoreAssignment`.

Variants:

F100027, E100027, W100027, I100027

100028:

`--implicit-use-cds` option with `enableTakeAddress`

Description:

The `--implicit-use-cds` option is provided on the command line, and the

configuration contains a port that has `enableTakeAddress` set and that is used for implicit access to a data element. The use of `--implicit-use-cds` with `enableTakeAddress` is not supported.

How to Solve:

Disable `enableTakeAddress` for the port, or remove the `--implicit-use-cds` option from the RTEGen command line.

Variants:

F100028, E100028, W100028, I100028

100029:

`RteOsTaskChain` detected

Description:

The supplied input configuration contains at least one `RteOsTaskChain`.

How to Solve:

(none)

Variants:

F100029, E100029, W100029, I100029

100030:

Name collision while using `--buffers-use-data-name`

Description:

Because `--buffers-use-data-name` is enabled, RTA-RTE is attempting to name data channels according to the names of the corresponding `DataPrototype`s. This error indicates that the same name would be applied to more than one data channel, which could lead to incorrect assignment of data.

How to Solve:

Consider the following options:

- Remove duplicate `DataPrototype` names,
- don't share the same `DataPrototype` with more than one communication graph,
- don't use multiple instantiation, or
- remove the option `--buffers-use-data-name` from the RTEGen command line.

Variants:

F100030, E100030, W100030, I100030

100031:

Circular `RteOsTaskChain`s detected

Description:

The supplied configuration of `RteOsTaskChain`s defines a circular chain. This violates AUTOSAR R20-11 `[SWS_Rte_CONSTR_04559]`.

How to Solve:

Remove the circular chain of `RteOsTaskChain`s from the configuration.

Variants:

F100031, E100031, W100031, I100031

100032:

`OsTask` referenced by multiple `RtePredecessorOsTaskRef`s

Description:

The supplied configuration contains an `OsTask` that is referenced by more than one `RtePredecessorOsTaskRef`. This violates AUTOSAR R20-11 `[SWS_Rte_CONSTR_04558]`.

How to Solve:

Remove all but one `RtePredecessorOsTaskRef` referencing the `OsTask` in the configured `RteOsTaskChain`s.

Variants:

F100032, E100032, W100032, I100032

100033:

Multiplicity check failure for `RtePredecessorOsTaskRef`s

Description:

An `RteOsTaskChain` in the supplied configuration does not contain exactly one `RtePredecessorOsTaskRef`. This violates the multiplicity rules for `RtePredecessorOsTaskRef`s, see AUTOSAR R20-11 `[ECUC_Rte_09136]`.

How to Solve:

Ensure each `RteOsTaskChain` contains one `RtePredecessorOsTaskRef`.

Variants:

F100033, E100033, W100033, I100033

100034:

Multiplicity check failure for `RteSuccessorOsTaskRef`s

Description:

An `RteOsTaskChain` in the supplied configuration does not contain exactly one `RteSuccessorOsTaskRef`. This violates the multiplicity rules for `RteSuccessorOsTaskRef`s, see AUTOSAR R20-11 `[ECUC_Rte_09137]`.

How to Solve:

Ensure each `RteOsTaskChain` contains one `RteSuccessorOsTaskRef`.

Variants:

F100034, E100034, W100034, I100034

100035:

Circular `RteOsTaskChain` detected

Description:

An `RteOsTaskChain` in the supplied configuration references the same `OsTask` in both the `RtePredecessorOsTaskRef` and the `RteSuccessorOsTaskRef` forming a circular `RteOsTaskChain`. This violates AUTOSAR R20-11 `[SWS_Rte_CONSTR_04559]`.

How to Solve:

Update or remove the circular `RteOsTaskChain` from the configuration.

Variants:

F100035, E100035, W100035, I100035

100037:

No code will be generated to support the LdCom `TP` API

Description:

RTA-RTE does not support the use of the LdCom `TP` API, and even though this configuration is acceptable for an `LdComIPdu` that transports an `ISignal` mapped to a RIPS data element, no code will be generated to support this configuration, save for the inclusion of the file `Rte_LdComTp_Cbk.h` from within `Rte_Cbk.h` to allow the system integrator to provide the required `LDCOM_TP` callback function prototypes to the RIPS plugin implementer.

How to Solve:

Either continue to use LdCom `TP` and implement the callbacks within the RIPS plugin or use the LdCom `IF` API, by setting the LdComApiType to `LDCOM_IF` in the referenced `LdComIPdu`.

Variants:

F100037, E100037, W100037, I100037

100038:

LdCom for implicit communication is not supported

Description:

The supplied configuration contains a data element that is implicitly written, a runnable has `DataWriteAccess` to it, and the same data element is transmitted via LdCom, the data element is mapped to an `ISignal` that is referenced by an `ISignalToIPduMapping` that is itself referenced by an `LdComIPdu`. This configuration is not supported by RTA-RTE.

How to Solve:

Amend the configuration so that the data element is not both accessed implicitly and transmitted via LdCom. Consider changing the runnable to

access the data element explicitly using `DataSendPoint` or `DataReceivePoint`

Variants:

F100038, E100038, W100038, I100038

100039:

`OsTask` referenced by multiple `RteSuccessorOsTaskRef`s

Description:

The supplied configuration contains an `OsTask` that is referenced by more than one `RteSuccessorOsTaskRef`. This violates AUTOSAR R20-11 `[SWS_Rte_CONSTR_04558]`.

How to Solve:

Remove all but one `RteSuccessorOsTaskRef` referencing the `OsTask` in the configured `RteOsTaskChain`s.

Variants:

F100039, E100039, W100039, I100039

100040:

RIPS implicit inter-ECU sender/receiver communication is not supported

Description:

The supplied configuration contains a RIPS-relevant data element implicitly accessed inter-ECU in a sender/receiver communication graph. This configuration is not supported by RTA-RTE.

How to Solve:

Amend the configuration so that the implicitly accessed data element is not part of an inter-ECU sender/receiver communication. Consider using explicit access instead.

Variants:

F100040, E100040, W100040, I100040

100041:

`ExternalTriggerOccurredEvent` does not correctly reference trigger

Description:

The supplied configuration contains an `ExternalTriggerOccurredEvent` that does not correctly reference a trigger as the source of the event.

How to Solve:

Ensure that the `ExternalTriggerOccurredEvent` has a `TriggerIRef` which contains a `ContextRPortRef` and a `TargetTriggerRef`

Variants:

F100041, E100041, W100041, I100041

100042:

Floating point data conversion of composite data element members is not supported

Description:

The supplied configuration contains a composite data element that is used in inter-ECU communication and floating point data conversion is configured for one of the members of this data element.

How to Solve:

Ensure that data conversion is not configured for any floating point members of composite data elements.

Variants:

F100042, E100042, W100042, I100042

100043:

Comment output is suppressed

Description:

RTA-RTE has detected the `--disable-comments` option so some comments in the output C code are suppressed.

How to Solve:

(none)

Variants:

F100043, E100043, W100043, I100043

100045:

A command-line option is deprecated

Description:

This warning is raised when a deprecated command-line option was presented to RTA-RTE, or it was presented in a deprecated form.

How to Solve:

(none)

Variants:

F100045, E100045, W100045, I100045

100046:

`RteUseComShadowSignalApi` detected

Description:

The supplied input configuration contains the legacy parameter `RteUseComShadowSignalApi`.

How to Solve:

(none)

Variants:

F100046, E100046, W100046, I100046

100048:

Unsupported use of `WaitPoint`s with inter-`OsApplication` asynchronous client-server

Description:

RTA-RTE does not support the use of `WaitPoint`s in the context of inter-`OsApplication` asynchronous client-server communication.

How to Solve:

Remove the `WaitPoint` from the `RunnableEntity` started by the `AsynchronousServerCallReturnsEvent` or choose a different communication paradigm.

Variants:

F100048, E100048, W100048, I100048

100049:

Unsupported use of `timeout`s with inter-`OsApplication` asynchronous client-server

Description:

RTA-RTE does not support the use of `timeout`s with inter-`OsApplication` asynchronous client-server communication.

How to Solve:

Set the `timeout` attribute of the relevant `AsynchronousServerCallPoint`s to zero or choose a different communication paradigm.

Variants:

F100049, E100049, W100049, I100049

100050:

`Operation` and `Port` referenced by multiple `AsynchronousServerCallPoint`s

Description:

RTA-RTE does not support the use of multiple `AsynchronousServerCallPoint`s referencing the same `ClientServerOperation` in the same `PortPrototype` of a `SoftwareComponent` instance.

How to Solve:

Modify the `AsynchronousServerCallPoint`s to use different `PortPrototype`s or move them into different `SoftwareComponent` instances.

Variants:

F100050, E100050, W100050, I100050

100051:

Unsupported mapping of `AsynchronousServerCallReturnsEvent` to extended `OsTask`

Description:

RTA-RTE does not support the mapping of `AsynchronousServerCallReturnsEvent`s to extended `OsTask`s in the context of inter-`OsApplication` asynchronous client-server communication.

How to Solve:

Map the `AsynchronousServerCallReturnsEvent` to an unextended `OsTask` or configure the `OsTask` to use forced-basic semantics.

Variants:

F100051, E100051, W100051, I100051

100052:

Inter-`OsApplication` asynchronous client-server activation limit mismatch

Description:

The configuration for the specified inter-`OsApplication` asynchronous client-server task defines a specific activation limit. The code has been generated using this activation limit. However RTA-RTE has calculated that it requires a different limit.

How to Solve:

Increase the activation limit such that it is equal to or larger than the total number of incoming and outgoing asynchronous client-server communications mapped to the `OsTask`.

Variants:

F100052, E100052, W100052, I100052

100054:

Invalid `queueLength` value in Inter-`OsApplication` asynchronous client-server

Description:

The configuration for the specified inter-`OsApplication` asynchronous client-server `PPort ServerComSpec` defines a queue length greater than one however RTA-RTE currently does not support queued inter-`OsApplication` asynchronous client-server communication.

How to Solve:

Set the `queueLength` field of the invalid `PortPrototype ServerComSpec` to one or use a different communication paradigm.

Variants:

F100054, E100054, W100054, I100054

100055:

Unsupported use of `ClientServerInterfaceMapping` with inter-`OsApp` async c/s

Description:

RTA-RTE does not support the use of `ClientServerInterfaceMapping`s in the context of inter-`OsApplication` asynchronous client-server communication.

How to Solve:

Use the same or compatible `ClientServerInterface`s that don't require a `ClientServerInterfaceMapping`.

Variants:

F100055, E100055, W100055, I100055

100056:

Unsupported `AsynchronousServerCallReturnsEvent`s with inter-`OsApp` async c/s

Description:

RTA-RTE does not support the use of multiple `AsynchronousServerCallReturnsEvent`s within a single SWC instance mapped to multiple `OsApplication`s in the context of inter-`OsApplication` asynchronous client-server communication (even when the command line option `--deviate-split-swci-support` is enabled).

How to Solve:

Map the `AsynchronousServerCallReturnsEvent`s within the SWC instance to a single `OsApplication`

Variants:

F100056, E100056, W100056, I100056

100058:

Extended task generated

Description:

This message provides information as to why RTA-RTE generated an extended task.

How to Solve:

No action needed, however, if extended tasks cannot be used then modify the input configuration to ensure that an extended task is not required or consider the use of force-basic semantics, see the *RTA-RTE User Guide* sections `24.4` and `24.5` for more information.

Variants:

F100058, E100058, W100058, I100058

100059:

Inter-ECU client server communication graph involving transformation detected

Description:

The supplied input configuration contains an inter-ECU client server communication graph that makes use of data transformation.

How to Solve:

(none)

Variants:

F100059, E100059, W100059, I100059

100060:

Unsupported use of `--buffers-use-provider-name` with multiple IOC channels

Description:

Because `--buffers-use-provider-name` is enabled for an inter-`OsApplication` communication graph, RTA-RTE is attempting to rename the IOC channels according to the provider. This error indicates that the same name would be applied to more than one IOC channel, which could lead to incorrect assignment of data.

How to Solve:

Consider the following options:

- Ensure that any inter-`OsApplication` communication graphs do not generate multiple IOC channels. For example, communication graphs with at least two requirers mapped to different `OsApplication`s will lead to multiple IOC channels being generated, or
- remove the option `--buffers-use-data-name` from the RTEGen command line.

Variants:

F100060, E100060, W100060, I100060

100062:

Incompatible `SwBaseType`s for `ImplementationDataType`s

Description:

When validating compatibility for two `ImplementationDataType`s it has been detected that there are incompatible `SwBaseType`s

How to Solve:

Consider adding a `PortInterfaceMapping` to force compatibility if you are sure that having two data types with different `SwBaseType`s is okay, otherwise, ensure that the two `SwBaseType`s are compatible as per the AUTOSAR

compatibility rules

Variants:

F100062, E100062, W100062, I100062

100063:

`InstantiationTimingEventProps` referencing events other than `TimingEvents`

Description:

An `InstantiationTimingEventProps` can only reference an instance of a `TimingEvent`

How to Solve:

Check that the event referenced by the `InstantiationTimingEventProps` is an instance of `TimingEvent` or remove the `InstantiationTimingEventProps`.

Variants:

F100063, E100063, W100063, I100063

100065:

`RteEventToTaskMapping` referencing a `BSWEvent` and not an `RTEevent`

Description:

An `RteEventToTaskMapping` can only reference an instance of a `RTEEvent`

How to Solve:

Check that the event referenced by the `RteEventToTaskMapping` is an instance of `RTEEvent` or use an `RteBswEventToTaskMapping`.

Variants:

F100065, E100065, W100065, I100065

100066:

`RteBswEventToTaskMapping` referencing an `RTEEvent` and not a `BSWEvent`

Description:

An `RteBswEventToTaskMapping` can only reference an instance of a `BSWEvent`

How to Solve:

Check that the event referenced by the `RteBswEventToTaskMapping` is an instance of `BSWEvent` or use an `RteEventToTaskMapping`.

Variants:

F100066, E100066, W100066, I100066

100067:

Inconsistent `InstantiationTimingEventProps` / `TimingEvent` context

Description:

An `InstantiationTimingEventProps` cannot reference a `TimingEvent` contained in a different `ApplicationSwComponentType` than the one that types

the `SwComponentPrototype` referenced by the `InstantiationTimingEventProps`.

How to Solve:

Ensure that the `InstantiationTimingEventProps` and `TimingEvent` correspond to the same `ApplicationSwComponentType`.

Variants:

F100067, E100067, W100067, I100067

100068:

Invalid `LdComCbkTriggerTransmit` function name

Description:

The `LdComCbkTriggerTransmit` function name configured for an `LdComIPdu` does not match the name specified by AUTOSAR R20-11 `[SWS_Rte_01408]`.

How to Solve:

Ensure that when `LdComCbkTriggerTransmit` is configured for an `LdComIPdu` the function name is `Rte_LdComCbkTriggerTransmit_<sn>` where `<sn>` is the `shortName` of the relevant `LdComIPdu`.

Variants:

F100068, E100068, W100068, I100068

100069:

`Com` missing from the `EcucValueCollection`

Description:

`Com` is configured but no `EcucModuleConfigurationValues` referencing an `EcuModuleDef` with `shortName Com` is present in the `EcucValueCollection`

How to Solve:

If Com is required then add the appropriate `EcuModuleDef` into the `EcucValueCollection`

Variants:

F100069, E100069, W100069, I100069

100070:

`LdComIPdu` with LdComApiType set to `LDCOM_TP` is not used

Description:

`LdComIPdu` with LdComApiType set to `LDCOM_TP` is not mapped to an `ISignal` that has a data element mapped to the referenced `SystemSignal`, so it will be ignored by RTA-RTE.

How to Solve:

Remove the unused `LdComIPdu`

Variants:

F100070, E100070, W100070, I100070

100072:

Transforming `ISignal` aggregates `NetworkRepresentationProps`

Description:

RTA-RTE does not support `ISignal`s that request data transformation and also aggregate a `NetworkRepresentationProps`.

How to Solve:

Ensure that any `ISignal` that requests data transformation does not also aggregate a `NetworkRepresentationProps`.

Variants:

F100072, E100072, W100072, I100072

100078:

`CpSoftwareCluster` detected

Description:

RTA-RTE is generating in cluster mode for the specified `CpSoftwareCluster`.

How to Solve:

(none)

Variants:

F100078, E100078, W100078, I100078

100079:

`CpSoftwareCluster` is incompatible with `RootSwCompositionPrototype`

Description:

The `System`'s `CpSoftwareCluster` must reference the same `CompositionSwComponentType` as the `System`'s `RootSwCompositionPrototype`.

How to Solve:

Modify the `CpSoftwareCluster` to reference the same `CompositionSwComponentType` as the `RootSwCompositionPrototype`.

Variants:

F100079, E100079, W100079, I100079

100080:

`swComponentPrototypeAssignment` does not reference `RootSwCompositionPrototype`

Description:

All `swComponentPrototypeAssignment`s aggregated by the `System`'s `CpSoftwareCluster` must reference the `System`'s `RootSwCompositionPrototype`.

How to Solve:

Modify the `swComponentPrototypeAssignment` to reference the

RootSwCompositionPrototype.

Variants:

F100080, E100080, W100080, I100080

100082:

Incorrect Rte_Cs_TransactionHandleType definition

Description:

The input configuration defines an ImplementationDataType with C symbol Rte_Cs_TransactionHandleType but its definition does not match the AUTOSAR definition, see AUTOSAR R20-11 [SWS_Rte_02649].

How to Solve:

Do not define the inter-ECU client-server transaction handle type in the input configuration. RTA-RTE defines Rte_Cs_TransactionHandleType in file{Rte.h}.

Variants:

F100082, E100082, W100082, I100082

100084:

Incorrect event to SWC mapping

Description:

The input configuration defines an event that is mapped to a SwComponentPrototype and that event is also mapped to an RteSwComponentInstance that references a different SwComponentPrototype.

How to Solve:

Ensure that any mapped events and RteSwComponentInstances are mapped to the same SwComponentPrototype.

Variants:

F100084, E100084, W100084, I100084

100085:

Unsupported use of floating point NetworkRepresentation for data conversion

Description:

RTA-RTE does not support data conversion using ISignals with a NetworkRepresentation referencing a SwBaseType with the attribute baseTypeEncoding set to IEEE754.

How to Solve:

Ensure that any ISignal used for data conversion does not use a floating point NetworkRepresentation.

Variants:

F100085, E100085, W100085, I100085

100086:

Unsupported `CpSoftwareCluster category`

Description:

The `CpSoftwareCluster category` provided in the input configuration is not supported.

How to Solve:

When generating for a `CpSoftwareCluster` the supported `category`s are `HOST_SOFTWARE_CLUSTER` and `APPLICATION_SOFTWARE_CLUSTER`.

Variants:

F100086, E100086, W100086, I100086

100087:

No support for receiving data from multiple sources unless handled by RIPS

Description:

RTA-RTE does not support RTE managed inter or intra-ECU reception reception of data in combination with inter-cluster reception of data.

How to Solve:

Either remove the RTE managed reception or add a RIPS plugin.

Variants:

F100087, E100087, W100087, I100087

100088:

Deprecate support for `System`s that do not have `category` `ECU_EXTRACT`

Description:

Support for configurations that describe a `System` that has a `category` that is not `ECU_EXTRACT` when running RTEGen in generation phase is deprecated and will be removed in a future version of RTA-RTE.

How to Solve:

Update the configuration to make use of a `System` with a `category` of `ECU_EXTRACT`.

Variants:

F100088, E100088, W100088, I100088

100089:

Unsupported nested `CompositionSwComponentType`s when generating for clusters

Description:

Nested `CompositionSwComponentType`s have been detected in the `System` when generating for a software cluster.

How to Solve:

Flatten the system and provide RTA-RTE with an `ECU_EXTRACT` that does not contain nested `CompositionSwComponentTypes`.

Variants:

F100089, E100089, W100089, I100089

100090:

Invalid multiplicity of `CpSoftwareClusters` in the `System`

Description:

RTA-RTE is generating for a cluster and the `System` references more than one `CpSoftwareCluster`.

How to Solve:

Configure exactly one `CpSoftwareCluster` in the `System`.

Variants:

F100090, E100090, W100090, I100090

100092:

No support for referencing an inner port with a cross cluster `FlatInstanceDescriptor`

Description:

RTA-RTE does not support referencing the inner port of a communication graph with a cross cluster `FlatInstanceDescriptor`.

How to Solve:

Do not reference an inner port with a `FlatInstanceDescriptor` that aggregates an `RtePluginProps` that references an `associatedCrossSwClusterComRtePlugin`.

Variants:

F100092, E100092, W100092, I100092

100093:

No support for `WaitPoints` in application cluster mode

Description:

The input configuration describes an application cluster and contains `WaitPoints`. This violates AUTOSAR R21-11 `[SWS_Rte_CONSTR_80031]`.

How to Solve:

Ensure that there are no `WaitPoints` present when generating for an application cluster.

Variants:

F100093, E100093, W100093, I100093

100094:

Invalid multiple `FlatInstanceDescriptor`s for cross-cluster data

Description:

It is invalid to have multiple `FlatInstanceDescriptor`s that reference an `associatedCrossSwClusterComRtePlugin` and reference the same data element.

How to Solve:

Remove all but one of the `FlatInstanceDescriptor`s.

Variants:

F100094, E100094, W100094, I100094

100095:

Missing `PortElementToCommunicationResourceMapping`

Description:

RTA-RTE is generating for a cluster and a `FlatInstanceDescriptor` that aggregates an `RtePluginProps` that references an `associatedCrossSwClusterComRtePlugin` references a data element that is not mapped to a `CpSoftwareClusterCommunicationResource`.

How to Solve:

Either add the missing `PortElementToCommunicationResourceMapping` or remove the cluster `FlatInstanceDescriptor`.

Variants:

F100095, E100095, W100095, I100095

100096:

Missing `CpSoftwareClusterCommunicationResource`

Description:

RTA-RTE is generating for a software cluster and the `PortElementToCommunicationResourceMapping` aggregated by the `SystemMapping` does not reference a `CpSoftwareClusterCommunicationResource`.

How to Solve:

Either add the `CpSoftwareClusterCommunicationResource` or remove the `CpSoftwareCluster` referenced by the `System` if not intending to generate for a cluster.

Variants:

F100096, E100096, W100096, I100096

100102:

Wrong plugin type referenced by cluster `FlatinstanceDescritor`

Description:

A `FlatInstanceDescriptor` has an `associatedCrossSwClusterComRtePluginRef` that references a RIPS plugin that has the parameter `RteRipsPluginCommunicationScope` that is not set to `RTE_RIPS_CROSS_SW_CLUSTER_COM`. This indicates the wrong type of plugin is being referenced.

How to Solve:

Change the `associatedCrossSwClusterComRtePluginRef` to reference a cross cluster RIPS plugin or set the value of the parameter `RteRipsPluginCommunicationScope` in the referenced plugin to `RTE_RIPS_CROSS_SW_CLUSTER_COM`.

Variants:

F100102, E100102, W100102, I100102

100103:

Wrong plugin type referenced by RIPS `FlatinstanceDescritor`

Description:

A `FlatInstanceDescriptor` has an `associatedRtePluginRef` that references a RIPS plugin that has the parameter `RteRipsPluginCommunicationScope` that is specified and not set to `RTE_RIPS_LOCAL_SW_CLUSTER_COM`. Note that when this parameter is omitted, a value of `RTE_RIPS_LOCAL_SW_CLUSTER_COM` is assumed. This indicates the wrong type of plugin is being referenced.

How to Solve:

Change the `associatedRtePluginRef` to reference a local RIPS plugin or set the value of the parameter `RteRipsPluginCommunicationScope` in the referenced plugin to `RTE_RIPS_LOCAL_SW_CLUSTER_COM`.

Variants:

F100103, E100103, W100103, I100103

100104:

Cross-cluster RIPS `FlatInstanceDescriptor` detected outside of cluster mode

Description:

RTA-RTE is not generating for a cluster however the input configuration contains a cross-cluster RIPS `FlatInstanceDescriptor`.

How to Solve:

Configure a `CpSoftwareCluster` in the `System` or remove the cross-cluster RIPS `FlatInstanceDescriptor`.

Variants:

F100104, E100104, W100104, I100104

100105:

Violation of `SWS_Rte_CONSTR_80200`

Description:

A `NonQueuedReceiverComSpec` aggregated by the referenced `RPortPrototype` is inconsistently configured with respect to the related `CpSoftwareClusterCommunicationResource.comProps.sendIndication` violating `SWS_Rte_CONSTR_80200@R21-11`

How to Solve:

Ensure that the `NonqueuedReceiverComSpec` aggregated by the referenced `RPortPrototype` meets all the following criteria: `aliveTimeout` value set to 0; `handleNeverReceived` set to `FALSE`; `handleDataStatus` set to `FALSE`; `handleOutOfRange` set to any value other than `none`, `enableUpdate` set to `FALSE` or set the `CpSoftwareClusterCommunicationResource.comProps.sendIndication` to `anySendOperation`.

Variants:

F100105, E100105, W100105, I100105

100106:

Inconsistent transformer error handling in a cluster configuration

Description:

The cluster communication graph marked by the referenced `FlatInstanceDescriptor` contains `PortPrototypes` referenced by `PortAPIOptions` with inconsistent `transformerErrorHandling` resulting in calls to the same RIPS cluster-plugin both with and without the optional parameter `transformerError`.

How to Solve:

Either enable or disable `errorHandling` on all connected ports in the cluster communication graph.

Variants:

F100106, E100106, W100106, I100106

100107:

Incomplete `NvBlockDescriptor` configuration in application cluster mode

Description:

The input configuration describes an application cluster containing an `NvBlockDescriptor` that does not aggregate a `romBlock`.

How to Solve:

Either remove the `NvBlockDescriptor` or configure a `romBlock`.

Variants:

F100107, E100107, W100107, I100107

100108:

No support for EA method `ALL_INTERRUPT_BLOCKING` in application cluster mode

Description:

The input configuration describes an application cluster with `RteExclusiveAreaImplMechanism` set to `ALL_INTERRUPT_BLOCKING` which is not supported.

How to Solve:

Do not configure the exclusive area implementation method `ALL_INTERRUPT_BLOCKING` when generating for an application cluster.

Variants:

F100108, E100108, W100108, I100108

100109:

Inter-OsApp IRV accessed by `RunnableEntity` not mapped to an `OsTask`

Description:

An `InterRunnableVariable` that is accessed by `Runnableentity`s that are mapped to more than one `OsApplication` is accessed by a `RunnableEntity` that is not mapped to any `OsTask`.

How to Solve:

Either map the `Event` that activates the `RunnableEntity` to an `OsTask`, or ensure that the `InterRunnableVariable` is accessed by `RunnableEntity`s that are all mapped to the same `OsApplication`

Variants:

F100109, E100109, W100109, I100109

100110:

Detected `EcucPartition` mapping via `NvMBlockEcucPartitionRef`

Description:

An `NvMBlockDescriptor` is mapped to an `EcucPartition` via an `NvMBlockEcucPartitionRef`.

How to Solve:

(none)

Variants:

F100110, E100110, W100110, I100110

100112:

Fallback `EcucPartition` mapping in absence of `NvMBlockEcucPartitionRef`

Description:

In the absence of an `NvMBlockEcucPartitionRef`, an `NvMBlockDescriptor` is mapped to the `EcucPartition` with `EcucPartitionBswModuleExecution`

enabled.

How to Solve:

(none)

Variants:

F100112, E100112, W100112, I100112

100113:

Multiple `RteNvRamAllocation`s reference the same `NvBlockDescriptor`

Description:

There can only be one `RteNvRamAllocation` per `NvBlockDescriptor`.

How to Solve:

Remove the extra `RteNvRamAllocation`s that reference the same `NvBlockDescriptor`.

Variants:

F100113, E100113, W100113, I100113

100114:

Connected structure data types are not AUTOSAR compatible

Description:

An `ApplicationRecordElement` specified in a structure used by the `PPort` did not match any `ApplicationRecordElement`s found in a structure used by the `RPort`. A match with the same name and compatible type was expected. This behavior may emerge from usage of `PortInterfaceMapping`.

How to Solve:

Make sure that the connected structures in the `PPort` and `RPort` are compatible.

Variants:

F100114, E100114, W100114, I100114

100117:

Unsupported mapping of ASCREs/OIEs in cross cluster communication graphs

Description:

RTA-RTE does not support the mapping of `AsynchronousServerCallReturnsEvent`s or `OperationInvokedEvent`s to a non periodic `OsTask` in the context of a cross cluster communication graph.

How to Solve:

Map the `AsynchronousServerCallReturnsEvent` / `OperationInvokedEvent` to a different `OsTask` or add a `TimingEvent` to the existing `OsTask`.

Variants:

F100117, E100117, W100117, I100117

100118:

Invalid inter-cluster synchronous client

Description:

Inter-cluster client-server communication must be asynchronous.

How to Solve:

Replace the offending `SynchronousServerCallPoint` with an `AsynchronousServerCallPoint`.

Variants:

F100118, E100118, W100118, I100118

100121:

Detected LdCom with `aliveTimeout`

Description:

The configuration contains the reception of data via LdCom where the relevant `RPortPrototype` aggregates a `NonqueuedReceiverComSpec` with an `aliveTimeout`.

How to Solve:

Either remove the `aliveTimeout`, do not use LdCom, or use the command line option `--deviate-ldcom-alivetimeout`.

Variants:

F100121, E100121, W100121, I100121

100122:

LdCom with `aliveTimeout` below the minimum supported

Description:

The configuration contains the reception of data via LdCom with an `aliveTimeout` that is less that the period of the RTE MainFunction. As the RTE MainFunction is used to set the relevant tick counter this is the minimum supported `aliveTimeout` with LdCom.

How to Solve:

Either remove the `aliveTimeout`, increase the `aliveTimeout`, or decrease the RTE MainFunction period (via the command line option `--period`).

Variants:

F100122, E100122, W100122, I100122

100123:

LdCom with `aliveTimeout` above the maximum supported

Description:

The configuration contains the reception of data via LdCom with an

`aliveTimeout` that is greater than RTA-RTE supports. RTA-RTE converts the `aliveTimeout` into ticks based on the period of the RTE MainFunction. The calculated value must be less than 2 to the power of 32.

How to Solve:

Either remove the `aliveTimeout`, decrease the `aliveTimeout`, or increase the RTE MainFunction period (via the command line option `--period`).

Variants:

F100123, E100123, W100123, I100123

100124:

Unsupported PRPorts configuration

Description:

In addition to Nv use cases when the command line option `--extended-prports-support` is enabled RTA-RTE allows limited support for communication graphs involving only `PRPortPrototype`s that are aggregated in `ApplicationSoftwareComponentType`s.

How to Solve:

Do not use inter-`OsApplication` communication, `DataReceivedEvent`s or implcit access.

Variants:

F100124, E100124, W100124, I100124

100126:

Queued LdCom is only supported with `--deviate-cbk-xfrm` enabled

Description:

The configuration includes a queued data element being received via LdCom (`LDCOM_IF`). This is only supported when the command line option `--deviate-cbk-xfrm` is enabled.

How to Solve:

Either remove the communication, use unqueued, or enable the command line option `--deviate-cbk-xfrm`.

Variants:

F100126, E100126, W100126, I100126

100127:

Unsupported `TransformationTechnology`

Description:

The `XfrmImplementationMapping` references an unsupported `TransformationTechnology`. The specific reason is in the error text.

How to Solve:

The `TransformationTechnology` must have a `TransformerClass` of `SERIALIZER`

and `needsOriginalData` must be `FALSE`.

Variants:

F100127, E100127, W100127, I100127

100128:

Exclusive Area detected in split SWCI/BSW

Description:

RTA-RTE has detected that one or more `ExclusiveArea`s are used via the 'runsInside' role within a split SWCI/BSW module on the target ECU.

How to Solve:

Split SWCI/BSW modules are not permitted to use Exclusive Areas in the 'runsInside' role.

Variants:

F100128, E100128, W100128, I100128

100129:

RIPS plugins must have a unique INDEX

Description:

The configuration includes RIPS plugins with either undefined or duplicate INDEX values. **Note** if only one RIPS plugins is configured and no INDEX is defined then RTEGen accepts the configurateion and assumes the INDEX value of `0`.

How to Solve:

Ensure that each `RteRipsPluginConfigurationRef` defines an INDEX value and that the value is unique.

Variants:

F100129, E100129, W100129, I100129

100130:

Unsupported `ExclusiveArea` detected in split SWCI/BSW module

Description:

RTA-RTE has detected that one or more `ExclusiveArea`s are used via an unsupported `RteExclusiveAreaImplMechanism` in the `canEnter` role within a split SWCI/BSW module on the target ECU.

How to Solve:

Split SWCI/BSW modules are only permitted to use the `RteExclusiveAreaImplMechanism OS_SPINLOCK` when using `ExclusiveArea`s in the `canEnter` role.

Variants:

F100130, E100130, W100130, I100130

100131:

Invalid `RteComUser` configuration

Description:

The provided `RteComUser` has a `ComUserSignal`/`ComUserSignalGroup` that references an `ISignalToIPduMapping` which is not referenced by a `ComSignal` or `ComSignalGroup` provided in the input.

How to Solve:

Ensure that the `ISignalToIPduMapping` references a mapping that is also referenced from a `ComSignal` or `ComSignalGroup` provided in the input.

Variants:

F100131, E100131, W100131, I100131

100133:

Multiple periods for Rte_MainFunction

Description:

The period of the `Rte_MainFunction` is set via the `BswTimingEvent` mapped to the main function's `bswSchedulableEntity` and is implicity required to be the default value of 10ms by the existance of multiple COM alive timeouts for a single COM signal.

How to Solve:

Provide the `--period` command line option to set the period of the `Rte_MainFunction` to the same as the `BswTimingEvent` or change the period of that `BswTimingEvent` to 10ms.

Variants:

F100133, E100133, W100133, I100133

100134:

Rte_MainFunction has a period via command line different from input config

Description:

The period of the `Rte_MainFunction` is set via both a `BswTimingEvent` and the `--period` command line options and they are set to different values.

How to Solve:

Either set the time periods as equal or do not set the period for the main function via a `BswTimingEvent` or the `--period` command.

Variants:

F100134, E100134, W100134, I100134

100135:

Scaled CompuMethod contains a CompuScale with missing limit

Description:

`CompuScale`s aggregegated by scaled `CompuMethod`s must contain both an

`UpperLimit` and `LowerLimit`.

How to Solve:

Ensure that a `CompuScale.upperLimit` and `CompuScale.lowerLimit` exists for each `CompuScale` aggregated by the scaled `CompuMethod`.

Variants:

F100135, E100135, W100135, I100135

100136:

Scaled CompuMethod contains a CompuScale with inconsistent limits

Description:

A `CompuScale` aggregated by the scaled `CompuMethod` contains a `LowerLimit` that is larger than its corresponding `UpperLimit`.

How to Solve:

Ensure that the `CompuScale.lowerLimit` is no larger than the corresponding `CompuScale.upperLimit`.

Variants:

F100136, E100136, W100136, I100136

100137:

CompuMethod.compuScale is outside the range of limits of the associated AutosarDataType

Description:

The limits of a `CompuScale` fall outside the limits of the assocaited `AutosarDataType`.

How to Solve:

Ensure that the `CompuScale.upperLimit` and `CompuScale.lowerLimit` are within the range of limits of the associated `AutosarDataType`.

Variants:

F100137, E100137, W100137, I100137

100138:

`ISignalToIPduMapping` maps multiple `ComSignal(Group)`s in bi-directional Com

Description:

The input configuration mapped more than two `ComSignal`s or `ComSignalGroup`s to the same `ISignalToIPduMapping` when the referenced `ISignal` is also referenced by an `ISignalTriggering` and the mapped `SystemSignal` is involved in inter-ECU communication in the context of both a `PPortPrototype` and an `RPortPrototype`. The following Info 1001 messages give details of the `ComSignal`s or `ComSignalGroup`s that reference the `ISignalToIPduMapping`.

How to Solve:

Remove superfluous `ComSignal`s or `ComSignalGroup`s that reference the `ISignalToIPduMapping`.

Variants:

F100138, E100138, W100138, I100138

100139:

Multiple `ComSignal(Group)`s with the same direction in bi-directional Com

Description:

The input configuration mapped two or more `ComSignal`s or `ComSignalGroup`s that have the same direction (are referenced by `ComIPduDirection`s with the same value or have no `ComIPduDirection`s) to the same `ISignalToIPduMapping` when the referenced `ISignal` is also referenced by an `ISignalTriggering` and the mapped `SystemSignal` is involved in inter-ECU communication in the context of both a `PPortPrototype` and an `RPortPrototype`. The following Info 1001 messages give details of the `ComSignal`s or `ComSignalGroup`s of the same direction that reference the `ISignalToIPduMapping`.

How to Solve:

Remove superfluous `ComSignal`s or `ComSignalGroup`s of the same direction that reference the `ISignalToIPduMapping`.

Variants:

F100139, E100139, W100139, I100139

100141:

SenderReceiverToSignalGroupMapping: dataElement and typeMapping mismatch

Description:

RTA-RTE does not support the usage of an `ImplementationDataTypeElement` in the `SenderRecRecordElementMapping` when the `VariableDataPrototype` for the `SenderReceiverToSignalGroupMapping` is typed by an `ApplicationRecordDataType` or the usage of an `ApplicationRecordElement` when the `VariableDataPrototype` is typed by an `ImplementationDataType`.

How to Solve:

Align the use of `Implementation` or `Application DataType`s in the `SenderReceiverToSignalGroupMapping dataElement` and `typeMapping`.

Variants:

F100141, E100141, W100141, I100141

100143:

XfrmBufferSizeInBytes out of range

Description:

XfrmBufferSizeInBytes contains an ARformula that did not evaluate to an integer between 1 and 4294967295. This limit is beause the `BufferLength` argument for transformers is specified to be uint32.

How to Solve:

Correct the XfrmBufferSizeInBytes, for example by entering an integer literal between 1 and 4294967295.

Variants:

F100143, E100143, W100143, I100143

100147:

Unsupported VSA shape

Description:

RTA-RTE does not support variable sized arrays with a `DynamicArraySizeProfile` with a value other than `VSA_LINEAR`.

How to Solve:

Ensure that all variable sized arrays have their `DynamicArraySizeProfile` set to `VSA_LINEAR` or use RIPS to handle the communcation.

Variants:

F100147, E100147, W100147, I100147

100148:

ISignal.networkReperesentation.compuMethod ignored

Description:

RTA-RTE used a `SystemSignal.physicalProps.compuMethod` over an `ISignal.networkRepresentationProps.compuMethod`.

How to Solve:

Remove the `ISignal.networkRepresentationProps.compuMethod`

Variants:

F100148, E100148, W100148, I100148

100149:

ISignal.networkReperesentation.compuMethod used

Description:

RTA-RTE used an `ISignal.networkRepresentationProps.compuMethod` in the absence of a `SystemSignal.physicalProps.compuMethod`.

How to Solve:

Replace the `ISignal.networkRepresentationProps.compuMethod` with a `SystemSignal.physicalProps.compuMethod`

Variants:

F100149, E100149, W100149, I100149

100150:

Variable sized array contains nested variable sized array

Description:

RTA-RTE does not support variable sized arrays where each element of the VSA contains a VSA.

How to Solve:

Ensure that each element of all VSAs are statically sized.

Variants:

F100150, E100150, W100150, I100150

100155:

Data element referenced by more than one `MetaDataItemSet`

Description:

A `VariableDataPrototype` in the input configuration is referenced by more than one `MetaDataItemSet` violating `constr_10121@R23-11`.

How to Solve:

Remove the duplicate references to the `VariableDataPrototype`.

Variants:

F100155, E100155, W100155, I100155

100156:

Data element inconsistent MetaDataItemSet/SenderReceiverInterface

Description:

A `VariableDataPrototype` in the input configuration is referenced by a `MetaDataItemSet` and a `SenderReceiverInterface` but the `SenderReceiverInterface` does not aggregate that `MetaDataItemSet` violating `constr_10120@R23-11`.

How to Solve:

Only reference the `VariableDataPrototype` by a `MetaDataItemSet` in the same `SenderReceiverInterface`. The accompanying info messages 1001 show the `SenderReceiverInterface`(s).

Variants:

F100156, E100156, W100156, I100156

100157:

Initial value validation failure

Description:

RTA-RTE internal C code representation model has detected a validation failure.

How to Solve:

Validation Error, ensure all configured initial values are appropriate for their associated C type.

Variants:

F100157, E100157, W100157, I100157

100158:

`NvMBlockDescriptor` and `NvBlockDescriptor` are mapped to different partition

Description:

The `NvBlockDescriptor` of the `NvBlockSwComponentType` is mapped to a different `EcuCPartition` than the associated `NvMBlockDescriptor` (via the `NvMBlockEcucPartitionRef`).

How to Solve:

Map the `NvMBlockDescriptor` and `NvBlockDescriptor` to the same partition.

Variants:

F100158, E100158, W100158, I100158

100160:

`DataSendCompletedEvent` references data element in non-1:1 communication

Description:

A `DataSendCompletedEvent` references a data element that is involved in a 1:N or N:1 communication graph, where N > 1.

How to Solve:

Only reference a data element in a 1:1 communication graph from a `DataSendCompletedEvent`

Variants:

F100160, E100160, W100160, I100160

100162:

`AutosarDataPrototype` associated with incompatible signal

Description:

RTA-RTE has detected that an `AutosarDataPrototype` is mapped to a primitive `ISignal` but it is not a primitive type, a byte array or associated with `DataTransformation`.

How to Solve:

A `SignalGroup` should be used via a `SenderReceiverToSignalGroupMapping`.

Variants:

F100162, E100162, W100162, I100162

100168:

Range of result of data conversion may not fit into the destination's range

Description:

RTA-RTE has detected that, after data conversion, the range of values of the result of data conversion may not fit into the possible range of values of the destination type.

How to Solve:

Ensure that the destination type is able to contain the possible range of values of the converted data.

Variants:

F100168, E100168, W100168, I100168

100170:

ASW `ExclusiveArea` not referenced by an `RteExclusiveAreaRef`

Description:

An ASW `ExclusiveArea` used by a `RunnableEntity` exists in the configuration but no `RteExclusiveAreaImplementation` or `RteExclusiveAreaRef` which references the `ExclusiveArea` could be found in the ECUC. RTA-RTE will default to using the `RteExclusiveAreaImplMechanism ALL_INTERRUPT_BLOCKING` for any `Rte_Enter` and `Rte_Exit` APIs generated to support this `ExclusiveArea`.

How to Solve:

Ensure that all ASW `ExclusiveArea`s in the configuration are referenced by a valid `RteExclusiveAreaRef` within an `RteExclusiveAreaImplementation` in the provided ECU configuration if a different `RteExclusiveAreaImplMechanism` is required.

Variants:

F100170, E100170, W100170, I100170

100171:

BSW `ExclusiveArea` not referenced by an `RteBswExclusiveAreaRef`

Description:

A BSW `ExclusiveArea` used by a `BswSchedulableEntity`, `BswCalledEntity` or `BswInterruptEntity` exists in the configuration but no `RteBswExclusiveAreaImpl` or `RteBswExclusiveAreaRef` which references the `ExclusiveArea` could be found in the ECUC. RTA-RTE will default to using the `RteExclusiveAreaImplMechanism ALL_INTERRUPT_BLOCKING` for any `SchM_Enter` and `SchM_Exit` APIs generated to support this `ExclusiveArea`.

How to Solve:

Ensure that all BSW `ExclusiveArea`s in the configuration are referenced by a valid `RteBswExclusiveAreaRef` within an `RteBswExclusiveAreaImpl` in the provided ECU configuration if a different `RteExclusiveAreaImplMechanism` is required.

Variants:

F100171, E100171, W100171, I100171

100172:

Duplicate non-splitable containers detected

Description:

RTA-RTE has detected duplicate non-splitable containers. RTA-RTE accepts the input if the duplicates are identical when ignoring comments, whitespace and `UUID`s. **Note** comparing duplicates may strongly impact performance.

How to Solve:

Either remove the duplicates or accept the warning (keeping in mind the performance cost).

Variants:

F100172, E100172, W100172, I100172

100173:

Unable to access or create folder

Description:

The specified folder name is not valid or the current user does not have the correct permissions to access or create the folder.

How to Solve:

Check the name and the user permissions of the folder.

Variants:

F100173, E100173, W100173, I100173

100175:

`OS_SPINLOCK` BSW `ExclusiveArea` does not specify an `OsSpinlock`

Description:

A BSW `ExclusiveArea` with the `RteExclusiveAreaImplMechanism` `OS_SPINLOCK` used by a `BswSchedulableEntity`, `BswCalledEntity` or `BswInterruptEntity` in the `canEnter` role exists in the configuration which does not reference an `OsSpinlock` via the ECUC parameter `RteBswExclusiveAreaOsSpinlockRef`.

How to Solve:

Ensure that all `OS_SPINLOCK` BSW `ExclusiveArea`s in the configuration which are used in the `canEnter` role reference a pre-existing `OsSpinlock` via the parameter `RteBswExclusiveAreaOsSpinlockRef` within the relevant `RteBswExclusiveAreaImpl` in the provided ECUC configuration. Accompanying info messages 1001 show the `BswSchedulableEntity`(s), `BswCalledEntity`(s) or `BswInterruptEntity`(s)which reference the `ExclusiveArea` in the `canEnter` role.

Variants:

F100175, E100175, W100175, I100175

100176:

`OS_SPINLOCK` BSW `ExclusiveArea` does not specify an `OsSpinlock`

Description:

A BSW `ExclusiveArea` with the `RteExclusiveAreaImplMechanism` `OS_SPINLOCK` used as a result of the `RteBswEnforceExclusiveAreaApisGeneration` option exists in the configuration which does not reference an `OsSpinlock` via the ECUC parameter `RteBswExclusiveAreaOsSpinlockRef`.

How to Solve:

When using the `RteBswEnforceExclusiveAreaApisGeneration` option, ensure that all `OS_SPINLOCK` BSW `ExclusiveArea`s in the configuration reference a pre-existing `OsSpinlock` via the parameter `RteBswExclusiveAreaOsSpinlockRef` within the relevant `RteBswExclusiveAreaImpl` in the provided ECUC configuration.

Variants:

F100176, E100176, W100176, I100176

100177:

`BswCalled/InterruptEntity` `ExclusiveArea` does not specify an `OsResource`

Description:

A BSW `ExclusiveArea` with the `RteExclusiveAreaImplMechanism` `OS_RESOURCE` used by a `BswCalledEntity` or `BswInterruptEntity` in the `canEnter` role exists in the configuration which does not reference an `OsResource` via the ECUC parameter `RteBswExclusiveAreaOsResourceRef`. RTA-RTE cannot determine the task context of `OsResources` used by `BswCalledEntity`s or `BswInterruptEntity`s so any `OsResources` used in an `ExclusiveArea` by this form of `ExecutableEntity` must be provided by the user.

How to Solve:

Ensure that all `OS_RESOURCE` BSW `ExclusiveArea`s used by a `BswCalledEntity` or `BswInterruptEntity` in the configuration reference a pre-existing `OsResource` via the parameter `RteBswExclusiveAreaOsResourceRef` within the relevant `RteBswExclusiveAreaImpl` in the provided ECUC configuration.

Variants:

F100177, E100177, W100177, I100177

100178:

Server runnable for Nv service callback not mapped to task in split SWCI context

Description:

Attempted to invoke a server runnable whose OperationInvokedEvent is not

mapped to an OsTask for NvM service callback NotifyJobFinished or NotifyInitBlock and whose parent component is split across multiple EcuCPartitions leading to ambiguous partition context.

How to Solve:

The server runnable must be marked as re-entrant and must be mapped to an OsTask so that RTA-RTE can determine the partition context.

Variants:

F100178, E100178, W100178, I100178

100184:

Violation of `SWS_Rte_CONSTR_09096`

Description:

An `Rte_DRead` API is configured in the input with the respective `PortPrototype` referenced by a `PortAPIOption` which has the attribute `errorHandling` set to `transformerErrorHandling` violating `SWS_Rte_CONSTR_09096`.

How to Solve:

Disable `transformerErrorHandling` or, if `transformerErrorHandling` is required use `dataReceivePointByArgument`.

Variants:

F100184, E100184, W100184, I100184

100187:

ScaleLinearAndTextTable CompuMethod contains too many rational CompuScale

Description:

A CompuMethod of category ScaleLinearAndTextTable contains more than 1 CompuScales that aggregate CompuScaleRationalFormula. This contravenes `TPS_SWCT_01560`.

How to Solve:

Ensure that CompuMethod of category ScaleLinearAndTextTable aggregate no more than 1 CompuScale that aggregates a CompuScaleRationalFormula.

Variants:

F100187, E100187, W100187, I100187

100188:

Multiple internal behaviors for a `BswModule` mapped to same `SWC`

Description:

A `BswModuleDescription` contains multiple `InternalBehaviors` that are mapped to the same `AtomicSwComponentType` via `SwcBswMappings`.

How to Solve:

Ensure that no two `InternalBehaviors` from a single `BswModuleDescription`

are mapped to the same `AtomicSwComponentType`.

Variants:

F100188, E100188, W100188, I100188

100189:

Invalid `OsResourceProperty` for ExclusiveArea

Description:

An `OsResource` used in a `SchM_Enter` and `SchM_Exit` API provided via the ECUC parameter `RteBswExclusiveAreaOsResourceRef` has an `OsResourceProperty` set to `INTERNAL`. Any pre-defined `OsResource`s used within a `SchM_Enter` and `SchM_Exit` API must have their `OsResourceProperty` set to `STANDARD`.

How to Solve:

Ensure that any `OsResource`s used within `SchM_Enter` and `SchM_Exit` APIs have their `OsResourceProperty` set to `STANDARD`.

Variants:

F100189, E100189, W100189, I100189

100192:

Conflicting API names for a BSW module

Description:

The input configuration resulted in the generation of two or more APIs with the same C symbol.

How to Solve:

API names for *Basic Software Module*s supporting multiple instances of the same *Basic Software Module* must contain the infix `bsnp[_vi_ai]` (see `SWS_Rte_70036@R23-11`). To ensure that all APIs for a given *Basic Software Module* with multiple `InternalBehavior`s have different names, either unique `BswScheduleNamePrefix`s or `VendorId`s and `VendorApiInfix`s must be provided (or both).

Variants:

F100192, E100192, W100192, I100192

100194:

Unmapped `BswInterruptEvent` detected

Description:

RTA-RTE has detected a `BswInterruptEvent` that is not mapped to an `ISR2` violating `[SWS_Rte_08924]@R23-11`.

How to Solve:

Either remove the `BswInterruptEvent` from the input or map it to an `ISR2` using an `RteBswEventToIsrMapping`.

Variants:

F100194, E100194, W100194, I100194

100195:

Mismatching `NetworkRepresentationProps` for `ISignal` `DataPrototype`

Description:

An `ISignal`, aggregating `TransformationISignalProps`, has a mismatch between the number of leaf-elements of the associated `AutosarDataPrototype` and the `NetworkRepresentationProps`s. This error corresponds to AUTOSAR constraint `constr_3319`.

How to Solve:

Ensure that for each leaf element of the associated `AutosarDataPrototype` there is a matching number of `NetworkRepresentationProps` or that there are no `NetworkRepresentationProps`.

Variants:

F100195, E100195, W100195, I100195

100197:

Mismatching `DataPrototypeTransformationProps` for `ISignal` `DataPrototype`

Description:

An `ISignal`, aggregating `TransformationISignalProps`, has a mismatch between the number of leaf-elements of the associated `AutosarDataPrototype` and the `DataPrototypeTransformationProps`s. This error corresponds to AUTOSAR constraint `constr_3319`.

How to Solve:

Ensure that for each leaf element of the associated `AutosarDataPrototype` there is a matching number of `DataPrototypeTransformationProps` or that there are no `DataPrototypeTransformationProps`.

Variants:

F100197, E100197, W100197, I100197

100198:

Missing `NetworkRepresentationProps.``baseType` reference for an `ISignal` aggregating a `TransformationISignalProps`

Description:

For all leaf elements, a reference to a `SwBaseType` is required when specifiying the network representation for an `ISignal`.

How to Solve:

Ensure that the `ISignal.``transformationISignalProps``.``dataPrototypeTransformationProps``.``networkRepr` references a `SwBaseType`.

Variants:

F100198, E100198, W100198, I100198

900000:

Internal Error

Description:

All error numbers from 900000 upwards relate to internal errors: RTA-RTE encountered an unexpected internal state and behavior is now outside defined bounds

How to Solve:

Contact your provider

Variants:

F900000, E900000, W900000, I900000

3 ETAS Contact Addresses

3.1 Technical Support

Technical support is available to all RTA-RTE users with a valid support contract. If you do not have a valid support contract, please contact your regional sales office (see below).

The best way to get technical support is by email. Any problems or questions about the use of RTA-RTE should be sent to: rta.hotline@etas.com

If you prefer to discuss your problem with the technical support team, you can call the support hotline on: +44 (0)1904 562624. The hotline is available during normal office hours (0900-1730 GMT/BST).

In either case, when contacting technical support, it is helpful if you can provide as much of the following information (where relevant):

- Your support contract number.
- The versions of the ETAS tools you are using.
- The version of the compiler tool chain you are using.
- The command line (or reproduction of steps) that result in an error message.
- The error messages or return codes you received (if any).
- Your .xml and .arxml files.
- Your OS configuration files.
- The diagnostic.dmp file (if it was generated).

3.2 General Enquiries

3.2.1 ETAS Global Headquarters

ETAS GmbH
Borsigstrasse 24
70469 Stuttgart
Germany

Phone: +49 711 3423-0
Fax: +49 711 3423-2106
www: https://www.etas.com

3.2.2 ETAS Local Sales and Support Offices

Contact details for your local sales office and local technical support team (where available) can be found on the ETAS website:

ETAS subsidiaries: https://www.etas.com/en/contact.php
ETAS technical support: https://www.etas.com/en/hotlines.php